A32NX flyPad failures

Clearer A32NX failure names and new failures with their FCOM alerts: transponders, RMPs and VHF radios, FWC 1 and 2, engine and APU fires with a real fire detection model and fire detection unit failures, wing tank pumps, the X FEED valve, the centre tank transfer valves, engine flameouts and seizures with the in-flight relight envelope, and the engine oil, ignition and start, stall, EGT, overspeed, vibration, FADEC, thrust lever and reverser failures with the FCOM start sequence, the fan blocked failure with ENG LOW N1, and the THR LEVER ABV IDLE and SAT ABOVE FLEX TEMP alerts; then engine and APU fire visual effects, the electrical (battery, BCL, IDG) and pneumatic (bleed valves, leaks, BMC) failures, the FCOM ELEC, AIR bleed, HYD, APU shutdown and F/CTL alerts fixes (FADEC relight, automatic ignition, gravity feed, IR alignment), the DC ESS supply in DC BUS 1+2 fault and the LEAP-1A N1 / N2 red lines; and the crew recovery procedures after failures (FAC reset, AC BUS 1 transfer, SEC test, IDG, manual cabin pressure, ignition indications, engine fire run-on, A/THR without V2); then the A32NX parts (FAC actuator, ADR / IR, brake control unit, cargo and avionics smoke, APU ECB, SFCC channel, HP fuel valve, FADEC fault, LP valve, FQI, engine anti-ice valve, probe and window heat and TCAS computer failures with their alerts; alerts on existing data, PFD FMA messages, SD advisory mode, CVR power, CPC landing elevation, BLEED page valves, gravity feed ceiling, GPWS self-test, reactive windshear, FM/GPS and LS tuning disagree), the A380X smoke detection function with its NSS zone, the shared relight EGT watch, the sidestick priority / intermediate RA / WINDSHEAR voices, the FWC stall aural priority and TCAS voice hold, the exceedance marks fix and the 5 s DC BUS 1 transfer

Aircraft	A32NX (the fire visual effects, the smoke detection function and the anti-ice systems part also A380X)
Where to find it	flyPad > Failures (ATA 21, 22, 23, 24, 26, 27, 28, 29, 30, 31, 32, 34, 36, 49, 72, 73, 74, 77, 78, 79, 80); flyPad > Settings > Sim Options; E/WD, STATUS and SD pages; PFD FMA; pedestal ATC and RMP panels; overhead FIRE, CARGO SMOKE, FUEL, ELEC, AIR COND, HYD, ADIRS, ANTI ICE, RCDR and ENG MAN START panels; glareshield GPWS pb; A/SKID & N/W STRG and autobrake; ENG MASTER and ENG MODE controls; thrust levers; exterior model
Branch	`core/failures-a32nx`, based on `core/gps` (stacked: needs core/surveillance, core/a380x-mfd-fms and core/gps merged first), with `core/fire` merged in (the fire failures use its fire warnings, and the engine failures use its ENG FIRE pb LP valve starvation). Since it also merges `core/fcom-audit-fixes` and `core/a32nx-air-conditioning`: the new FWC alerts build on their code (F/CTL law lines, APU memos, GPWS lamp test, air conditioning STATUS lines). Merge them into this branch when they change. Only the features below belong to this branch; the GPS, fire warning, FCOM audit and air conditioning features are described in the core/gps, core/fire, core/fcom-audit-fixes and core/a32nx-air-conditioning manuals. Some engine commits also carry A380X files (they are on `core/failures-a380x`); this manual describes only what the A32NX gets, except the engine and APU fire visual effects, whose A380X part is on this branch too (the fire model of both aircraft lives here). It now also carries the A380X smoke detection function (SDF) and the A380X anti-ice systems and cockpit part, whose A380X alerts are on `core/a380x-elec-fuel-alerts`: merge both.
Status	Transponder, RMP/VHF, FWC, fire and fuel failures tested in the sim by the user: OK; the fire detection unit failures, the FIRE TEST light logic, the APU fire horn and the centre tank transfer valve failures too (the horn itself is silent in MSFS 2024); the engine flameout and seizure failures: OK, tests; their LEAP-1A relight envelope: sim-tested OK (recorded); the in-flight start without N2 overshoot: OK (recorded). Engine failures A2-A5 (test flight and retest, recorded): oil failures, ground starts with a failed igniter and the dry crank, stall, EGT overtemperature and vibration, thrust lever resolvers and FADEC FAULT, normal reverse thrust: OK; the other engine failures, the later tuning and the T.O CONFIG NORMAL / WING TK LO LVL fixes: not sim-tested. Clearer names: not sim-tested (labels only). Engine alerts (feature branch): SAT ABOVE FLEX TEMP OK (recorded); THR LEVER

	ABV IDLE and the fan blocked / LOW N1 OK (recorded). The fuel filter clog and the stall during a relight fix: not sim-tested. Later commits, tested by the user on a combined test build (recorded): fire visual effects OK; cabin pressure fixes OK; FADEC relight and the 10 min IR alignment OK, but the automatic continuous ignition FAILED (no IGNITION memo); ELEC alerts OK; electrical failures OK; pneumatic failures OK; AIR bleed (BLEED 1 OFF part) OK; HYD and APU alerts OK; F/CTL alerts OK; DC ESS supply OK; LEAP-1A red lines OK. Nothing is sim-tested on this branch itself. Recovery procedures (checked against the code and the FCOM): FAC reset OK, manual cabin V/S OK, the others per the two last sections. The later FCOM alert additions: not sim-tested; unit/Rust tests only. The A380X NSS smoke zone and the shared relight EGT watch (placed 2026-10-18): not sim-tested on the core branch; Rust tests only.
Pictures	No pictures: the failures page, the ECAM alerts and the panels are the stock FBW displays.

1. Clearer failure names
2. Transponder failures
3. RMP and VHF failures
4. FWC 1 and FWC 2 failures
5. Fire detection model and fire failures
6. Fire detection unit failures, FIRE TEST light logic, APU fire horn
7. Wing tank pump and X FEED valve failures
8. Centre tank transfer valve failures and fuel follow-ups
9. Engine flameout and seizure failures, in-flight relight
10. Engine oil failures: leak, filter clog, overheat
11. Engine fuel filter clog
12. Engine ignition and start failures, FCOM start sequence
13. Engine stall, EGT overtemperature, overspeed and vibration failures
14. FADEC, thrust lever and reverser failures
15. T.O CONFIG NORMAL and FUEL WING TK LO LVL fixes
16. ENG THR LEVER ABV IDLE and ENG SAT ABOVE FLEX TEMP
17. Engine fan blocked failure and ENG LOW N1
18. Engine and APU fire visual effects
19. FCOM fixes: DIRECT LAW trim, GPWS mode 2, CAB PRESS and oil temperature indications
20. FCOM fixes: FADEC relight, automatic ignition, start faults, gravity feed, IR alignment
21. ELEC bus, ESS bus and TR alerts; DC ESS supply in DC BUS 1+2 fault
22. Electrical failures: battery thermal runaway, BCL, IDG oil
23. Pneumatic failures with BMC leak and bleed monitoring
24. AIR bleed cautions
25. HYD and APU shutdown alerts
26. F/CTL surface cautions and STATUS
27. FAC actuator failures, YAW DAMPER / RUD TRIM / RUD TRV LIM cautions
28. ADR and IR failures, ADIRU power supplies, NAV ADR / IR / RA alerts
29. alerts on existing data (CAB PR, THRUST LOCKED, fuel, WAI, doors, L/G, GPWS)
30. PFD FMA messages, AVAIL, SD FUEL imbalance, SD BLEED crossbleed valve
31. SD advisory mode
32. CVR automatic power and CVR TEST
33. CPC landing elevation without FMGS data, EXCESS CAB ALT threshold
34. brake and steering control units (BSCU, ABCU), BRAKES alerts
35. APU automatic shutdowns, cargo and avionics smoke, cargo extinguishing
36. A380X smoke detection function (systems part)
37. A380X NSS smoke zone and the SMOKE light of the NSS MASTER sw
38. Relight EGT watch in the shared engine start (no change on the A32NX)
39. BLEED page HP valve and bleed valve disagree
40. HP fuel valve, FADEC fault failures, pre-start motoring
41. gravity feed ceiling, LP valve and FQI failures
42. engine anti-ice valves, probe and window heat, anti-ice key events
43. SFCC flap and slat channel failures
44. GPWS self-test, TCAS computer failure, reactive windshear
45. NAV FM/GPS POS DISAGREE and NAV LS TUNING DISAGREE
46. Voices: sidestick priority, intermediate RA callouts, WINDSHEAR
47. FWC aural priority: stall warning, TCAS voices, C chord in a TCAS RA
48. Engine exceedance marks only on displayed values
49. DC BUS 2 takes over DC BUS 1 after 5 s
50. Recovery after electrical failures: FAC reset, AC BUS 1 transfer, SEC test, IDG
51. Crew procedures: manual cabin pressure, ignition indications, engine fire, A/THR without V2

1. Clearer failure names

> **Aircraft:** A32NX | **Where:** flyPad > Failures | **Status:** Not sim-tested (labels only).

What it is for

The failure names were audited ATA by ATA against what the systems code fails. Three names were misleading: the ATA 21 valve is the hot air pressure regulating valve, the FMGC failures only fail the flight guidance part of the FMGC,

and the RA failures are antenna failures. Only the labels change.

ATA	Before	Now
21	Trim Air Pressure Regulating Valve	Hot Air Pressure Regulating Valve
22	FMGC 1 / FMGC 2	FMGC 1 (flight guidance) / FMGC 2 (flight guidance)
34	RA 1 Interrupted / RA 2 Interrupted	RA 1 antenna interrupted / RA 2 antenna interrupted
34	RA 1 Direct Coupling / RA 2 Direct Coupling	RA 1 antenna direct coupling / RA 2 antenna direct coupling

How to use

1. Open flyPad > Failures and pick the failure by its new name (table above).

How to test

#	Do this	Expected result
1	Open flyPad > Failures > ATA 21 Air Conditioning.	"Hot Air Pressure Regulating Valve" is listed.
2	Open ATA 22 Auto Flight.	"FMGC 1 (flight guidance)" and "FMGC 2 (flight guidance)".
3	Open ATA 34 Navigation.	"RA 1 antenna interrupted", "RA 2 antenna interrupted", "RA 1 antenna direct coupling", "RA 2 antenna direct coupling".

2. Transponder failures

> **Aircraft:** A32NX | **Where:** flyPad > Failures > ATA 34 Navigation; pedestal ATC/TCAS panel; E/WD, STATUS | **Status:** Tested in the sim by the user: OK.

What it is for

Two new failures, **ATC/XPDR 1** and **ATC/XPDR 2**. MSFS has a single transponder: it stands for the transponder selected on the ATC panel (SYS 1 / SYS 2). When the selected one fails, the sim transponder stops replying and the ATC FAIL light comes on; selecting the other system brings the replies back. With both lost, the TCAS goes to standby. The ECAM alerts and STATUS follow the A320 FCOM PRO-ABN-NAV procedures.

Fix on the way: the ATC/XPDR STBY indication read a second sim transponder that does not exist; the standby state of each transponder now comes from the system selection and the mode selector.

What you should see

Situation	What you should see
ATC/XPDR 1 failed, SYS 1 selected	ATC FAIL light; no transponder reply. E/WD amber **NAV ATC/XPDR 1 FAULT** with -ATC/XPDR ..... SYS 2 (only while XPDR 1 is selected and XPDR 2 works). STATUS INOP SYS: ATC/XPDR 1, ADS-B RPTG 1.
SYS 2 selected	ATC FAIL off, replies again; the procedure line is done.
ATC/XPDR 2 failed	The same with NAV ATC/XPDR 2 FAULT, -ATC/XPDR ..... SYS 1, INOP SYS ATC/XPDR 2, ADS-B RPTG 2.
Both failed	**NAV ATC/XPDR 1+2 FAULT** (crew awareness); STATUS INOP SYS: TCAS, ATC/XPDR 1, ATC/XPDR 2, ADS-B RPTG 1, ADS-B RPTG 2; the TCAS goes to standby.

A transponder loses its bus (AC ESS SHED for XPDR 1, AC 2 for XPDR 2)	The transponder is lost, but no ATC/XPDR FAULT: a bus loss is an ELEC alert.

Limitations and why

Limitation	Why	What to do
The ATC/XPDR alerts are inhibited during the takeoff and below 800 ft in approach to the landing roll.	The FCOM gives the inhibition as a figure: the one of the RA faults is used (flight phases 3, 4, 5, 7, 8).	Test at the gate or in cruise.

How to use

1. Open flyPad > Failures > ATA 34 Navigation and activate **ATC/XPDR 1** or **ATC/XPDR 2**.
2. Follow the ECAM: select the other system on the ATC/TCAS panel.

How to test

#	Do this	Expected result
1	A32NX at the gate, powered, ATC/TCAS panel: SYS 1, mode AUTO (or ON).	No ATC FAIL light.
2	Activate **ATC/XPDR 1**.	ATC FAIL light on; master caution and single chime; E/WD NAV ATC/XPDR 1 FAULT with -ATC/XPDR ..... SYS 2. ATC (or a traffic tool) no longer sees your transponder.
3	Open the STATUS page.	INOP SYS: ATC/XPDR 1, ADS-B RPTG 1.
4	Select SYS 2 on the ATC panel.	ATC FAIL off, the transponder replies again, the SYS 2 line is done.
5	Activate **ATC/XPDR 2** as well.	NAV ATC/XPDR 1+2 FAULT; STATUS INOP SYS: TCAS, ATC/XPDR 1, ATC/XPDR 2, ADS-B RPTG 1, ADS-B RPTG 2; the TCAS is in standby.
6	Deactivate both failures.	The alerts clear; the TCAS works again.

3. RMP and VHF failures

> **Aircraft:** A32NX | **Where:** flyPad > Failures > ATA 23 Communication; pedestal RMP 1 and RMP 2 | **Status:** Tested in the sim by the user: OK.

What it is for

Five new ATA 23 failures: **RMP 1**, **RMP 2**, **VHF 1**, **VHF 2** and **VHF 3** (same ids as the A380X ATA 23 failures). A failed RMP goes blank and stops tuning; the other RMP still tunes every VHF (A320 FCOM DSC-23-10-20: "If one RMP fails, the remaining one controls all the VHF transceivers"). A failed VHF switches its MSFS COM radio off: it neither receives nor transmits.

What you should see

Situation	What you should see
RMP 1 (or 2) failed	Both frequency displays blank, the green key lights and SEL off, the keys and knobs do nothing (FCOM DSC-23-60 failure cases).

The other RMP	Still tunes VHF 1, 2 and 3.
VHF 1 (2, 3) failed	COM 1 (2, 3) silent: no reception, no transmission. The RMPs still show and change its frequency.

Limitations and why

Limitation	Why	What to do
No ECAM alert for these failures.	Not part of this change.	Check the RMP and the radio.
A NAV key left on on a failed RMP still holds the FMGC navaid tuning.	As with an RMP switched off: the navaid tuner (shared with the A380X) reads the NAV key only.	Avoid the NAV key before failing an RMP.
No RMP 3 and no audio control panel / AMU failure.	RMP 3 is not in the A32NX cockpit, and the MSFS radios have no audio channel to fail.	None.

How to use

1. Open flyPad > Failures > ATA 23 Communication and activate RMP 1, RMP 2, VHF 1, VHF 2 or VHF 3.
2. With an RMP failed, tune the radios from the other RMP.

How to test

#	Do this	Expected result
1	A32NX powered, an ATIS frequency on VHF 1, audible.	The ATIS is heard.
2	Activate **RMP 1**.	RMP 1 goes blank, its key lights go off, its knobs do nothing. The ATIS is still heard (VHF 1 itself works).
3	On RMP 2, press the VHF 1 key and tune another frequency, transfer it.	VHF 1 changes frequency: RMP 2 tunes VHF 1.
4	Deactivate RMP 1. Activate **VHF 1**.	COM 1 goes silent; transmitting on VHF 1 is not possible. RMP 1 still shows the frequency.
5	Deactivate VHF 1.	The ATIS is heard again.

4. FWC 1 and FWC 2 failures

> **Aircraft:** A32NX | **Where:** flyPad > Failures > ATA 31 Indicating / Recording System; E/WD, STATUS | **Status:** Tested in the sim by the user: OK.

What it is for

Two new failures, **FWC 1** and **FWC 2** (A320 FCOM DSC-31-05-30, PRO-ABN-FWS). With one FWC failed, the other one keeps the warning system working and shows the fault. With both failed, the ECAM alerts, memos, STATUS, MASTER WARN / CAUT lights, aurals (including the STALL aural) and the altitude callouts are lost, and the E/WD shows the FWC 1+2 FAULT display. Each FWC also follows its own electrical supply.

What you should see

Situation	What you should see
FWC 1 failed	E/WD **FWS FWC 1 FAULT** (shown by FWC 2), level 1: no MASTER CAUT, no chime. STATUS: CAT 3 SINGLE ONLY; INOP SYS: CAT 3 DUAL, FWC 1.
FWC 2 failed	The same with FWS FWC 2 FAULT and FWC 2.

Both failed	E/WD: **FWS FWC 1+2 FAULT**, -MONITOR SYS, -MONITOR OVERHEAD PANEL; right side NOT AVAIL: ECAM WARN, ALTI ALERT, STATUS, A/CALL OUT, MEMO. No other alert, memo, STATUS, STS reminder, master light or aural.
One FWC restored	Every active alert is shown again, as a new one.

Limitations and why

Limitation	Why	What to do
FWS FWC 1(2) FAULT is level 1 and inhibited during the takeoff and the landing (phases 3, 4, 5, 7, 8).	The FCOM gives the level and the inhibition in figures only: those of the other computer faults are used.	Test at the gate or in cruise.

How to use

1. Open flyPad > Failures > ATA 31 Indicating / Recording System and activate FWC 1 and / or FWC 2.
2. With both failed, monitor the systems and the overhead panel yourself.

How to test

#	Do this	Expected result
1	A32NX at the gate, powered, no alert. Activate **FWC 1**.	E/WD: FWS FWC 1 FAULT, no master caution, no chime.
2	Open the STATUS page.	CAT 3 SINGLE ONLY; INOP SYS: CAT 3 DUAL, FWC 1.
3	Activate **FWC 2** as well.	E/WD: FWS FWC 1+2 FAULT, -MONITOR SYS, -MONITOR OVERHEAD PANEL, and NOT AVAIL: ECAM WARN, ALTI ALERT, STATUS, A/CALL OUT, MEMO. The memos are gone.
4	Press the STS key on the ECAM control panel.	No STATUS page (STATUS is in the NOT AVAIL list).
5	Deactivate FWC 2.	FWS FWC 1 FAULT again; the active alerts come back as new ones.

5. Fire detection model and fire failures

> **Aircraft:** A32NX | **Where:** flyPad > Failures > ATA 26 Fire Protection; overhead FIRE panel; ENG MASTER panel; E/WD, STATUS | **Status:** Tested in the sim by the user: OK. An agent putting out an MSFS failures menu fire: not sim-tested (Rust fire protection tests).

What it is for

Nine new failures: **Engine 1 fire**, **Engine 2 fire**, **APU fire**, and **Engine 1 fire detection loop A**, **Engine 1 fire detection loop B**, **Engine 2 fire detection loop A**, **Engine 2 fire detection loop B**, **APU fire detection loop A**, **APU fire detection loop B**.

Each zone now has the two detection loops and the fire detection unit of the aircraft (A320 FCOM DSC-26-20-10). A fire warning comes when both loops detect the fire, when one loop detects it and the other one is failed, or when both loops break within 5 s (flame effect). The logic moved from the A380X into the common code; the A380X behaviour is unchanged. The fires of the MSFS failures menu still give the warnings, through the loops; an AGENT discharge (or the automatic APU discharge) can also put such a fire out, which ends the ENG / APU FIRE warning, and the fire stays out while the MSFS failure stays on (both aircraft). The FIRE TEST is unchanged.

- The ENG MASTER panel FIRE light comes on with the engine fire warning (before: FIRE TEST only).

- APU fire on the ground: the APU shuts down and the APU AGENT discharges automatically 3 s after the warning (FCOM DSC-26-20-20).
- Each AGENT discharge has one chance in two to put a flyPad fire (or an MSFS failures menu fire) out.
- Loop supplies (FCOM PRO-ABN-ELEC INOP SYS): DC BUS 2 for ENG 1 LOOP B and ENG 2 LOOP A, DC ESS BUS for ENG 1 LOOP A and ENG 2 LOOP B, DC BAT BUS for both APU loops. An unpowered loop is a failed loop.

What you should see

Situation	What you should see
Engine 1 fire	ENG 1 FIRE warning (from `core/fire`), red ENG 1 FIRE pushbutton light and red FIRE light on the ENG MASTER panel.
APU fire on the ground	APU FIRE warning; the APU shuts down; 3 s later the APU AGENT discharges (DISCH light).
AGENT discharged	The fire goes out, or not (one chance in two per discharge).
One loop failed	Amber **ENG 1 FIRE LOOP A FAULT** (LOOP B, ENG 2, APU FIRE LOOP A / B); STATUS INOP SYS ENG 1 LOOP A (ENG 1 LOOP B, ENG 2 LOOP A / B, APU LOOP A / B). A fire is still detected by the other loop.
Both loops of a zone failed	**ENG 1 FIRE DET FAULT** (ENG 2, APU FIRE DET FAULT) instead of the two loop cautions; STATUS INOP SYS FIRE DET 1 (FIRE DET 2, APU FIRE DET).

Limitations and why

Limitation	Why	What to do
FIRE DET FAULT also comes from a failed fire detection unit.	See the next feature, "Fire detection unit failures".	-
The LOOP and DET FAULT cautions are inhibited during the takeoff and the landing (phases 3, 4, 5, 7, 8).	The FCOM gives the inhibition as a figure: that of the other crew awareness cautions is used.	Test at the gate or in cruise.
A full MSFS restart is needed after an update.	The systems WASM module and the cockpit behaviour XML change.	Restart the sim after installing.

How to use

1. Open flyPad > Failures > ATA 26 Fire Protection and activate a fire or a loop failure.
2. Fight a fire with the ECAM procedure (FIRE pushbutton, AGENT 1, AGENT 2). Deactivating the failure also ends the fire.

How to test

#	Do this	Expected result
1	A32NX on the ground, engines running. Activate **Engine 1 fire**.	ENG 1 FIRE warning, master warning, continuous chime, red light in the ENG 1 FIRE pushbutton and on the ENG MASTER 1 panel.
2	Follow the ECAM: ENG MASTER 1 OFF, push the ENG 1 FIRE pushbutton, press AGENT 1.	AGENT 1 DISCH; the fire goes out or not (one chance in two). If not, AGENT 2 gives another chance.
3	Deactivate the failure, reset the FIRE pushbutton. With the APU running, activate **APU fire**.	APU FIRE warning; the APU shuts down; about 3 s later the APU AGENT discharges by itself.

4	Deactivate it. Activate **Engine 2 fire detection loop A**.	Amber ENG 2 FIRE LOOP A FAULT; STATUS INOP SYS ENG 2 LOOP A.
5	Activate **Engine 2 fire** with loop A still failed.	ENG 2 FIRE warning: loop B alone detects the fire.
6	Deactivate the fire, activate **Engine 2 fire detection loop B** as well.	The loop cautions are replaced by ENG 2 FIRE DET FAULT; STATUS INOP SYS FIRE DET 2.
7	Optional: start an engine fire from the MSFS failures menu.	The ENG FIRE warning still comes (through the loops).
8	Optional (not sim-tested): with that MSFS menu fire still on, push the ENG FIRE pb and press AGENT 1 (AGENT 2 if the FIRE pb stays red).	One agent in two puts the fire out: the ENG FIRE warning and the FIRE pb light go, and stay off while the MSFS failure stays on.

6. Fire detection unit failures, FIRE TEST light logic, APU fire horn

> **Aircraft:** A32NX | **Where:** flyPad > Failures > ATA 26 Fire Protection; overhead FIRE panel; ENG MASTER panel; E/WD, STATUS | **Status:** Tested in the sim by the user: OK; the horn sound itself is silent in MSFS 2024. Rust tests (fire_protection), unit tests: FireDetectionFaults.spec.ts, EngineMasterFireLight.spec.ts.

What it is for

Three follow-ups of the fire detection model above:

- **Fire detection unit failures:** three new failures, **Engine 1 fire detection unit (FDU)**, **Engine 2 fire detection unit (FDU)** and **APU fire detection unit (FDU)**. A failed FDU gives the FIRE DET FAULT caution of its zone even with healthy loops, and detects no fire. Clearing the failure with a fire present gives the fire warning at once.
- **ENG MASTER panel FIRE light during the FIRE TEST:** the A320 FCOM (DSC-26-20-20, ENG 1(2) FIRE TEST pb) says that the FIRE lights do not come on with an engine 1 loop A fault or an engine 2 loop B fault. The light now stays dark in those cases (a loop fault is its failure or the loss of its DC ESS supply).
- **APU fire external horn:** an APU fire on the ground sounds the horn in the nose gear bay (FCOM DSC-26-20-20 APU FIRE light, DSC-49-10-20), on the same sound as the MECH call; not during the FIRE TEST.

What you should see

Situation	What you should see
Engine 1 FDU failed	Amber **ENG 1 FIRE DET FAULT** (ENG 2 FIRE DET FAULT, APU FIRE DET FAULT for the other units), no LOOP FAULT caution (a failed FDU cannot tell which loop is failed); STATUS INOP SYS FIRE DET 1 (FIRE DET 2, APU FIRE DET).
A fire in a zone with its FDU failed	No fire warning.
FDU failure cleared while the fire burns	The fire warning comes at once.
ENG 1 FIRE TEST with Engine 1 fire detection loop A failed	The ENG MASTER 1 FIRE light stays dark; the rest of the test as before.
ENG 2 FIRE TEST with Engine 2 fire detection loop B failed	The ENG MASTER 2 FIRE light stays dark.
APU fire on the ground	The external horn variable is set (L:A32NX_FIRE_APU_EXTERNAL_HORN); the horn sound plays at the nose gear.

Limitations and why

Limitation	Why	What to do

The APU fire horn is not heard in MSFS 2024.	The Asobo horn sound it uses is silent in MSFS 2024 (the MECH call too).	Check the L:var with a SimConnect / variable tool.
No APU SHUT OFF pb and no APU FIRE light on the external power panel.	The exterior model has no such panel.	None.
A full MSFS restart is needed after an update.	The systems WASM module, the cockpit behaviour XML and the sound.xml change.	Restart the sim after installing.

How to use

1. Open flyPad > Failures > ATA 26 Fire Protection and activate an "... fire detection unit (FDU)" failure.
2. Follow the ECAM (crew awareness); deactivate the failure to restore the unit.

How to test

#	Do this	Expected result
1	A32NX at the gate, powered, no failure. Activate **Engine 1 fire detection unit (FDU)**.	Amber ENG 1 FIRE DET FAULT, no ENG 1 FIRE LOOP A / B FAULT; STATUS INOP SYS FIRE DET 1.
2	Also activate **Engine 1 fire**.	No ENG 1 FIRE warning.
3	Deactivate **Engine 1 fire detection unit (FDU)**.	The ENG 1 FIRE warning comes at once. Deactivate the fire.
4	Repeat with **Engine 2 fire detection unit (FDU)** and **APU fire detection unit (FDU)**.	ENG 2 FIRE DET FAULT / STATUS FIRE DET 2; APU FIRE DET FAULT / STATUS APU FIRE DET.
5	Activate **Engine 1 fire detection loop A**, then press and hold the ENG 1 FIRE TEST pb.	The ENG 1 FIRE pushbutton lights as in the normal test, but the FIRE light on the ENG MASTER 1 panel stays dark.
6	Deactivate it, activate **Engine 2 fire detection loop B**, hold the ENG 2 FIRE TEST pb.	The ENG MASTER 2 FIRE light stays dark. With loop A of engine 2 failed instead, it comes on.
7	On the ground with the APU running, activate **APU fire**.	APU FIRE warning; L:A32NX_FIRE_APU_EXTERNAL_HORN is 1 (the horn itself is silent in MSFS 2024).

7. Wing tank pump and X FEED valve failures

> **Aircraft:** A32NX | **Where:** flyPad > Failures > ATA 28 Fuel; overhead FUEL panel; SD FUEL | **Status:** Tested in the sim by the user: OK, after the fix that holds the MSFS pump / valve switch.

What it is for

Five new failures: **L TK pump 1**, **L TK pump 2**, **R TK pump 1**, **R TK pump 2** and **X FEED valve jammed in place**. The FUEL panel pushbuttons now set a selection, and the systems host moves the MSFS pump or valve switch from it: a failed pump is held off, the jammed valve is held where it was. The FAULT lights, the SD FUEL page and the cautions compare the selection with the real state (A320 FCOM PRO-ABN-FUEL).

- SD FUEL page: amber LO only for a pump that is ON with a low pressure; the X FEED valve is amber when it disagrees with its pushbutton (FCOM colours).
- The FUEL X FEED memo now needs the X FEED pushbutton ON and the valve not fully closed.

What you should see

Situation	What you should see
L TK pump 1 failed, pushbutton ON	FAULT light; amber LO on the SD FUEL page (page called). E/WD **FUEL L TK PUMP 1 LO PR** with -L TK PUMP 1 ..... OFF (shown while the pushbutton is ON). STATUS INOP SYS L TK PUMP 1.
Both pumps of one wing failed	**FUEL L TK PUMP 1+2 LO PR** (R for the right wing): .IF NO FUEL LEAK: -FUEL X FEED ON (above FL 150 only), -ENG MODE SEL IGN, -L TK PUMP 1 OFF, -L TK PUMP 2 OFF, .WHEN L TK FUEL RQRD: L TK FEED GRVTY ONLY, -FUEL X FEED OFF (below FL 150, X FEED ON), PROC:GRVTY FUEL FEEDING, AVOID NEGATIVE G FACTOR. Each action line goes once done. STATUS INOP SYS L TK PUMPS.
X FEED valve jammed, pushbutton moved	The valve stays: amber on the SD FUEL page, **FUEL X FEED VALVE FAULT** (crew awareness), STATUS INOP SYS FUEL X FEED. With the valve stuck closed, no FUEL X FEED memo.

Limitations and why

Limitation	Why	What to do
The cautions are inhibited during the takeoff and the landing (phases 3, 4, 5, 7, 8).	The FCOM gives the inhibition as a figure: that of the other single system faults is used.	Test at the gate or in cruise.
The centre tank has no pump failure.	The A320neo centre tank has no pumps: its fuel is moved by the jet pumps through the CTR TK L / R XFR valves, whose jams are a feature of their own (below).	None.

How to use

1. Open flyPad > Failures > ATA 28 Fuel and activate a pump or the X FEED valve failure.
2. Follow the ECAM with the overhead FUEL panel; check the SD FUEL page.

How to test

#	Do this	Expected result
1	A32NX at the gate, APU or engines running, all fuel pumps ON. Activate **L TK pump 1**.	After a few seconds: FAULT light in the L TK PUMP 1 pushbutton, SD FUEL page with amber LO, master caution, E/WD FUEL L TK PUMP 1 LO PR with -L TK PUMP 1 ..... OFF.
2	Set the L TK PUMP 1 pushbutton OFF. Open STATUS.	The FAULT light and the OFF line go; no LO on the SD; STATUS INOP SYS L TK PUMP 1.
3	Set it ON again and also activate **L TK pump 2**.	FUEL L TK PUMP 1+2 LO PR with its procedure (on the ground, below FL 150: .IF NO FUEL LEAK:, -ENG MODE SEL IGN, -L TK PUMP 1 OFF, -L TK PUMP 2 OFF, ...); STATUS INOP SYS L TK PUMPS.
4	Deactivate both. With the X FEED valve closed, activate **X FEED valve jammed in place**, then set the X FEED pushbutton ON.	The valve stays closed: amber X FEED valve on the SD, FUEL X FEED VALVE FAULT, STATUS INOP SYS FUEL X FEED, no FUEL X FEED memo.
5	Deactivate the failure.	The valve opens; the FUEL X FEED memo comes; the caution clears.

8. Centre tank transfer valve failures and fuel follow-ups

> **Aircraft:** A32NX | **Where:** flyPad > Failures > ATA 28 Fuel; overhead FUEL panel; SD FUEL; flyPad checklists | **Status:** Tested in the sim by the user: OK. Unit tests: FuelPumpValveFaults.spec.ts, FuelPumpsAndValves.spec.ts, FuelPumpValveIndications.spec.ts, CheckItemStates.spec.ts.

What it is for

Two new failures, **CTR TK L XFR valve jammed in place** and **CTR TK R XFR valve jammed in place**: the centre tank transfer valve of that side stays open or closed, whatever its CTR TK XFR pushbutton commands. The FWC raises the A320 FCOM PRO-ABN-FUEL cautions: FUEL CTR L(R) XFR FAULT (valve not fully closed, or not fully open) and, with both valves stuck the same way, FUEL CTR L + R XFR FAULT, which replaces the two single ones. The SD FUEL page shows the valve amber and boxes the centre tank quantity.

- The centre tank fuel only transfers while an inner tank pump of the same wing runs: the jet pumps are driven by the wing tank pumps (FCOM DSC-28-10). Switching the wing pumps off stops the centre tank transfer.
- flyPad checklist: the securing / parking checklist item **FUEL PUMPS ... OFF** now reads the four wing tank pump pushbuttons, not whether the pumps run (the CTR TK XFR pushbuttons are items of their own in the FCOM).

What you should see

Situation	What you should see
L valve stuck open (CTR TK L XFR pb OFF)	E/WD **FUEL CTR L XFR FAULT**: -CTR TK L XFR ... OFF, -FUEL X FEED ... ON, -L TK PUMP 1 ... OFF, -L TK PUMP 2 ... OFF, .WHEN CTR TK EMPTY: -L TK PUMP 1 ... ON, -L TK PUMP 2 ... ON, -FUEL X FEED ... OFF (each action line goes once done). STATUS INOP SYS CTR TK L XFR.
L valve stuck closed (CTR TK L XFR pb ON)	**FUEL CTR L XFR FAULT**: -FUEL MODE SEL ... MAN, .IF UNSUCCESSFUL: -FUEL X FEED ... ON, -L TK PUMP 1 / 2 ... OFF, .WHEN CTR TK EMPTY: pumps ON, X FEED OFF. STATUS INOP SYS CTR TK L XFR.
Both valves stuck open	**FUEL CTR L+R XFR FAULT**: -CTR TK L XFR ... OFF, -CTR TK R XFR ... OFF; STATUS INOP SYS CTR TK XFR.
Both valves stuck closed	**FUEL CTR L+R XFR FAULT**: -FUEL MODE SEL ... MAN, .IF UNSUCCESSFUL: CTR AVAIL BY GRAVITY, 2T (4400LBS) UNUSABLE; STATUS: CTR TK USABLE BY GRAVITY, 2T (4400LBS) UNUSABLE, INOP SYS CTR TK XFR.
SD FUEL page	The jammed valve amber; the centre tank quantity boxed.

Limitations and why

Limitation	Why	What to do
The cautions are inhibited during the takeoff and the landing (phases 3, 4, 5, 7, 8).	The FCOM gives the inhibition as a figure: that of the other fuel cautions is used.	Test at the gate or in cruise.
No gravity transfer from the centre tank.	Not simulated: with both valves stuck closed, the centre tank fuel stays there.	Plan with the centre tank fuel as unusable.

How to use

1. Open flyPad > Failures > ATA 28 Fuel and activate **CTR TK L XFR valve jammed in place** or **CTR TK R XFR valve jammed in place**.
2. Move the CTR TK XFR pushbutton (or let the automatic control move the valve), then follow the ECAM with the overhead FUEL panel.

How to test

#	Do this	Expected result
1	A32NX at the gate with fuel in the centre tank, engines or APU running, all FUEL pushbuttons ON. Activate **CTR TK L XFR valve jammed in place** while the valve is open, then set the CTR TK L XFR pushbutton OFF.	The valve stays open: master caution, E/WD FUEL CTR L XFR FAULT with -CTR TK L XFR ... OFF done, -FUEL X FEED ... ON, -L TK PUMP 1 ... OFF, -L TK PUMP 2 ... OFF; SD FUEL with the valve amber.
2	Open the STATUS page.	INOP SYS: CTR TK L XFR.
3	Also activate **CTR TK R XFR valve jammed in place** and set the CTR TK R XFR pushbutton OFF.	The two single cautions are replaced by FUEL CTR L+R XFR FAULT; STATUS INOP SYS CTR TK XFR.
4	Deactivate both failures, CTR TK XFR pushbuttons ON. Set L TK PUMP 1 and 2 OFF.	The left centre tank transfer stops (no jet pump drive) while the right one goes on.
5	Set every wing tank pump pushbutton OFF and open the flyPad checklist with the FUEL PUMPS ... OFF item.	The item is checked from the four wing tank pump pushbuttons.

9. Engine flameout and seizure failures, in-flight relight

> **Aircraft:** A32NX | **Where:** flyPad > Failures > ATA 72 Engine; ENG MASTER 1 / 2, ENG MODE selector; E/WD, STATUS, SD ENG | **Status:** Tested in the sim by the user: OK; the LEAP-1A relight envelope (a27ba6c35): sim-tested OK (recorded). Rust tests (engine_failure.rs of systems and a320_systems, incl. the LEAP-1A envelope corners and two test-bed relights; full-aircraft tests incl. the ENG FIRE pb LP valve starvation), unit tests: EngineFailAlerts.spec.ts, A32NX_FADEC.spec.ts, Eng.spec.ts, a320.spec.ts. The end of an in-flight start without N2 overshoot: sim-tested OK (relight, recorded), native test engine_start_throttle_hold_test. The engine fuel valve hold: not sim-tested on the A32NX (Rust tests sim_fuel_valve.rs). The stall during a relight fix: not sim-tested (installed, no visible effect expected; native test). The FADEC automatic relight after a flameout: OK on a combined test build (recorded). Not sim-tested on this branch.

What it is for

Four new failures in a new flyPad chapter, ATA 72 Engine. A **flameout** cuts the engine fuel once: the crew can relight the engine with the ENG MASTER OFF then ON, inside the in-flight relight envelope. A **seizure** also stops the core: no relight while the failure is active. The FWC gives the A320 FCOM ENG 1(2) FAIL and ENG 1(2) SHUT DOWN alerts with their STATUS, and the flyPad failures page now describes the ATA 28 Fuel and ATA 72 Engine chapters like the others.

Id	flyPad failure (ATA 72 Engine)	What the code does
72000	Engine 1 flameout (crew relight possible)	Cuts the engine 1 fuel (latched). The engine runs down and windmills. In flight the FADEC makes one automatic relight attempt for 30 s (see "FCOM fixes: FADEC relight, automatic ignition..."): it relights the engine inside the relight envelope once the continuous ignition is on, without an ENG MASTER cycle. Otherwise ENG MASTER 1 OFF then ON inside the relight envelope relights it; it then runs normally while the failure stays listed. Deactivate and activate it again for a new flameout.
72001	Engine 2 flameout (crew relight possible)	The same for engine 2.
72010	Engine 1 seizure (no relight)	Cuts the fuel; the core (N2) stops in about 3 s while N1 windmills. No relight while the failure is active; once it is cleared the engine relights by the normal procedure.

72011	Engine 2 seizure (no relight)	The same for engine 2.

The **relight envelope** comes from the LEAP-1A chart ENG RELIGHT IN FLIGHT of the Air Arabia A320 QRH (18-Aug-21, ABN-19.06A, A6-ATA to ATF, CFM LEAP-1A engines, the engines FBW models). It replaces the PW1100G chart of the IndiGo A320neo FCOM, used while no LEAP-1A chart was available. Vc is the indicated airspeed, Zp the pressure altitude.

Zone	Where	Needs
3, stabilized windmill relight (N2 > 8 %)	270-350 kt up to 27 000 ft; the ceiling falls from 27 000 ft at 330 kt to 24 000 ft at 350 kt	ENG MODE IGN, ENG MASTER OFF then ON
1, starter assisted relight	135-270 kt up to 15 000 ft; 200-270 kt from 15 000 to 27 000 ft	ENG MODE IGN, ENG MASTER OFF then ON, and starter air (APU BLEED, or X BLEED OPEN with the other engine)
2, windmill quick relight (t < 20 s)	230-270 kt up to 20 000 ft	ENG MASTER OFF then ON within 20 s, with N2 above 15 % (design choice, not on the LEAP-1A chart), at any ENG MODE position

- Above these limits (e.g. in cruise at FL 350) no relight lights up.
- A windmilling engine (no combustion, in flight) keeps N2 = 8 % x IAS / 270 kt and N1 = 1.5 x N2 (at 300 kt: N2 8.9 %, N1 13.3 %); on the ground it runs down to 0. Its fuel flow shows 0 and its oil pressure falls with N2.
- During an in-flight relight both igniters are supplied (A320 FCOM DSC-70-80-30), whatever the ENG MODE selector: the SD ENG page shows A B.
- The SD ENG page reads the N2 the FADEC shows, so a seized core reads 0 there as on the E/WD.
- ENG ALL ENGINES FAILURE gets its FCOM flight phase inhibition (phases 1-4 and 8-10). Its OPT RELIGHT SPD line reads 270/.77, the LEAP-1A value of the same QRH (ABN-19.01A ALL ENG FAIL; the CFM56 aircraft use 300/.77); the removal of the unused 280/300/260 lines is on core/fcom-audit-fixes.
- **End of an in-flight start (fix):** an engine started or relit in flight used to overshoot to 110-114 % N2 for 3-7 s at the end of the start, because the thrust loop wound the MSFS throttle up while the engine was out. In flight, the MSFS throttle is now held at idle while the engine is out or starting, until the engine runs at idle; then the engine accelerates to the lever or A/THR demand (A320 FCOM DSC-70-80-40, QRH ENG RELIGHT IN FLIGHT "When idle reached"). Ground starts are unchanged, and reverse thrust passes through this hold (a first version of it blocked the reverse thrust).
- **Stall during a relight (fix):** the hold ends when the commanded N1 reaches idle. With the "Engine 1(2) compressor stall" failure active at the end of a relight, the stall N1 loss (up to 15 %) was counted in the commanded N1 and the hold ended about 0.8 s early; the stall N1 loss is now taken out of that check (native : a relight with a stall).
- **ENG MODE selector (fix):** the FADEC read a selector variable that does not exist (always NORM), so it never saw IGN/START; it now reads the real selector.
- **Engine fuel valve hold (fix):** MSFS can reopen the engine fuel valve (valves 13 and 14 of the A32NX engine fuel cut) on its own after the systems closed it. The systems now command the valve again whenever MSFS disagrees, and once when a flight is loaded.

What you should see

Situation	What you should see

Flameout in flight	N2, EGT and FF run down, FF 0, N1 / N2 settle at the windmill values. Amber **ENG 1 FAIL**, single chime, MASTER CAUT, SD ENG page: -ENG MODE SEL ..... IGN, -THR LEVER 1 ..... IDLE, .IF NO RELIGHT AFT 30S:; 30 s after the alert -ENG MASTER 1 ..... OFF, .IF DAMAGE: -ENG 1 FIRE P/B ..... PUSH, -AGEN AFTER 10S ..... DISCH, .IF NO DAMAGE: -ENG 1 RELIGHT ..... CONSIDER (A320 FCOM PRO-ABN-ENG ENG 1(2) FAIL). LAND ASAP amber. The ENG 1 FAULT light stays off (FCOM: a flameout does not light it).
ENG MASTER 1 OFF in flight (after a failure or not)	ENG 1 FAIL goes, **ENG 1 SHUT DOWN** comes: .IF NO FUEL LEAK:, -IMBALANCE ..... MONITOR, -TCAS MODE SEL ..... TA (with the wing anti-ice on: the pack of the affected side OFF and X BLEED OPEN; with the ENG FIRE pb pushed: X BLEED SHUT, WING ANTI ICE OFF, AVOID ICING CONDITIONS). STATUS: IF PERF PERMITS: X BLEED ..... OPEN, IF NO ENG 1 DAMAGE: CONSIDER ENG 1 RELIGHT, CAT 3 SINGLE ONLY, ONE PACK ONLY IF WAI ON; INOP SYS: CAT 3 DUAL, ENG 1 BLEED, PACK 1, MAIN GALLEY, GEN 1, G ENG 1 PUMP (WING A. ICE with the ENG FIRE pb pushed).
Relight inside the envelope	Light-up: FF, N2 and EGT rise, A B igniters on the SD ENG page, AVAIL; the engine alerts and LAND ASAP clear.
Relight outside the envelope	No light-up: N2 stays at the windmill value, FF 0.
Seizure	N2 to 0 in about 3 s, N1 windmills, oil pressure 0 (red), ENG 2 FAIL as above; GEN 2 and the yellow engine pump lost.
Both engines failed	Red ENG ALL ENGINES FAILURE; the single ENG 1 / ENG 2 FAIL alerts do not show while it is active. Relighting one engine brings the single alert of the other one.
On the ground	ENG 1 FAIL with the ground lines (-THR LEVER 1 ..... IDLE, -ENG MASTER 1 ..... OFF, .IF DAMAGE: ..., -AGENT 1 ..... DISCH, .IF NO DAMAGE: ...), no LAND ASAP; N1 / N2 run down to 0. ENG MASTER OFF gives no ENG SHUT DOWN in flight phases 1-2 and 9-10 (FCOM: phases 3-8 only).

Limitations and design choices

Limitation	Why	What to do
The relight envelope applies to every in-flight relight, also after a crew shutdown without any failure.	Design choice: MSFS would otherwise relight anywhere; the QRH chart is the limit of any relight.	Relight inside the envelope (descend and adjust the speed).
The envelope corners are read off the QRH chart grid.	The chart gives no table; the calibrated airspeed is taken as the indicated airspeed.	-
The quick relight also needs N2 above 15 %.	Design choice kept from the PW1100G chart (the LEAP-1A chart only says t < 20 s): the sign of an engine that was running before the MASTER cycle.	An engine that has windmilled for a while needs ENG MODE IGN (zone 1 or 3).
The windmill N1 / N2 values are design values.	The chart only marks zone 3 "N2 > 8 %"; N2 is taken proportional to the airspeed, 8 % at 270 kt, and N1 = 1.5 x N2.	-
The seizure stops the core in about 3 s.	Design choice: the FCOM gives no rate.	-

ENG 1(2) FAIL does not come back after the ENG MASTER OFF, even if the relight does not light up.	Design choice: the MASTER OFF disarms the alert; ENG SHUT DOWN is shown meanwhile.	After a failed attempt (no light-up within 30 s), MASTER OFF and try again.
While ENG ALL ENGINES FAILURE is active, the single engine alerts are not shown.	Design choice: its procedure replaces them.	-
In flight, the HP fuel valve opening at N2 > 15 % is not modelled.	The fuel comes back when the engine lights up inside the envelope (on the ground the start sequence of "Engine ignition and start failures" opens the fuel at 22 % N2).	-
The idle hold after an in-flight start ends after a fixed time if MSFS settles below the FBW idle.	Design choice (no FCOM value): MSFS can end its start slightly below the FBW idle core speed.	-
A full MSFS restart is needed after an update.	The systems WASM module, the FADEC WASM module and the fbw WASM module (throttle hold) change.	Restart the sim after installing.

How to use

1. Open flyPad > Failures > ATA 72 Engine and activate a flameout or a seizure.
2. Follow the ECAM (ENG FAIL, then ENG SHUT DOWN). To relight, get inside the envelope (table above), ENG MODE IGN, ENG MASTER OFF for 30 s, then ON. For a quick relight in zone 2, set the MASTER OFF then ON at once (within 20 s, N2 above 15 %, 230-270 kt up to 20 000 ft).
3. Clear the failure in the flyPad after the test; a seizure must be cleared before a relight.

How to test

Setups: S1 cruise 60 t, FL 350, M.78; S2 60 t, 10 000 ft, 300 kt; S3 60 t, 15 000 ft, 210 kt, APU running with APU BLEED ON; S4 on the ground, both engines at idle.

#	Do this	Expected result
1	Activate **Engine 1 flameout (crew relight possible)**.	ENG 1 FAIL with its lines, LAND ASAP, windmill N1 / N2, FF 0, oil pressure falling with N2; the 30 s lines come after 30 s. ENG MODE IGN and THR LEVER 1 IDLE clear their lines.
2	After 30 s, ENG MASTER 1 OFF; then MASTER ON at FL 350.	ENG 1 SHUT DOWN with its STATUS and INOP SYS; at FL 350 no light-up (outside the envelope), A B igniters on the SD ENG page.
3	Descend to 20 000 ft or below at 280-300 kt; ENG MODE IGN, MASTER 1 OFF 30 s, ON.	Light-up at once; the engine back at idle in about 30-60 s; all engine 1 alerts and LAND ASAP clear.
4	Flameout; MASTER 1 OFF; APU BLEED OFF, X BLEED not open; 30 s; MASTER 1 ON.	No light-up (210 kt is below zone 3, no starter air). MASTER OFF, APU BLEED ON, 30 s, MASTER ON: the start valve opens and the engine lights up.

5	At 245 kt below 13 000 ft, ENG MODE NORM: flameout, then at once MASTER 1 OFF then ON within 20 s.	Quick relight at NORM; the ENG MODE knob stays at NORM and still works afterwards. Waiting until N2 is below 15 % before the MASTER OFF / ON: no light-up. Since the code makes the FADEC relight a flameout in this quick relight zone by itself, at once and at NORM, before the MASTER cycle (not sim-tested): the crew quick relight is then best seen with the next step (MASTER OFF / ON without a failure).
6	Activate **Engine 2 seizure (no relight)**.	N2 to 0 in about 3 s, N1 windmills, oil pressure 0, ENG 2 FAIL; MASTER 2 OFF 30 s / ON at 300 kt: no light-up. Clear the seizure, MASTER 2 OFF 30 s / ON: light-up.
7	Optional: activate both flameouts.	ENG ALL ENGINES FAILURE without the single ENG FAIL alerts; relighting engine 1 brings ENG 2 FAIL.
8	No failure, MASTER 1 OFF at FL 350, then ON with ENG MODE IGN.	ENG 1 SHUT DOWN; no light-up at FL 350; at FL 200 / 290 kt, MASTER OFF 30 s then ON: light-up.
9	Activate **Engine 1 flameout (crew relight possible)** on the ground.	ENG 1 FAIL with the ground lines, no LAND ASAP, N1 / N2 to 0; MASTER 1 OFF gives no ENG 1 SHUT DOWN; a normal start works.
10	Normal flight (starts, single-engine taxi, takeoff to descent, shutdown); on the ground at idle, push the ENG 1 FIRE pb with MASTER 1 ON.	No ENG FAIL / SHUT DOWN alert in the normal flight. With the FIRE pb: ENG 1 SHUT DOWN at once, no ENG 1 FAIL; the engine stops after about 100 s on its residual fuel (LP valve starvation of core/fire).

LEAP-1A envelope re-test (recorded over SimConnect, OK): two points the old PW1100G envelope refused. L1 setup: FL 150, 250 kt, AP and A/THR ON, APU OFF, ENG MODE NORM, both engines running. L2 setup: climbed to FL 265 at 290 kt (279 kt at the relight), AP and A/THR ON, APU OFF, X BLEED AUTO, ENG MODE NORM.

#	Do this	Expected result
1	L1 (quick relight at FL 150): ENG MASTER 1 OFF, count 5 s, ENG MASTER 1 ON.	Engine 1 fuel flow comes back within a few seconds, N2 and EGT rise, engine 1 back at idle; no ENG 1 FAIL stays; the ENG MODE selector stays at NORM.
2	L2 (windmill relight at FL 265): activate **Engine 1 flameout (crew relight possible)**; THR LEVER 1 IDLE; ENG MODE IGN/START; ENG MASTER 1 OFF, count 10 s; ENG MASTER 1 ON. At idle: ENG MODE NORM, THR LEVER 1 CL, deactivate the failure.	Engine 1 fuel flow appears within a few seconds, N2 and EGT rise, engine 1 at idle within about 60 s, ENG 1 FAIL goes away (the aircraft may sink a few hundred feet meanwhile).

End of the in-flight start (recorded, OK). Setup: FL 150, 250 kt, AP and A/THR ON, APU OFF, ENG MODE NORM, both engines running, both thrust levers in CL.

#	Do this	Expected result
1	ENG MASTER 1 OFF, count 5 s, ENG MASTER 1 ON. Leave both levers in CL and watch engine 1 N2 for 40 s.	Engine 1 relights within about 10 s; its N2 rises to idle (about 70 %) and holds there until the engine runs, then rises in about 5 s to the N2 of engine 2 (about 93 %). Engine 1 N2 never goes more than 1-2 % above engine 2 N2 (before the fix: 110-114 %).

10. Engine oil failures: leak, filter clog, overheat

> **Aircraft:** A32NX | **Where:** flyPad > Failures > ATA 79 Oil; SD ENGINE and CRUISE pages; E/WD | **Status:** Sim-tested by the user, data-checked OK (re: no false oil alert in a climb to FL 300; the three oil failures together in cruise: overheat 81 to 158 °C with ENG 2 OIL HI TEMP, leak to the red oil pressure below 13 PSI with ENG 1 OIL LO PR, then the shutdown). The later tuning (gradual cool-down after the overheat, overheat times): not sim-tested on the A32NX. Rust tests oil_failure.rs, native tests engine_oil_failures_test and oil_system_test, unit tests EngineOilAlerts.spec.ts, EngOilIndications.spec.ts.

What it is for

Six new failures: **Engine 1 oil leak**, **Engine 2 oil leak**, **Engine 1 oil filter clog**, **Engine 2 oil filter clog**, **Engine 1 oil overheat** and **Engine 2 oil overheat**, with the oil indications and the ECAM alerts of the A320 FCOM (DSC-70-90-40, PRO-ABN-ENG).

The FADEC computes the oil quantity, pressure and temperature again: its oil model had been dead since a FADEC migration and is restored (from the A380X FADEC, with the A32NX engine imbalance effect on the oil pressure). The failures act on that model:

- **Leak:** the oil system empties at 2 qt per minute of simulation time (the SD tank reading falls about 1.6 qt/min). The pump keeps its normal pressure down to 2.5 qt in the whole system and gives none at 0.6 qt, so the pressure falls once the tank reads nearly empty.
- **Filter clog:** CLOG on the SD ENGINE page and the ENG OIL FILTER CLOG caution, no other effect.
- **Overheat:** the oil goes to 135 °C at idle and 175 °C at 100 % N2 (about 165 °C at cruise), with a 60 s time constant: ENG OIL HI TEMP after about 2 minutes in cruise; THR LEVER IDLE brings it below 155 °C within about 30 s. Once the failure is cleared the oil cools down gradually to its normal temperature.
- **SD ENGINE page:** the FCOM 0-22 QT scale; the quantity pulses below 3.25 QT (until 4.75 QT); the oil pressure red below 13 PSI, pulsing between 13 and 16 PSI with N2 above 75 % (until it is back above 20 PSI); amber CLOG under the oil pressure. The CRUISE page shows the same FADEC oil quantity.
- **Fixes found in the sim tests:** the SD oil quantity read nearly empty in every climb (the oil gulping formula got the thrust in newtons instead of pounds), and the pressure followed the tank reading, which gave a false ENG OIL LO PR in a normal climb: the pump pressure now follows the oil of the whole system. The overheat stopped far below its target because MSFS pulls the oil temperature back to its own model: the FADEC now keeps its own overheat temperature.
- The start N2 and fuel flow of the FADEC model also blend smoothly at low N2 now (they overshot below about 10 % N2).

What you should see

Situation	What you should see
Oil leak, after several minutes	Oil quantity falling, then pulsing below 3.25 QT; the oil pressure falls and turns red below 13 PSI; red **ENG 1 OIL LO PR**, MASTER WARN and continuous chime: -THR LEVER 1 ... IDLE, -ENG MASTER 1 ... OFF.
Oil filter clog	Amber **ENG 2 OIL FILTER CLOG** (title only, single chime) and amber CLOG on the SD ENGINE page.
Oil overheat	The oil temperature rises from about 85 °C, pulses above 140 °C and is amber above 155 °C; amber **ENG 2 OIL HI TEMP**: -THR LEVER 2 ... IDLE, -ENG MASTER 2 ... OFF. It also comes after more than 15 minutes above 140 °C, and the SD value then turns amber too (fix).
Lever at idle	The -THR LEVER line goes.
ENG MASTER OFF	The oil alert ends and ENG SHUT DOWN takes over (FCOM associated procedure).

Limitations and why

Limitation	Why	What to do
Inhibitions: OIL LO PR in flight phases 1 and 10, OIL HI TEMP in 4, 5, 7, 8, OIL FILTER CLOG in 3, 4, 5, 7, 8.	The FCOM figures.	Test in cruise.
The oil alerts and CLOG only show with the engine running and its ENG MASTER ON.	Design choice: a stopped engine has no oil pressure, and the procedures end with the ENG MASTER OFF.	-
The leak rate, the pressure loss and the overheat temperatures are design values.	The FCOM gives no rates; the overheat cause is not modelled either.	-
The leak rate is per simulation minute.	At 4x sim rate the oil goes four times faster on the real clock.	-
The upper oil pressure scale stays the FBW one (130 PSI, advisory from 129 PSI).	The FCOM 0-100 PSI scale is the CFM56 one; the FBW LEAP-1A oil pressure reaches about 90 PSI at take-off thrust.	-
A full MSFS restart is needed after an update.	The systems and FADEC WASM modules change.	Restart the sim after installing.

How to use

1. Call the SD ENGINE page (ECAM control panel ENG key).
2. Open flyPad > Failures > ATA 79 Oil and activate a leak, a filter clog or an overheat.
3. Follow the ECAM (THR LEVER IDLE, then ENG MASTER OFF if needed). Deactivate the failure after the test.

How to test

Setup: normal departure, takeoff and climb at CL with A/THR ON, then cruise at FL 300-390.

#	Do this	Expected result
1	Climb normally and watch the oil indications.	No OIL LO PR, no pulsing oil quantity; oil quantity about 10-15 qt in the climb.
2	In cruise, SD ENGINE page. Activate **Engine 2 oil filter clog**, **Engine 2 oil overheat** and **Engine 1 oil leak**.	Amber ENG 2 OIL FILTER CLOG and CLOG on the SD; engine 2 oil temperature rises past 155 °C within about 3 minutes, then amber ENG 2 OIL HI TEMP with -THR LEVER 2 ... IDLE, -ENG MASTER 2 ... OFF.
3	Deactivate **Engine 2 oil filter clog** and **Engine 2 oil overheat**.	Both engine 2 alerts and CLOG go; the temperature falls gradually.
4	Wait for the leak (about 6 to 10 minutes after it began).	Engine 1 oil quantity falls about 2 QT/min and pulses below 3.25 QT; the pressure falls and turns red below 13 PSI; red ENG 1 OIL LO PR with MASTER WARN.
5	THR LEVER 1 IDLE, ENG MASTER 1 OFF. Deactivate **Engine 1 oil leak**.	The lever line goes; after the MASTER OFF, ENG 1 SHUT DOWN replaces the oil alert.

11. Engine fuel filter clog

> **Aircraft:** A32NX | **Where:** flyPad > Failures > ATA 73 Engine - Fuel And Control; SD ENGINE page; E/WD | **Status:** Not sim-tested (offline: Rust fuel_filter_failure.rs tests, EngFuelFilterIndication.spec, EngineFuelFilterAlerts.spec, failure list specs). The A380X version of the same code path was sim-tested.

What it is for

Two new failures, **Engine 1 fuel filter clog** and **Engine 2 fuel filter clog**: "the pressure loss across the fuel filter is excessive" (FCOM DSC-70-90-40). Amber **CLOG** below FUEL USED on the SD ENGINE page while the engine runs, and the **ENG 1(2) FUEL FILTER CLOG** caution (crew awareness, "short term engine operation is not affected"). No effect on the engine: the filter is bypassed.

Limitations and why

Limitation	Why	What to do
Inhibited in flight phases 3, 4, 5, 7 and 8.	The FCOM procedure page.	Test on the ground or in cruise.
No fuel system contamination failure.	Not modelled; the clog only gives the indication.	-

How to use

1. Call the SD ENGINE page (ECAM control panel ENG key).
2. Open flyPad > Failures > ATA 73 and activate **Engine 1 fuel filter clog** with engine 1 running.
3. Deactivate it after the test.

How to test

On the ground with engine 1 running at idle (flight phase 1 or 2), SD ENGINE page.

#	Do this	Expected result
1	Activate **Engine 1 fuel filter clog**.	Amber CLOG below FUEL USED of engine 1; amber ENG 1 FUEL FILTER CLOG with MASTER CAUT. Engine 1 keeps running normally.
2	Deactivate it.	CLOG and the caution go.

12. Engine ignition and start failures, FCOM start sequence

> **Aircraft:** A32NX | **Where:** flyPad > Failures > ATA 74 Ignition, ATA 80 Starting; ENG MODE, ENG MASTER, ENG MAN START; E/WD, STATUS, SD ENGINE | **Status:** Sim-tested by the user: OK (test flight S7, recorded: a ground start with Engine 1 igniter A failed, the automatic dry crank and the new start with both igniters; the dry crank with ENG MAN START). The other start failures: not sim-tested (Rust tests). Rust tests engine_start.rs, native test start_sequence_test, unit tests EngineStartAlerts.spec.ts, StartIndications.spec.ts.

What it is for

The FBW FADEC ran the engine model but not what a real FADEC does during a start. The systems now run the start sequence of the A320 FCOM (DSC-70-80-30, DSC-70-80-40):

- **Igniters:** one igniter for a first automatic start on the ground, alternated between starts; both for a new attempt, a manual start and an in-flight start. Continuous ignition with the ENG MODE selector at IGN/START (one igniter, both if it has failed), with the green IGNITION memo. Since the FADEC also selects the continuous ignition by itself

while the engine anti-ice is on (see "FCOM fixes: FADEC relight, automatic ignition..."). The SD igniter indication now comes from this sequence.

- **Automatic start:** ignition from 16 % N2, fuel (HP valve) at 22 % N2, ignition off and start valve closed at 50 % N2. On the ground, a start that does not light up within 18 s, stalls, hangs or goes above 725 °C EGT is aborted and tried again with both igniters, up to three attempts; then the ENG FAULT light (ENG MASTER panel) until ENG MASTER OFF. A start that does not light up or hangs is first dry cranked for 30 s; after a stall or an EGT above the limit the FADEC runs a new start first and dry cranks only when no new start is left.
- **ENG MAN START pushbuttons:** manual start, dry crank (ENG MODE CRANK, MAN START ON, MASTER OFF) and wet crank. The MSFS starter turns the core while the sequence keeps the fuel cut, with no EGT rise. A manual start lights up from 20 % N2.
- **Starter shaft shear in flight**: a starter that fails during an in-flight relight gives ENG 1(2) START FAULT STARTER SHAFT SHEAR with WINDMILL START ONLY, for the rest of the flight.
- The start code shared with the A380X got two settings (A380X engine alerts); the A32NX keeps its values (no light-up after 18 s also in flight, LO START AIR PRESS for a ground start only): nothing changes for the A32NX.
- The SD ENGINE page shows the start indications at CRANK too.
- Tuning: an engine that is OFF only shows running once MSFS really burns fuel in it.

flyPad failure (both engines)	ATA	What the code does
Engine 1(2) igniter A / igniter B	74 Ignition	The igniter does not light the engine: ENG IGN A (B) FAULT; a start that uses it does not light up.
Engine 1(2) start valve stuck closed / stuck open	80 Starting	The start valve stays as it is whatever the FADEC commands: ENG START VALVE FAULT.
Engine 1(2) hot start (start EGT over limit)	80 Starting	The start EGT overshoots: START FAULT EGT OVERLIMIT, abort and new attempts.
Engine 1(2) hung start (N2 stops below idle)	80 Starting	N2 hangs below idle: START FAULT HUNG START.
Engine 1(2) stall during the start	80 Starting	N2 hangs and the EGT overshoots: START FAULT STALL.
Engine 1(2) starter shaft shear	80 Starting	The starter gets air but does not turn the engine: no start, no starter assisted relight; START FAULT STARTER SHAFT SHEAR.

What you should see

Situation	What you should see
Start fault on the ground (automatic start)	Amber **ENG 1 START FAULT** with the cause (IGNITION FAULT, STALL, EGT OVERLIMIT, HUNG START, STARTER SHAFT SHEAR, LO START AIR PRESS or THR LEVER NOT AT IDLE), then AUTO CRANK IN PROGRESS during the dry crank, NEW START IN PROGRESS during the next attempt (after a STALL or EGT OVERLIMIT: NEW START IN PROGRESS first, AUTO CRANK IN PROGRESS only after the last new start), and -ENG MASTER 1 ... OFF once the start is aborted.
Start fault, manual start or in flight	Manual start on the ground: -ENG MASTER 1 ... OFF, -MAN START 1 ... OFF. In flight: -ENG MASTER 1 ... OFF. Starter shaft shear in flight: WINDMILL START ONLY, and STATUS ENG 1 WINDMILL START ONLY. Low start air: -BLEED AIR SUPPLY ... CHECK; lever not at idle: -THR LEVER 1 ... IDLE.
Start valve stuck open after the start	**ENG 1 START VALVE FAULT**, START VALVE NOT CLOSED: -APU BLEED ... OFF (engine 1), -X BLEED ... SHUT; in flight -ENG 1 BLEED ... OFF, -MAN START 1 ... OFF, -WING ANTI ICE ... OFF, AVOID ICING CONDITIONS (with its STATUS); on the ground -ENG MASTER 1 ... OFF.

Start valve stuck closed at a start	ENG 1 START VALVE FAULT, START VALVE NOT OPEN: -X BLEED ... OPEN (other engine running), -APU BLEED ... ON (APU available, below FL 200), .IF UNSUCCESSFUL: MAN START / ENG MASTER OFF.
Igniter failed	**ENG 1 IGN A FAULT** (IGN B FAULT) crew awareness, STATUS INOP SYS ENG 1 IGN A (B); both failed: ENG 1 IGN A+B FAULT, AVOID ADVERSE CONDITIONS, INOP SYS ENG 1 IGN.
ENG MODE IGN/START with the engines running	Green IGNITION memo.

Limitations and why

Limitation	Why	What to do
The numbers are those of the CFM56-5B FCOM.	FBW models the LEAP-1A, for which no FCOM is available.	-
The 30 s automatic dry crank is a design value.	The FCOM gives no length (30 s is its minimum ventilation).	-
No starter time limit.	The A320 FCOM gives none.	-
The SD still shows the selected igniter when that igniter has failed.	Design choice: the FCOM shows the igniter energized.	-
Inhibitions: the three cautions in flight phases 3, 4, 5, 7, 8; IGN A or B FAULT also in 6; START FAULT also in 6 when due to the lever not at idle.	The FCOM figures.	Test on the ground or in cruise.
The "EEC control of start valve failed" lines are not modelled.	They are for IAE and PW engines.	-
An aborted automatic start takes about 2.5 minutes in the sim.	Up to three attempts with 30 s dry cranks.	Use the igniter test below; the other abort paths are covered by Rust tests.
A full MSFS restart is needed after an update.	The systems and FADEC WASM modules and the cockpit behaviour XML (ENG FAULT light) change.	Restart the sim after installing.

How to use

1. Open flyPad > Failures > ATA 74 Ignition or ATA 80 Starting and activate a failure before the start.
2. Start the engine as usual (ENG MODE IGN/START, ENG MASTER ON) and follow the ECAM; the FADEC aborts and retries a failed automatic start on the ground by itself.
3. For a dry crank: ENG MODE CRANK, ENG MAN START ON with the ENG MASTER OFF.

How to test

Setup: A32NX cold and dark at the gate, freshly loaded, external power or batteries ON, APU AVAIL, APU BLEED ON, parking brake ON, thrust levers IDLE.

#	Do this	Expected result

1	flyPad > Failures > ATA 74 Ignition: activate **Engine 1 igniter A**. ENG MODE IGN/START, ENG MASTER 1 ON. Wait 75 s.	About 20 s after MASTER ON: amber ENG 1 START FAULT, IGNITION FAULT, AUTO CRANK IN PROGRESS and amber ENG 1 IGN A FAULT; SD ENG: N2 about 20-30 %, no EGT rise, start valve open, igniter A shown then gone.
2	Keep waiting.	About 50 s after MASTER ON: NEW START IN PROGRESS, SD igniters A B, EGT rises, engine 1 starts to idle, START FAULT goes; the ENG 1 FAULT light stays off. Deactivate the failure.
3	ENG MODE CRANK; ENG MAN START 2 pb ON (overhead, guarded) for 20 s, then OFF; ENG MODE NORM.	SD ENG: engine 2 start valve open, N2 about 20-25 % and steady, no fuel flow, no EGT rise, no igniter; engine 2 does not start. Then the valve closes and N2 runs down. No ENG 2 START FAULT.
4	Optional (not sim-tested): activate **Engine 2 starter shaft shear**, ENG MODE IGN/START, ENG MASTER 2 ON.	N2 does not rise; ENG 2 START FAULT with STARTER SHAFT SHEAR, -ENG MASTER 2 ... OFF, -MAN START 2 ... OFF. ENG MASTER 2 OFF and deactivate the failure.
5	Optional (not sim-tested): engine 1 off, activate **Engine 1 start valve stuck open**, then start engine 1 (ENG MODE IGN/START, ENG MASTER 1 ON).	Above 50 % N2 the valve stays open: ENG 1 START VALVE FAULT, START VALVE NOT CLOSED with its ground lines (-APU BLEED ... OFF, -X BLEED ... SHUT, -ENG MASTER 1 ... OFF).

13. Engine stall, EGT overtemperature, overspeed and vibration failures

> **Aircraft:** A32NX | **Where:** flyPad > Failures > ATA 72 Engine, ATA 77 Engine Indicating; E/WD, SD ENGINE | **Status:** Sim-tested by the user: OK (test flight, recorded: stall in the climb, EGT overtemperature with high vibration). VIB N2 then stayed at 4.6 at idle, above the 4.3 advisory: the vibration tuning that brings it below at idle is not sim-tested. The LEAP-1A N1 / N2 red lines and the retuned overspeed indication: tested in the sim by the user on a combined test build: OK (TOGA take-off without an N2 caution, N2 105.3 %; the overspeed failure gives ENG N1 OVER LIMIT with THR LVR BELOW LIMIT; recorded). Rust tests engine_malfunction.rs, unit tests EngineParameterAlerts.spec.ts, EngineLimits.spec.ts, EngVibration.spec.ts, Eng.spec.ts.

What it is for

Eight new failures; the engine keeps running:

flyPad failure (both engines)	ATA	What the code does
Engine 1(2) compressor stall (above 60 % N1)	72 Engine	Above 60 % N1 command the engine stalls: it loses N1 (about 15 %), the EGT rises (about 100 °C or more), N1 and N2 fluctuate, N1 vibrates. It recovers with the lever at IDLE and stalls again above the threshold (QRH: reduce thrust below it).
Engine 1(2) EGT overtemperature	72 Engine	The EGT is higher in proportion to N1 (about 100 °C at climb thrust): ENG EGT OVER LIMIT at climb thrust.
Engine 1(2) N1/N2 overspeed indication	72 Engine	The N1 indication reads 20 % high, up to 101.9 %, and the N2 indication 6 % high, up to 116.9 % (retuned for the LEAP-1A red lines): ENG N1 OVER LIMIT at take-off and climb thrust (above the 101 % red line, below the 102.8 % second band). Reducing the thrust below about 84 % N1 ends the alert. The thrust is unchanged; at idle the N1 reads about 24 % instead of 20 %.
Engine 1(2) high vibration	77 Engine Indicating	N1 and N2 vibrations growing with the square of the rotor speed: above the SD advisory (6 / 4.3) at climb and cruise thrust, below it at reduced thrust and clearly below at idle. No ECAM alert (QRH procedure).

- **E/WD limits** (now in one shared place): EGT amber above 940 °C at TOGA/FLX, 905 °C at CLB/MCT, 725 °C without a rated limit or in the start; red above 975 °C, with a red mark at the highest EGT reached (it stays until the next engine start on the ground); these EGT values are the CFM56-5B TCDS values the user chose. N1 amber above its limit, red above the LEAP-1A red line 101 % (red arc and red mark); N2 red above the LEAP-1A 116.5 %, with a red cross. The N1 / N2 red lines come from the CFM LEAP-1A General Familiarization, page 34 (before, the CFM56 104 % / 105 % of the FCOM, which gave ENG N2 OVER LIMIT at every TOGA take-off, as the LEAP-1A runs above 105 % N2 there).
- **SD ENGINE page:** separate N1 and N2 vibrations, pulsing above the advisory.

What you should see

Situation	What you should see
Stall in flight	Amber **ENG 1 STALL**: -THR LEVER 1 ... IDLE, -ENG 1 PARAMETERS ... CHECK, -ENG 1 STALL PROC ... APPLY; SD ENGINE page. On the ground: -THR LEVER 1 ... IDLE, -ENG MASTER 1 ... OFF.
A parameter above its limit	Amber **ENG 1 EGT OVER LIMIT** (N1, N2 OVER LIMIT; the title names the parameter): -THR LVR 1 .. BELOW LIMIT; once the EGT went above 975 °C (or N1 above 102.8 %, N2 above 117.3 %): -THR LEVER 1 ... IDLE, -ENG MASTER 1 ... OFF. N1 is over its limit above 101 %, N2 above 116.5 %.
TOGA take-off, no failure	No ENG N2 OVER LIMIT (the LEAP-1A N2 stays below 116.5 %).
High vibration at climb thrust	VIB N1 6 or more and VIB N2 4.3 or more, pulsing green.

Limitations and why

Limitation	Why	What to do
What each failure does (N1 loss, EGT and indication offsets, vibration levels) is a design choice.	The FCOM gives the symptoms, not values.	-
The EGT limits are the CFM56-5B TCDS values.	The LEAP-1A document gives 1060 °C "To Be Confirmed", so it is not used (user decision); the FCOM text gives 950 / 915 °C.	-
The second bands of OVER LIMIT (N1 102.8 %, N2 117.3 %) are design values.	The LEAP-1A document gives no second band: each keeps the FCOM gap above the red line (N1 1.8 %, N2 0.8 %).	-
Inhibitions: STALL in flight phases 3, 4, 5, 7, 8; OVER LIMIT in 4 and 8.	The FCOM figures.	Test the stall in the climb above 1 500 ft or in cruise.
A full MSFS restart is needed after an update.	The systems, FADEC and fbw WASM modules change.	Restart the sim after installing.

How to use

1. Open flyPad > Failures > ATA 72 Engine or ATA 77 Engine Indicating and activate a failure at climb or cruise thrust.
2. Follow the ECAM (THR LEVER IDLE or below the limit); deactivate the failure after the test.

How to test

Setup: climbing through FL 150 at about 280 kt, A/THR ON at CLB, AP ON, both engines running.

#	Do this	Expected result

1	Activate **Engine 1 compressor stall (above 60 % N1)**. Wait 20 s.	Amber ENG 1 STALL with its three lines; SD ENGINE page; N1 1 fluctuates and settles about 15 % below N1 2; EGT 1 rises by about 100 °C or more.
2	Thrust lever 1 to IDLE, wait 20 s.	Within about 10 s the alert goes; N1 1, EGT 1 and the vibration back to normal.
3	Thrust lever 1 back to CL, wait 10 s; deactivate the failure.	The stall comes back, then ends with the failure.
4	At CL, activate **Engine 1 EGT overtemperature** and **Engine 1 high vibration**. Wait 30 s.	EGT 1 about 100 °C above EGT 2, amber above 905 °C with ENG 1 EGT OVER LIMIT and -THR LVR 1 .. BELOW LIMIT (above 975 °C: red, -THR LEVER 1 ... IDLE, -ENG MASTER 1 ... OFF, and a red mark stays); SD VIB N1 6 or more and VIB N2 4.3 or more, pulsing.
5	Thrust lever 1 to IDLE, wait 30 s; deactivate both failures, lever back to CL.	The alert goes; EGT 1 green; both vibrations steady green below the advisory (VIB N2 stayed above it at idle before the tuning). A red EGT mark stays.

LEAP-1A red lines and overspeed (recorded OK on a combined test build). Setup: A32NX on the runway, V2 entered in the MCDU (A/THR armed at TOGA).

#	Do this	Expected result
1	TOGA take-off, A/THR armed; climb to the acceleration altitude.	N2 about 105 %, not red; no ENG N2 OVER LIMIT (before it came at every TOGA take-off).
2	In the climb at CL, activate **Engine 1 N1/N2 overspeed indication**.	N1 1 reads about 101.9 %, red; amber ENG 1 N1 OVER LIMIT with -THR LVR 1 .. BELOW LIMIT (no ENG MASTER OFF line).
3	Thrust lever 1 back below about 84 % N1, then deactivate the failure.	The alert goes; N1 1 back to its normal value (a red mark stays).

14. FADEC, thrust lever and reverser failures

> **Aircraft:** A32NX | **Where:** flyPad > Failures > ATA 73 Engine - Fuel And Control, ATA 78 Exhaust; thrust levers; E/WD, STATUS, SD | **Status:** Sim-tested by the user: OK (test flight, recorded: Thrust lever 1 resolvers (both), then Engine 2 FADEC channel A and B). The reverser fault test of that flight found no reverse thrust at all (the in-flight start hold clamped it): fixed, normal reverse on the ground data-checked OK; the reverser failures were not re-tested after the fix. FADEC overheat, resolver disagree, reverser unlocked and pressurized: not sim-tested. Rust tests engine_control_failure.rs and hydraulic, native tests fadec_failure_inputs_test and reverse_throttle_test, unit tests FadecReverserAlerts.spec.ts, A32NX_FADEC.spec.ts.

What it is for

flyPad failure (both engines / levers / reversers)	ATA	What the code does
Engine 1(2) FADEC channel A / channel B	73	One channel lost: the other takes control (ENG FADEC A (B) FAULT). Both lost: ENG FADEC FAULT, the engine data are lost (XX on the E/WD and SD), the A/THR disconnects, no reverser; the engine keeps following its lever.
Engine 1(2) FADEC overheat	73	ENG FADEC HI TEMP (ECU above 105 °C).
Thrust lever 1(2) resolvers (both)	73	The FADEC selects idle, CLB or frozen takeoff thrust by the FCOM rules: in flight it keeps CLB (or the takeoff thrust if the lever failed in TO/FLX) until the slats are out, then idle; idle on the ground; no reverser.

Thrust lever 1(2) resolver disagree	73	ENG THR LEVER DISAGREE: idle on the ground, CLB as the maximum in cruise, idle in approach.
Reverser 1(2) fault (does not deploy)	78 Exhaust	The reverser stays stowed; its engine stays at reverse idle: ENG REVERSER FAULT.
Reverser 1(2) unlocked (engine at idle)	78 Exhaust	The doors leave their stowed and locked position (no reverse thrust); the FADEC sets the engine at idle: ENG REVERSE UNLOCKED.
Reverser 1(2) pressurized (shutoff valve open)	78 Exhaust	ENG REV PRESSURIZED.

- Reverse thrust stays at reverse idle until the reverser is fully deployed (FCOM DSC-70-70): a reverser that does not deploy no longer gives forward thrust with the lever in reverse. The hydraulic reversers report their status to the FADECs and stop deploying without a FADEC command.
- Fix: the in-flight start throttle hold (feature "Engine flameout ...") clamped the MSFS throttle at 0, which blocked all reverse thrust on the ground; the reverse throttle now reaches MSFS.

What you should see

Situation	What you should see
One FADEC channel failed	Amber **ENG 1 FADEC A FAULT** (B), crew awareness.
Both channels failed	**ENG 2 FADEC FAULT**: in flight -THR LEVER 2 ... IDLE, -ENG 2 PARAMETERS ... CHECK, .IF ABN ENG OPERATION: -ENG MASTER 2 ... OFF (on the ground THR LVR 2 NOT ABV IDLE); engine 2 N1, EGT, N2, FF XX; A/THR OFF; STATUS THR LVR 2 NOT ABV IDLE.
FADEC overheat	**ENG 1 FADEC HI TEMP**, FADEC OVHT; in flight -ENG 1 PARAMETERS ... CHECK, .IF ABN ENG OPERATION: -THR LEVER 1 ... IDLE, -ENG MASTER 1 ... OFF; on the ground also -ENG MODE SEL ... NORM, -FADEC GND PWR ... CHK OFF.
Thrust lever resolvers lost, in flight at CL	**ENG 1 THR LEVER FAULT** with -A/THR ... KEEP ON (A/THR engaged) or ENG 1 HI PWR IN MAN THR; LAND ASAP amber; STATUS WHEN SLATS OUT: ENG 1 AT IDLE, INOP SYS REVERSER 1, ENG 1 THR. With the slats out: ENG 1 AT IDLE.
Thrust lever resolvers lost, on the ground	ENG 1 THR LEVER FAULT with ENG 1 IDLE POWER ONLY (-THR LEVER 1 ... IDLE with the lever above idle), as in the FCOM (fix, not sim-tested).
Resolver disagree	**ENG 1 THR LEVER DISAGREE**: on the ground ENG 1 IDLE POWER ONLY; in cruise AVAIL MAX POWER : CLB with -A/THR ... KEEP ON (or ON); in approach ENG 1 AT IDLE. STATUS ENG 1 AVAIL MAX PWR:CLB, GND ENG 1 MAX PWR: IDLE, INOP SYS ENG 1 THR.
Reverser fault	Amber **ENG 1 REVERSER FAULT** (title only); STATUS INOP SYS REVERSER 1; with the levers in reverse, REV only on the other engine.
Reverser unlocked in flight	**ENG 1 REVERSE UNLOCKED**: ENG 1 AT IDLE, -THR LEVER 1 ... IDLE, -MAX SPEED ... 300/.78, .IF BUFFET: -MAX SPEED ... 240 KT, -ENG MASTER 1 ... OFF; STATUS MAX SPEED 300/.78.
Reverser pressurized	**ENG 1 REV PRESSURIZED**: in flight -THR LEVER 1 ... IDLE; on the ground THR LVR 1 NOT ABV IDLE.

Limitations and why

Limitation	Why	What to do

Inhibitions (FCOM figures): FADEC A(B) FAULT and FADEC HI TEMP phases 3, 4, 5, 7, 8; FADEC FAULT 4, 5, 7, 8; REV PRESSURIZED 4, 5, 8; REVERSE UNLOCKED 4, 5, 8; REVERSER FAULT 3, 4, 5; THR LEVER FAULT / DISAGREE 4, 5. Some are not inhibited when the FADEC sets the engine at idle.	The FCOM gives them as figures.	Test on the ground or in cruise.
The flyPad reverser fault gives the title only.	It is not a position fault: the FCOM lines are for a position fault with the reverser pressurized.	-
A reverser that really deploys in flight is not modelled.	The unlocked doors stay in the stowed position.	-
FADEC OVHT is always shown with ENG FADEC HI TEMP.	Design choice: the flyPad overheat is an ECU above 105 °C.	-
A full MSFS restart is needed after an update.	The systems and fbw WASM modules change.	Restart the sim after installing.

How to use

1. Open flyPad > Failures > ATA 73 Engine - Fuel And Control or ATA 78 Exhaust and activate a failure.
2. Follow the ECAM and STATUS; deactivate the failure after the test.

How to test

Check the flyPad throttle calibration (reverse and idle detents) first.

#	Do this	Expected result
1	On the runway, both engines at idle, parking brake ON. Activate **Reverser 1 fault (does not deploy)**.	Amber ENG 1 REVERSER FAULT; STATUS INOP SYS REVERSER 1.
2	Both thrust levers to full reverse for 5 s, then IDLE. Deactivate the failure.	REV green on engine 2 only; engine 2 N1 rises, engine 1 N1 stays near idle; at IDLE the reverser stows, no other caution.
3	After a landing (no failure): both levers to full reverse until 70 kt, then IDLE.	REV green on both engines, N1 well above idle, the aircraft decelerates; no forward thrust jump at IDLE (OK).
4	FL 100, 220 kt, flaps 0, AP and A/THR ON, levers in CL. Activate **Thrust lever 1 resolvers (both)**, then thrust lever 1 to IDLE.	ENG 1 THR LEVER FAULT with -A/THR ... KEEP ON and LAND ASAP; STATUS WHEN SLATS OUT: ENG 1 AT IDLE, INOP SYS REVERSER 1, ENG 1 THR; engine 1 N1 does not drop to idle.
5	FLAPS 1.	Engine 1 N1 goes to idle and stays there.
6	Activate **Engine 2 FADEC channel A**, then **Engine 2 FADEC channel B**.	ENG 2 FADEC FAULT; engine 2 N1, EGT, N2, FF XX on the E/WD and the SD ENGINE page; the A/THR disconnects; thrust lever 2 still changes engine 2 thrust.
7	Deactivate all failures, FLAPS 0, lever 1 to CL, A/THR ON.	All alerts clear.

15. T.O CONFIG NORMAL and FUEL WING TK LO LVL fixes

> **Aircraft:** A32NX | **Where:** ECAM control panel T.O CONFIG key; E/WD, STATUS | **Status:** Not sim-tested. Unit tests ToConfigNormal.spec.ts, WingTankLowLevel.spec.ts.

What it is for

Two deviations from the A320 FCOM found by the FCOM gap audit (the other fixes of that set are on `core/fcom-audit-fixes`). They use the wing tank pump alerts of this branch.

- **T.O CONFIG NORMAL** is no longer given with a FUEL L(R) TK PUMP 1+2 LO PR alert (FCOM DSC-31-15: the T.O CONFIG test triggers it, and NORMAL needs no such alert).
- **FUEL L(R) WING TK LO LVL** (inner tank below 750 kg for 30 s) asks for -FUEL MODE SEL ... MAN only if the centre tank is not empty, then .IF NO FUEL LEAK AND FUEL IMBALANCE: -FUEL X FEED ... ON, -L TK PUMP 1 / 2 ... OFF. It now gives its STATUS: INOP SYS L (R) TK PUMPS and, with fuel in the centre tank, CTR TK FEED: MAN ONLY (FCOM PRO-ABN-FUEL).

Limitations and why

Limitation	Why	What to do
FUEL WING TK LO LVL is inhibited in flight phases 3, 4, 5, 7, 8 and 9.	FBW inhibition, unchanged.	Test at the gate or in cruise.
A low level in one wing only needs a fuel tool.	The flyPad fuel page fills both wings alike.	Set the tank quantity with a SimConnect variable tool (FUEL TANK LEFT MAIN QUANTITY).

How to use

1. Nothing to set: the alerts follow the FCOM by themselves.
2. Press the T.O CONFIG key before takeoff as usual; follow FUEL WING TK LO LVL and its STATUS if it comes.

How to test

#	Do this	Expected result
1	A32NX at the gate, engines running, all FUEL pumps ON. flyPad > Failures > ATA 28 Fuel: activate **L TK pump 1** and **L TK pump 2**.	FUEL L TK PUMP 1+2 LO PR.
2	Press the T.O CONFIG key (ECAM control panel).	No T.O CONFIG NORMAL in the T.O memo. Deactivate the failures: T.O CONFIG gives T.O CONFIG NORMAL again (with a takeoff configuration set).
3	Centre tank empty; set the left inner tank to about 600 kg with a SimConnect tool (right inner tank above 750 kg). Wait 30 s.	FUEL L WING TK LO LVL without the -FUEL MODE SEL ... MAN line; STATUS INOP SYS L TK PUMPS.
4	Put fuel in the centre tank (FUEL MODE SEL at AUTO).	The -FUEL MODE SEL ... MAN line shows; STATUS also CTR TK FEED: MAN ONLY.

16. ENG THR LEVER ABV IDLE and ENG SAT ABOVE FLEX TEMP

> **Aircraft:** A32NX | **Where:** E/WD; thrust levers; MCDU PERF TAKE OFF page (FLEX TO TEMP) | **Status:** SAT ABOVE FLEX TEMP: tested in the sim by the user: OK, data-confirmed (on the feature branch). THR LEVER ABV IDLE: OK after a real landing (phase 9, lever 1 at CL with lever 2 in reverse: red warning and its line within 0.4 s; it stays with lever 2 back at idle and goes with lever 1 out of CL; recorded). Unit tests EngineThrustSettingAlerts.spec.ts.

What it is for

Two thrust setting alerts of the A320 FCOM (PRO-ABN-ENG) that the A32NX did not have. They need no failure: they come from the thrust levers and the FLEX TEMP.

- **ENG 1(2) THR LEVER ABV IDLE** (red, MASTER WARN, repetitive "RETARD-RETARD"; FCOM): during the landing roll (FWC phases 8 and 9), one thrust lever is above idle while the other one is in reverse, or stays above idle after the other one came out of reverse. Line: -THR LEVER 1(2) ... IDLE. The voice is heard above 40 kt and stops when all levers are at IDLE or REVERSE.
- **ENG SAT ABOVE FLEX TEMP** (amber, MASTER CAUT; FCOM): before take-off (phase 2 only) the outside air temperature is above the FLEX TEMP entered in the MCDU. Line: T.O DATA ... CHECK. No FLEX TEMP entered gives no alert.

Limitations and why

Limitation	Why	What to do
THR LEVER ABV IDLE only after a real landing.	The FWC goes to phase 9 only after a touchdown or a take-off aborted above 80 kt; a take-off power cut at a standstill goes back to phase 2.	Test it after a landing.
"At reverser deselection" has no time limit in the FCOM.	Design choice: the memory of a reverse selection is kept until both levers are at idle or the landing roll ends.	-

How to use

1. Nothing to set. Before take-off, enter a FLEX TEMP above the outside temperature (or none for TOGA).
2. After landing, bring both thrust levers to REVERSE or IDLE together; follow the RETARD call if it comes.

How to test

#	Do this	Expected result
1	A32NX at the gate, engines running, FWC phase 2. MCDU PERF TAKE OFF: enter a FLEX TO TEMP 10 °C below the outside temperature (SAT).	Within a second: amber ENG SAT ABOVE FLEX TEMP with T.O DATA ... CHECK, single chime, MASTER CAUT.
2	Enter a FLEX TO TEMP of 60.	The caution goes; the T.O memo is back.
3	After a landing, stopped on the runway or a taxiway (phase 9), both levers at IDLE, PARKING BRK ON: lever 2 to REVERSE IDLE, then lever 1 to the CL detent.	Red ENG 1 THR LEVER ABV IDLE with -THR LEVER 1 ... IDLE, MASTER WARN.
4	Lever 2 forward to IDLE (lever 1 stays at CL), then lever 1 to IDLE.	The warning stays while lever 1 is at CL, and goes once it is at IDLE.

17. Engine fan blocked failure and ENG LOW N1

> **Aircraft:** A32NX | **Where:** flyPad > Failures > ATA 72; ENG MASTER and ENG MODE; E/WD | **Status:** OK (recorded: N1 0 % while N2 rose; ENG 1 LOW N1 with .IF CONFIRMED and ENG MASTER 1 OFF 2.5 s after N2 35 %; core hung at 41.5 %, start aborted; alert gone at ENG MASTER OFF; normal restart without it). Unit tests EngineLowN1Alerts.spec.ts, a320.spec.ts; native test fan_blocked_start_test.cpp; Rust failure type.

What it is for

The A320 FCOM alert **ENG 1(2) LOW N1** (on ground, PRO-ABN-ENG) is for a start in which the fan does not turn (a blocked fan, for example ice). The sim cannot produce that by itself, so the branch adds a failure for it.

- New flyPad failures, ATA 72: **Engine 1 fan blocked (no N1 rotation at start)** and **Engine 2 fan blocked (no N1 rotation at start)** (ids 72020, 72021). During a start on the ground the FADEC holds the N1 at 0 and the core hangs below idle (about 41 % N2).
- ENG 1(2) LOW N1 (amber, MASTER CAUT, single chime): N2 at or above 35 % with N1 below 2 % for 2 s, on the ground; lines .IF CONFIRMED:, -THR LEVER 1(2) ... IDLE (only with the lever above idle), -ENG MASTER 1(2) ... OFF. It stays until the ENG MASTER is set OFF; inhibited in flight phases 4 to 9.

Limitations and why

Limitation	Why	What to do
The 35 % N2, 2 % N1 and 2 s values are not in the FCOM.	Design choices for "no N1 rotation".	-
ENG START FAULT may follow.	The FADEC aborts an automatic start whose core hangs, about 20 s later.	Both alerts are expected; LOW N1 stays.
A full MSFS restart is needed after an update.	The systems and FADEC WASM modules change.	Restart the sim after installing.

How to use

1. flyPad > Failures > ATA 72 Engine: activate "Engine 1 (2) fan blocked (no N1 rotation at start)" before the engine start.
2. Start the engine as usual; follow ENG LOW N1 (ENG MASTER OFF). Clear the failure before the next start.

How to test

#	Do this	Expected result
1	A32NX on the ground, PARKING BRK ON, APU BLEED ON, engine 1 OFF, both levers at IDLE. flyPad > Failures > ATA 72: activate **Engine 1 fan blocked (no N1 rotation at start)**.	No alert yet.
2	ENG MODE selector IGN/START, ENG MASTER 1 ON. Watch engine 1 for 60 s.	N2 rises, N1 stays at 0. Once N2 is above 35 % for 2 s: amber ENG 1 LOW N1 with .IF CONFIRMED: and -ENG MASTER 1 ... OFF, single chime, MASTER CAUT. N2 hangs near 41 %; ENG 1 START FAULT may follow.
3	ENG MASTER 1 OFF.	ENG 1 LOW N1 goes (and START FAULT with it).
4	Clear the failure, wait 30 s, ENG MASTER 1 ON.	A normal start: N1 rises with N2, no LOW N1.

18. Engine and APU fire visual effects

> **Aircraft:** A32NX, A380X | **Where:** Exterior model (engine and APU exhausts); flyPad > Settings > Sim Options; flyPad > Failures > ATA 26, ATA 79 | **Status:** A32NX: tested in the sim by the user on a combined test build: OK (tests F1 engine fire, F2 fire put out by the agent, F3 APU fire smoke). A380X: tests F4, O1, K1 and L1 passed (per the commit). The engine damage smoke (MSFS failures menu) and the flyPad switch on the A32NX: not sim-tested. Unit tests EngineFireEffects.spec.ts (both aircraft), sync.spec.ts; Rust fire protection tests. Not sim-tested on this branch.

What it is for

A fire you can see from outside: flames at the exhaust of an engine on fire, smoke at the APU exhaust for an APU fire, until the fire is out. Also smoke for an engine failed from the MSFS failures menu and for an engine oil leak. A flyPad switch turns all these effects off.

- **A32NX:** flames and smoke at the exhaust of the engine on fire (flyPad **Engine 1 fire** / **Engine 2 fire**, or a fire of the MSFS failures menu); smoke at the APU exhaust for an **APU fire**. The APU exhaust point of the model moved to the tail cone tip, where the real APU exhaust is.
- The effects follow the fire itself, not the detection: nothing during a FIRE TEST, and the effect stays after the engine shutdown until the fire is out (deactivated failure, or put out by an AGENT discharge).
- Smoke at the exhaust of an engine failed from the MSFS failures menu (engine damage), and of an engine with the flyPad **Engine 1(2) oil leak** (ATA 79) while the oil flows out.
- **A380X:** the same for engines 1 to 4 and the APU; the stock MSFS engine fire, engine damage and oil leak effects are turned off so they do not show twice; the engine 2 effects (heat haze, contrails, fire) no longer vanish at the medium model detail (LOD01).
- **Fire model (both aircraft):** an AGENT discharge, or the automatic APU discharge, can also put out a fire set from the MSFS failures menu, which then ends the ENG / APU FIRE warning (see "Fire detection model and fire failures").
- **flyPad > Settings > Sim Options > "Situation Visual Effects (Engine and APU Fire, Engine Damage, Oil Leak)"**: on by default; OFF hides all these effects (L:A32NX_CONFIG_SITUATION_FX, both flyPads).

Limitations and why

Limitation	Why	What to do
Each AGENT discharge puts the fire out one time in two.	The FBW fire model (user decision).	Use AGENT 2 if the FIRE pb stays red.
After the ENG FIRE pb the engine runs on for up to 2 min 30 s.	The FIRE pb closes the LP fuel valve; the engine stops once the fuel downstream is burnt (FCOM).	-
On this branch the A380X oil leak smoke has nothing to start it.	The A380X flyPad engine oil failures (ATA 79) are on `core/failures-a380x`; this branch has only the model part.	Use a build with core/failures-a380x merged.
The A380X engine 2 effects at LOD01 / LOD02 need the exhaust nodes of the wing LOD models.	The model files are in the large-files repository, not in the code repository.	Install the matching A380X wing models.
A full MSFS restart is needed after an update.	The exterior models, aircraft.cfg, the model behaviour XML and the systems WASM modules change.	Restart the sim after installing.

How to use

1. Keep flyPad > Settings > Sim Options > Situation Visual Effects ON (default).
2. Set a fire (flyPad > Failures > ATA 26, or the MSFS failures menu) and look at the aircraft from outside; fight the fire with the ECAM procedure.
3. Set the switch OFF to hide the effects (the fire itself and its warnings are not changed).

How to test

Setup: A32NX on a stand, both engines at idle, PARKING BRK ON, APU running; outside (drone) view of the left engine from behind.

#	Do this	Expected result

1	F1: flyPad > Failures > ATA 26: activate **Engine 1 fire**; wait 20 s. Then deactivate it; wait 10 s.	Flames and smoke at the left engine exhaust; gone within a few seconds of the deactivation, with the ENG 1 FIRE warning.
2	F2: activate **Engine 1 fire** again. ENG 1 FIRE pb: lift the guard and push it; AGENT 1; wait 30 s (AGENT 2 if the FIRE pb is still red). Then deactivate the failure, FIRE pb back in.	The flames stop within a few seconds of the agent that puts the fire out, together with the ENG 1 FIRE pb red light. Engine 1 runs on for up to 2 min 30 s (normal).
3	F3: view of the tail from the side. Activate **APU fire**; look 20 s; deactivate it.	Smoke from the tip of the tail cone; gone after the deactivation. On the ground the APU bottle discharges by itself and puts the fire out one time in two: smoke stopping early is correct.
4	Optional (not sim-tested): MSFS menu (Esc > Failures > Engines > Engine 1): fail engine 1. Then flyPad > Settings > Sim Options: Situation Visual Effects OFF, then ON.	Smoke at the engine 1 exhaust; gone with the switch OFF, back with it ON.
5	A380X: engines off, APU running. flyPad > Failures > ATA 26: **Fire - Engine 1**; wait 15 s; deactivate; activate again, then ENG 1 FIRE pb and AGENT 1 (AGENT 2 if still red).	Flames and smoke at the left outer engine; gone with the deactivation; they stop with the FIRE pb red light when the agent puts the fire out.

19. FCOM fixes: DIRECT LAW trim, GPWS mode 2, CAB PRESS and oil temperature indications

> **Aircraft:** A32NX | **Where:** E/WD, STATUS; SD CAB PRESS and ENGINE pages; GPWS; overhead RAT & EMER GEN MAN ON pb | **Status:** Tested in the sim by the user on a combined test build: OK, recorded: CAB PRESS pulses (redone with short taps), ENG SHUT DOWN with the reverser unlocked, oil temperature amber after 15 min (caution 900.6 s after 140 °C). DIRECT LAW trim line, GPWS mode 2 hold, THR LEVER FAULT ground lines and RAT tooltip: not sim-tested (unit tests PressIndications.spec.ts, EngOilIndications.spec.ts, EngineFailAlerts.spec.ts, FadecReverserAlerts.spec.ts, FlightControlLawAlerts.spec.ts; Rust egpws tests).

What it is for

Deviations from the A320 FCOM found by the FCOM gap audit:

- **F/CTL DIRECT LAW** no longer asks for MAN PITCH TRIM USE when the green and yellow hydraulic systems are both at low pressure: the trim cannot move (FCOM "EXCEPT IF HYD Y + G SYS LO PR").
- **GPWS mode 2:** after a PULL UP, leaving the warning envelope keeps the TERRAIN aural and the red PULL UP lights until the aircraft has climbed 300 ft (45 s at most).
- **SD CAB PRESS page:** the CAB ALT value pulses from 8800 ft until back at 8600 ft, the V/S value pulses above 1750 ft/min until back at 1650 ft/min, delta P pulses above 1.5 PSI in approach; the needles no longer pulse.
- **SD ENGINE page:** the oil temperature turns amber after 15 min between 140 and 155 °C (the timer restarted at every temperature change and never ran out).
- **ENG 1(2) SHUT DOWN** with the reverser unlocked: .IF BUFFET: -MAX SPEED ... 240 KT on the E/WD; STATUS MAX SPEED 300/.78 and the approach procedure (IF BUFFET:, FOR LDG USE FLAP 3, GPWS LDG FLAP 3 ON, APPR SPD VREF+10 KT, LDG DIST PROC APPLY). **ENG THR LEVER FAULT** on the ground: ENG 1(2) IDLE POWER ONLY.
- The RAT & EMER GEN MAN ON pushbutton tooltip no longer says "(Inop.)".

Limitations and why

Limitation	Why	What to do
The cabin altitude must stay below 9550 ft in the CAB PRESS test.	Above it the CAB PR EXCESS CAB ALT warning comes.	Use short taps of the MAN V/S CTL switch.

A full MSFS restart is needed after an update.	The systems WASM module (GPWS) and the cockpit XML (tooltip) change.	Restart the sim after installing.

How to use

1. Nothing to set: the alerts and the SD pages follow the FCOM by themselves.
2. Call the SD CAB PRESS (PRESS key) or ENGINE (ENG key) page to watch the pulsing values.

How to test

#	Do this	Expected result
1	Cruise at FL 250 or above, AP ON, pressurization AUTO. SD CAB PRESS page. MODE SEL pb (MAN light on); MAN V/S CTL UP in short taps until CAB V/S about +2000 ft/min; wait until CAB ALT 8900 ft; DN until about -500 ft/min; wait until CAB ALT 8500 ft; MODE SEL pb back.	The V/S digits pulse green above 1750 ft/min and stop at 1650 ft/min or less; the CAB ALT digits pulse from 8800 ft and stop at 8600 ft; the needles stay steady green.
2	Cruise above FL 200, A/THR ON. flyPad > Failures > ATA 78: **Reverser 1 unlocked (engine at idle)**; thrust lever 1 IDLE; ENG MASTER 1 OFF; STS key.	ENG 1 SHUT DOWN with .IF BUFFET: -MAX SPEED ... 240 KT after -TCAS MODE SEL ... TA (ENG 1 REVERSE UNLOCKED also shown). STATUS: MAX SPEED 300/.78, APPR PROC:, IF BUFFET:, FOR LDG USE FLAP 3, GPWS LDG FLAP 3 ON, APPR SPD VREF+10 KT, LDG DIST PROC APPLY. Deactivate the failure, relight.
3	On the ground, engines at idle, SD ENGINE page. flyPad > Failures > ATA 79: **Engine 1 oil overheat**; thrust lever 1 forward until the oil reads 145-150 °C; wait 16 min.	The oil temperature pulses green, then about 15 min after 140 °C turns steady amber together with ENG 1 OIL HI TEMP. Lever IDLE, deactivate the failure.
4	Optional (not sim-tested): hover over the RAT & EMER GEN MAN ON pb (overhead EMER ELEC PWR panel).	The tooltip has no "(Inop.)".

20. FCOM fixes: FADEC relight, automatic ignition, start faults, gravity feed, IR alignment

> **Aircraft:** A32NX | **Where:** ENG MODE selector, ENG MASTER, overhead ANTI ICE ENG pb, ENG MAN START; overhead FUEL and ADIRS panels; E/WD, SD ENGINE and FUEL | **Status:** Tested in the sim by the user on a combined test build, recorded: FADEC relight after a flameout OK (relight at the IGN selection, idle in 35 s); IR alignment 10 min OK (601 s). Automatic continuous ignition with the engine anti-ice: FAILED : L:A32NX_ENGINE_1_CONTINUOUS_IGNITION went to 1 but the E/WD IGNITION memo did not show; cause open. Start fault order, manual start at 20 % N2, starter shaft shear in flight and gravity feed: not sim-tested (Rust tests, native cross_feed_burn_test.cpp).

What it is for

Engine, fuel and ADIRS deviations from the A320 FCOM:

- **FADEC relight:** in flight, a flameout of a running engine starts a FADEC relight attempt at once (FCOM PRO-ABN-ENG ENG 1(2) FAIL: "Selection of continuous ignition confirms the immediate relight attempt made by the FADEC"). The attempt lasts 30 s from the flameout and lights up inside the relight envelope (see "Engine flameout and seizure failures") once the continuous ignition is on, with ENG MODE IGN or selected automatically: no ENG MASTER cycle. In the quick relight zone (230-270 kt up to 20 000 ft, N2 above 15 %) it lights up at once at any ENG MODE position (design choice: the time since the flameout stands for the MASTER OFF time). No light-up within 30 s: the engine stays out until a crew relight (MASTER OFF then ON). Not on the ground, not for a seizure.
- **Automatic continuous ignition:** the FADEC selects the continuous ignition by itself while the engine anti-ice is on (DSC-70-80-30); the IGNITION memo and the SD igniters should show it.

- **Start faults:** on the ground, after a stall or an EGT over limit the FADEC runs a new start first and dry cranks only when no new start is left; a manual start lights up from 20 % N2; a starter that fails during an in-flight relight gives ENG START FAULT STARTER SHAFT SHEAR with WINDMILL START ONLY for the rest of the flight (details in "Engine ignition and start failures").
- **Gravity feed:** with the X FEED open and all four wing tank pumps off, each engine burns the fuel of its own wing tank (before, the burned fuel was taken from no tank).
- **IR alignment:** a complete IR alignment takes 10 min at any latitude, as in the A320 FCOM (before, 5 to 17 min with the latitude, like the A380).
- The MCDU PROG REC MAX with one engine out is not on this branch (it is on `core/efb-redesign`).

Limitations and why

Limitation	Why	What to do
The IGNITION memo did not show with the automatic continuous ignition (sim).	Cause open: the ignition variable was set, and the FWC review found nothing in this code that removes the memo.	Select ENG MODE IGN for the memo; the relight itself works with the automatic ignition (Rust tests).
The automatic ignition only follows the engine anti-ice.	Design choice: the MSFS ENG ANTI ICE stands for the valve; the other conditions of the FCOM automatic selection are a figure, not in the text.	-
In the quick relight zone a flameout relights at once by itself.	Design choice of the FADEC attempt (see above).	Test the crew relight outside that zone.
A full MSFS restart is needed after an update.	The systems and FADEC WASM modules change.	Restart the sim after installing.

How to use

1. After an ENG FAIL in flight, follow the ECAM: ENG MODE SEL IGN within 30 s lets the FADEC relight the engine inside the envelope; after 30 s use the crew relight (ENG MASTER OFF then ON).
2. For the IR alignment, flyPad > Settings > Realism > ADIRS align time: Real; IR selectors to NAV.
3. For gravity feeding, FUEL X FEED pb ON with the wing tank pumps off.

How to test

#	Do this	Expected result
1	G9: A32NX cold and dark, ADIRS align time Real. IR 1, 2 and 3 selectors to NAV.	IRS IN ALIGN memo counting down; the alignment ends after about 10 min at any latitude (OK: 601 s).
2	G1: about FL 100, 290 kt, A/THR ON, ENG MODE NORM, engine anti-ice OFF. flyPad > Failures > ATA 72: **Engine 1 flameout (crew relight possible)**; wait 10 s; ENG MODE IGN/START.	Amber ENG 1 FAIL; within about 2 s of IGN/START engine 1 N2 rises back to idle without touching ENG MASTER 1, and ENG 1 FAIL clears (OK). Deactivate the failure, ENG MODE NORM.
3	G3 (FAILED): cruise, ENG MODE NORM, SD ENGINE page. Overhead ANTI ICE ENG 1 pb ON; wait 10 s; then OFF.	Expected: IGN A (or B) for engine 1 on the SD and the green IGNITION memo, gone at the pb OFF. In the test the IGNITION memo did not show.
4	Optional G2 (not sim-tested): as G1, but wait 35 s before IGN/START.	No relight; ENG MASTER 1 OFF then ON still relights.
5	Optional G4 (not sim-tested): ground, APU BLEED ON, ENG MODE IGN/START. ATA 80: **Engine 1 stall during the start**; ENG MASTER 1 ON.	ENG 1 START FAULT STALL with NEW START IN PROGRESS first, then AUTO CRANK IN PROGRESS, then -ENG MASTER 1 ... OFF.

6	Optional G5 (not sim-tested): ground, APU BLEED ON, ENG MODE IGN/START. ENG MAN START 1 ON; when N2 stops rising, ENG MASTER 1 ON.	Light-up (FF and EGT rise) at 20-21 % N2, no IGNITION FAULT.
7	Optional G6 (not sim-tested): FL 100, 200 kt, APU BLEED ON. **Engine 1 starter shaft shear**, then **Engine 1 flameout (crew relight possible)**; ENG MODE IGN/START; ENG MASTER 1 OFF, 30 s, ON; wait 12 s.	ENG 1 START FAULT STARTER SHAFT SHEAR; STATUS ENG 1 WINDMILL START ONLY.
8	Optional G8 (not sim-tested): note both wing tank quantities (SD FUEL). ATA 28: **L TK pump 1**, **L TK pump 2**, **R TK pump 1**, **R TK pump 2**; FUEL X FEED pb ON; wait 5 min.	Both wing tanks go down (before: neither moved).

21. ELEC bus, ESS bus and TR alerts; DC ESS supply in DC BUS 1+2 fault

> **Aircraft:** A32NX | **Where:** flyPad > Failures > ATA 24 Electrical Power; overhead ELEC panel; E/WD, STATUS, SD ELEC | **Status:** ELEC alerts: tested in the sim by the user on a combined test build: OK (no nuisance alert in a start-up; AC BUS 1 FAULT, DC EMER CONFIG and DC BUS 1+2 FAULT; recorded). DC ESS supply: OK (recorded). Unit tests ElecBusAlerts.spec.ts; Rust tests electrical/mod.rs.

What it is for

New ELEC alerts of the A320 FCOM (PRO-ABN-ELEC) with their procedure and STATUS lines, triggered by the Rust electrical network: **AC BUS 1(2) FAULT**, **AC ESS BUS FAULT**, **AC ESS BUS SHED**, **AC ESS BUS ALTN**, **DC BUS 1(2) FAULT**, **DC BUS 1+2 FAULT**, **DC ESS BUS FAULT**, **DC ESS BUS SHED**, **DC BAT BUS FAULT**, **DC EMER CONFIG**, **ESS BUSES ON BAT**, **TR 1(2) FAULT**, **GEN 1(2) OFF**, **BAT 1(2) OFF** and **EMER GEN 1 LINE OFF**.

- DC EMER CONFIG covers DC BUS 1, DC BUS 2, DC BUS 1+2 and DC ESS BUS FAULT / SHED; DC BUS 1+2 FAULT covers DC BUS 1 and DC BUS 2 FAULT; an ESS BUS FAULT covers its SHED alert; with the ESS buses on the batteries no SHED alert (the shedding is normal). In ELEC EMER CONFIG the bus losses that make that configuration are not shown again.
- **DC ESS supply (fix):** with DC BUS 1 and 2 lost (DC BUS 1+2 FAULT) the ESS TR now supplies the DC ESS BUS (FCOM DSC-24-10-30-30): FAC 1, FMGC 1 / AP 1, FCDC 1, FCU channel 1 and the other DC ESS users stay available; only FAC 2 / AP 2 / FMGC 2 are lost. Before, the DC ESS BUS was lost too.
- **No power break (fix):** the DC BAT BUS and the DC ESS BUS behind it lost power for one update each time a DC bus was lost or came back: the FCU restarted and dropped the AP and A/THR. They now stay powered.

Limitations and why

Limitation	Why	What to do
The flight phase inhibitions and alert levels are general Airbus values.	The FCOM gives them as figures that the text copy drops (design choice).	-
No FUEL CONSUMPT INCRSD / FMS PRED UNRELIABLE lines in these procedures.	The condition ("affects the aerodynamic characteristics") is not computed.	-
After DC BUS 1+2 is restored in flight, FAC 2 stays in fault.	As on the aircraft: the FCOM asks for FAC OFF then ON.	FAC 2 pb OFF then ON.
In DC EMER CONFIG the cabin climbs.	The packs and the cabin pressure controllers lose power (seen in the sim, as the FCOM gives).	Descend as the procedure asks.

The STATUS of DC BUS 1+2 FAULT does not page over two pages here.	The STATUS page paging needs the fork-only CLR / RCL rework, which this branch does not have.	-
The DC ESS test ran with a SEC power-up self-test fix that is not on this branch.	That fix needs code this branch does not have.	A SEC FAULT after the DC buses come back is not checked here.
A full MSFS restart is needed after an update.	The systems WASM module changes (DC ESS fix).	Restart the sim after installing.

How to use

1. Open flyPad > Failures > ATA 24 Electrical Power and activate a bus failure ("AC 1", "DC 1", "DC 2", "DC ESS"...).
2. Follow the ECAM and STATUS; deactivate the failure after the test (FAC OFF then ON if it stays in fault).

How to test

#	Do this	Expected result
1	E1: cold and dark at the gate. BAT 1 and 2 AUTO, EXT PWR ON; APU MASTER SW ON, START, APU BLEED ON at AVAIL; EXT PWR OFF; start engine 2, then engine 1; APU BLEED OFF, APU MASTER SW OFF.	Never an AC / DC BUS, ESS, SHED, TR, ESS BUSES ON BAT, APU BLEED FAULT, X BLEED FAULT, ENG BLEED NOT CLSD or BLEED n OFF alert.
2	E2: cruise, AP ON. ATA 24: **AC 1**; wait 10 s; VENT BLOWER pb OVRD; STS key. Then deactivate **AC 1**, BLOWER AUTO.	After about 1 s amber ELEC AC BUS 1 FAULT, chime, MASTER CAUT, SD ELEC, -BLOWER ... OVRD; then AC ESS BUS ALTN (no chime); never AC / DC ESS BUS FAULT. STATUS: LDG DIST PROC APPLY, SLATS SLOW, CAT 2 ONLY, INOP SYS. The AP may disconnect during the AC ESS transfer.
3	Cruise. ATA 24: **DC 1**, then **DC 2**, then **DC ESS**; wait 10 s; then deactivate the three.	Amber DC EMER CONFIG with -EMER ELEC PWR MAN ON and the DC BUS 1+2 lines, DC BAT BUS FAULT, amber LAND ASAP; no separate DC BUS 1 / DC BUS 2 / DC BUS 1+2 / DC ESS BUS FAULT.
4	Cruise, AP and A/THR ON. ATA 24: **DC 1** and **DC 2**; wait 30 s; then deactivate both.	ELEC DC BUS 1+2 FAULT; the DC ESS BUS stays powered (SD ELEC), FAC 1 healthy, normal law, AP 1 and A/THR stay engaged; only FAC 2 FAULT. After the restore: no AP OFF; FAC 2 FAULT until FAC 2 pb OFF then ON (OK).

22. Electrical failures: battery thermal runaway, BCL, IDG oil

> **Aircraft:** A32NX | **Where:** flyPad > Failures > ATA 24 Electrical Power; overhead ELEC panel (BAT, IDG, GALY & CAB pb); E/WD, STATUS, SD ELEC | **Status:** Tested in the sim by the user on a combined test build: OK (P1 battery thermal runaway, recorded; P2 IDG oil overheat, judged by the user, no IDG variable recorded). BCL, IDG oil leak, GEN OVERLOAD and STATIC INV FAULT: not sim-tested. Unit tests ElecSystemAlerts.spec.ts, IdgIndications.spec.ts; Rust tests.

What it is for

flyPad failure (ATA 24)	What the code does
Battery 1(2) (thermal runaway)	A cell short: the charging current keeps rising (2 A per second); the BCL sees it above 12 A for 2 s, opens the battery contactor and latches the fault: ELEC BAT 1(2) FAULT, BAT pb FAULT light. BAT pb OFF resets it.

BCL 1(2)	The battery charge limiter fails and keeps the battery contactor open: ELEC BCL 1(2) FAULT, BAT pb FAULT light.
IDG 1(2) oil leak	The drive loses 1 % of its oil each second while it turns: IDG LO PR about 50 s later (below half of the oil), then the oil heats up.
IDG 1(2) oil overheat	The oil cooling is lost: the IDG temperature rises 0.5 °C/s; ELEC IDG 1(2) OIL OVHT above 180 °C (about 3.5 min), and the IDG disconnects by itself at 200 °C (about 40 s later).

- Alerts (FCOM PRO-ABN-ELEC): **ELEC BAT 1(2) FAULT** and **BCL 1(2) FAULT** (crew awareness; STATUS APU BAT START NOT AVAIL, INOP SYS BAT 1(2) / BCL 1(2)); **ELEC IDG 1(2) OIL LO PR** / **OIL OVHT** (-IDG 1(2) ... OFF; STATUS CAT 3 SINGLE ONLY, INOP SYS MAIN GALLEY if the APU GEN is not online, GEN 1(2), CAT 3 DUAL); **ELEC GEN 1(2) / APU GEN OVERLOAD** (a generator above 100 % of its rated load: -GALY/CAB ... OFF, INOP SYS GALY/CAB); **ELEC STATIC INV FAULT** (the existing **Static Inverter** failure).
- Cockpit: the BAT, IDG and GALY & CAB pb FAULT lights.
- SD ELEC page: IDG LO PR, the IDG temperature pulsing green from 147 °C and amber above 180 °C. The normal IDG oil now reads about 50-95 °C (it read 160-210 °C at high power).

Limitations and why

Limitation	Why	What to do
Inhibitions: BAT, BCL, GEN OVERLOAD and STATIC INV FAULT in flight phases 3, 4, 5, 7, 8; IDG OIL LO PR / OVHT in 1, 4, 5, 7, 8, 10.	The FCOM figures (2019 FCOM PDF).	Test with the engines running at the gate or in cruise.
The runaway, leak and heating rates are design values.	The FCOM gives none.	-
A disconnected IDG cannot be reconnected in flight.	As on the aircraft (fusible device).	-
A full MSFS restart is needed after an update.	The systems WASM module changes.	Restart the sim after installing.

How to use

1. Open flyPad > Failures > ATA 24 Electrical Power and activate a battery, BCL or IDG failure with the engines running.
2. Follow the ECAM; after a battery fault set the BAT pb OFF then AUTO once the failure is cleared.

How to test

Setup: on the ground, both engines running, BAT 1 and 2 AUTO.

#	Do this	Expected result
1	P1: ATA 24: **Battery 1 (thermal runaway)**; wait 15 s; STS key.	Amber ELEC BAT 1 FAULT, chime, BAT 1 pb FAULT light, BAT 1 contactor open on the SD ELEC page. STATUS: APU BAT START NOT AVAIL, INOP SYS BAT 1.
2	Deactivate it; BAT 1 pb OFF, then AUTO.	The light and the alert go.
3	P2: ATA 24: **IDG 2 oil overheat**; watch the SD ELEC page for 5 min.	The IDG 2 temperature rises, pulses green from 147 °C, turns amber above 180 °C with ELEC IDG 2 OIL OVHT and the IDG 2 FAULT light, then the IDG disconnects by itself at 200 °C (DISC on the SD).

4	Optional (not sim-tested): ATA 24: **IDG 1 oil leak**; wait 1 min.	ELEC IDG 1 OIL LO PR, IDG 1 FAULT light, LO PR on the SD ELEC page.

23. Pneumatic failures with BMC leak and bleed monitoring

> **Aircraft:** A32NX | **Where:** flyPad > Failures > ATA 36 Pneumatic; overhead AIR COND panel; E/WD, SD BLEED | **Status:** Tested in the sim by the user on a combined test build: OK (left wing bleed leak; engine bleed overheat in the climb; recorded). The other ATA 36 failures: not sim-tested. Unit tests BleedLeakMonitoringAlerts.spec.ts, a320.spec.ts; Rust tests pneumatic.rs.

What it is for

flyPad failures (ATA 36)	What the code does
Engine 1(2) bleed valve stuck closed / stuck open (no regulation)	The valve stays closed / fully open whatever the BMC commands; stuck open: ENG BLEED FAULT by overpressure at climb power, ENG BLEED NOT CLSD at engine start, with APU BLEED ON or the engine stopped.
Engine 1(2) HP valve stuck closed	No HP bleed at low thrust: ENG HP VALVE FAULT, low bleed pressure at idle.
Engine 1(2) precooler fan air valve stuck closed (overheat) / stuck open (low temperature)	Closed: ENG BLEED HI TEMP, then FAULT above climb power. Open: ENG BLEED LO TEMP in flight with the wing anti-ice at low thrust.
X BLEED valve stuck closed / stuck open; APU bleed valve stuck closed / stuck open	The valve stays where it is: X BLEED FAULT or APU BLEED FAULT (see "AIR bleed cautions").
Engine 1(2) pylon bleed leak, Left / Right wing bleed leak, APU bleed duct leak	Hot air leaks while the duct is pressurised; the leak detection loops close the valves that feed it (the pylon loop at most thrust settings, not always at ground idle; the wing loops at any thrust; the APU loop with APU BLEED ON).
Engine 1(2) pylon, Left / Right wing loop A / B, APU leak detection loop	LEAK DET FAULT (a wing: both loops of that wing failed).
BMC 1 (2) (bleed monitoring computer)	BLEED MONIT SYS 1(2) FAULT; the other BMC keeps the overpressure, overheat and wing leak monitoring; the engine bleed valve of the failed BMC no longer closes by itself and its FAULT light is lost. Both: BLEED MONITORING FAULT.

- The BMCs now detect leaks, HP valve faults and an overpressure above 57 PSI, light the ENG BLEED and APU BLEED FAULT lights and close the affected valves (for a wing leak, the engine bleed and X BLEED valves).
- New AIR cautions (amber, SD BLEED page): **L(R) WING LEAK** and **ENG 1(2) BLEED LEAK** (-ENG 1(2) BLEED ... OFF, -APU BLEED ... OFF on the left side, -X BLEED ... SHUT, -WING ANTI ICE ... OFF, AVOID ICING CONDITIONS), **APU BLEED LEAK** (-APU BLEED ... OFF), **ENG 1(2) / L(R) WNG / APU LEAK DET FAULT**, **ENG 1(2) HP VALVE FAULT**, **BLEED MONIT SYS 1(2) FAULT** and **BLEED MONITORING FAULT**.

Limitations and why

Limitation	Why	What to do
No APU BLEED LEAK "APU LEAK FED BY ENG" variant, no LEFT / RIGHT LEAK variants of ENG 1+2 BLEED FAULT.	Not modelled: the Rust APU duct leak only blows APU air.	-
The flight phase inhibitions are the FCOM PDF figures.	Read on the FCOM PDF pages.	Test on the ground or in cruise.

A full MSFS restart is needed after an update.	The systems WASM module changes.	Restart the sim after installing.

How to use

1. Open flyPad > Failures > ATA 36 Pneumatic and activate a valve, leak, loop or BMC failure.
2. Follow the ECAM (bleed pb OFF, X BLEED SHUT...); deactivate the failure and set the bleed pb back ON.

How to test

#	Do this	Expected result
1	K1: on the ground, both engines at idle, APU off. ATA 36: **Left wing bleed leak**; wait 10 s.	Amber AIR L WING LEAK, ENG 1 BLEED FAULT light, engine 1 bleed valve and X BLEED closed (SD BLEED).
2	ENG 1 BLEED pb OFF; deactivate the leak; wait 15 s; ENG 1 BLEED pb ON.	The alert and the FAULT light go out.
3	K2: climbing above 1 500 ft at CLB thrust. ATA 36: **Engine 2 precooler fan air valve stuck closed (overheat)**; wait 20 s.	ENG 2 BLEED HI TEMP for a few seconds, then AIR ENG 2 BLEED FAULT, FAULT light on, engine 2 bleed valve closed.
4	ENG 2 BLEED pb OFF; deactivate the failure; wait 30 s; ENG 2 BLEED pb ON.	The FAULT clears and the valve reopens.

24. AIR bleed cautions

> **Aircraft:** A32NX | **Where:** Overhead AIR COND panel (ENG BLEED, APU BLEED pb, X BLEED selector); E/WD, STATUS, SD BLEED | **Status:** Tested in the sim by the user on a combined test build: OK (tests E1: no nuisance bleed alert at start-up, and B1: AIR BLEED 1 OFF; recorded). The other bleed cautions: tested through K1 / K2 of the pneumatic failures, otherwise not sim-tested. Unit tests BleedAlerts.spec.ts, ApuMemos.spec.ts.

What it is for

New AIR cautions of the A320 FCOM (PRO-ABN-AIR) with their procedures and STATUS, from the flags of the Rust pneumatic system: **ENG 1(2) BLEED FAULT** and **ENG 1+2 BLEED FAULT**, **ENG 1(2) BLEED LO TEMP** and **ENG 1+2 BLEED LO TEMP**, **ENG 1(2) BLEED HI TEMP**, **ENG 1(2) BLEED NOT CLSD**, **BLEED 1(2) OFF**, **APU BLEED FAULT** and **X BLEED FAULT**, with the X BLEED and APU BLEED STATUS lines.

- ENG BLEED FAULT: engine running with a bleed pressure above 57 PSI, or a bleed temperature above 257 °C for 55 s, 270 °C for 15 s or 290 °C for 5 s (the BMC overheat monitor). ENG BLEED HI TEMP: precooler outlet above 240 °C. ENG BLEED LO TEMP: bleed air below 150 °C in flight with the wing anti-ice ON.
- BLEED 1(2) OFF: the ENG BLEED pb OFF with the engine running, shown in flight phases 2 and 6 only.
- Fix: the APU BLEED memo and the bleed alerts read the APU BLEED pb variable the cockpit writes (L:A32NX_OVHD_PNEU_APU_BLEED_PB_IS_ON); the old name was never written, so the APU BLEED memo never showed.

Limitations and why

Limitation	Why	What to do
The FCOM EMER CANC and RCL behaviour of the same change is not on this branch.	It needs the fork-only E/WD CLR / RCL rework, which no core branch has: EMER CANC and RCL keep the FBW behaviour here.	-

The valve confirmation times (APU BLEED FAULT 10 s, X BLEED FAULT 5 s, BLEED NOT CLSD 10 s) are design values.	The FCOM gives none.	-
The flight phase inhibitions are the FCOM PDF figures.	Read on the FCOM PDF pages.	Test on the ground with the engines running, or in cruise.

How to use

1. Nothing to set: the cautions follow the bleed system. Use the ATA 36 failures (previous feature) to make the bleed faults.
2. Follow the ECAM and STATUS.

How to test

#	Do this	Expected result
1	On the ground, both engines running, no alert (FWC phase 2). ENG 1 BLEED pb OFF; wait 3 s.	Amber AIR BLEED 1 OFF, single chime, MASTER CAUT, SD BLEED page.
2	ENG 1 BLEED pb back ON.	The caution goes.
3	Optional (not sim-tested on its own): APU AVAIL, APU BLEED pb ON.	Green APU BLEED memo on the E/WD in place of APU AVAIL (it never showed before the fix).

25. HYD and APU shutdown alerts

> **Aircraft:** A32NX | **Where:** flyPad > Failures > ATA 29 Hydraulic Power; overhead HYD and APU panels, APU FIRE pb; E/WD, STATUS, SD HYD | **Status:** Tested in the sim by the user on a combined test build: OK (engine start without nuisance HYD alerts and APU EMER SHUT DOWN;: blue then yellow reservoir leak; recorded). Unit tests HydraulicAlerts.spec.ts, ApuShutdownAlerts.spec.ts, StatusMessages.spec.ts.

What it is for

- New HYD cautions and warnings of the A320 FCOM (PRO-ABN-HYD) with their procedure and STATUS lines: **B (Y) ELEC PUMP LO PR / OVHT**, **G ENG 1 (Y ENG 2) PUMP LO PR**, **B / G / Y RSVR LO AIR PR**, **RSVR LO LVL** and **SYS LO PR**, **PTU FAULT**, and the red **B+Y**, **G+B**, **G+Y SYS LO PR** warnings (with red LAND ASAP) that replace the single system alerts.
- The RSVR OVHT procedures show the "x SYS LO PR" lines; RSVR OVHT, RAT FAULT and PTU FAULT now have their STATUS (INOP SYS, APPR PROC). No pump caution on the ground with its engine stopped.
- New **APU AUTO SHUT DOWN** (an automatic shutdown other than a fire) and **APU EMER SHUT DOWN** (APU FIRE pb pushed, or an APU fire on the ground) cautions: -MASTER SW ... OFF, STATUS INOP SYS APU; they end when the APU has stopped after the MASTER SW OFF.
- A STATUS line given by two alerts is shown once.

Limitations and why

Limitation	Why	What to do
The flight phase inhibitions and alert levels are general Airbus values.	The FCOM gives them as figures that the text copy drops (design choice), except the notes the text keeps.	-

No SECONDARY FAILURES lines (*F/CTL, *WHEEL) and no option lines in STATUS.	Not modelled (design choice).	-
No APU EMER SHUT DOWN while the APU FIRE warning shows.	Design choice: the fire procedure has the MASTER SW OFF line too.	-
A reservoir leak does not refill.	The leaked fluid stays lost.	End or reload the flight after the test.
Watch items: an ENG FAIL in flight with the ENG MASTER still ON also brings the G ENG 1 (Y ENG 2) PUMP LO PR; AC BUS 1 FAULT also brings HYD B SYS LO PR.	The FCOM inhibits the pump caution only on the ground.	-

How to use

1. Open flyPad > Failures > ATA 29 Hydraulic Power and activate a reservoir or pump failure; or push the APU FIRE pb with the APU running.
2. Follow the ECAM and STATUS.

How to test

#	Do this	Expected result
1	Y1: cold and dark at the gate, APU running, APU BLEED ON. Start engine 2, then engine 1; ENG MODE NORM.	No HYD caution at any moment (no ENG PUMP, ELEC PUMP, SYS LO PR or PTU FAULT).
2	Push the APU FIRE pb (no fire); STS key; then APU MASTER SW OFF; APU FIRE pb back in.	Amber APU EMER SHUT DOWN with -MASTER SW ... OFF, no APU FIRE warning; STATUS INOP SYS APU. The line goes at MASTER SW OFF and the caution ends when the APU has stopped.
3	Y2 (last in the flight): cruise, AP ON. ATA 29: **Blue reservoir leak**; wait for the caution; BLUE ELEC PUMP pb OFF; STS key.	HYD B RSVR LO LVL, single chime, -BLUE ELEC PUMP ... OFF; no separate HYD B SYS LO PR. After the pb OFF: B SYS LO PR, FUEL CONSUMPT INCRSD, FMS PRED UNRELIABLE. STATUS: LDG DIST PROC APPLY, SLATS SLOW, CAT 3 SINGLE ONLY; INOP SYS BLUE HYD, B ELEC PUMP.
4	ATA 29: **Yellow reservoir leak**; wait for the warning.	Red HYD B+Y SYS LO PR with the continuous chime and red LAND ASAP, MAX SPEED 320/.77, MANEUVER WITH CARE; the B and Y RSVR LO LVL cautions go.

26. F/CTL surface cautions and STATUS

> **Aircraft:** A32NX | **Where:** flyPad > Failures > ATA 27 Flight Controls; E/WD, STATUS, SD F/CTL | **Status:** Tested in the sim by the user on a combined test build: OK (tests C1 left elevator and aileron lost, C2 ELAC 1+2 FAULT; recorded). Unit tests FlightControlSurfaceAlerts.spec.ts, FlightControlStatus.spec.ts.

What it is for

- New F/CTL cautions from the flight control computers: **L(R) ELEV FAULT**, **L(R) AIL FAULT**, **SPLR FAULT**, **ELEV SERVO FAULT**, **AIL SERVO FAULT** and **ELAC 1(2) PITCH FAULT**, with the FCOM phase inhibitions. A loss caused by a hydraulic system or a failed computer stays with the HYD or ELAC / SEC alert.

- Both ELACs failed: one **ELAC 1+2 FAULT** procedure instead of two ELAC FAULT alerts. ALTN LAW shows SPD BRK DO NOT USE for an L or R elevator fault, as in the FCOM. FCDC 1+2 FAULT is no longer inhibited in phase 3.
- STATUS: ALTN LAW, DIRECT LAW, ELAC, SEC, FCDC, elevator, aileron and spoiler faults show their FCOM limitations, approach procedure, information and INOP SYS lines.

Limitations and why

Limitation	Why	What to do
SPLR FAULT, AIL / ELEV SERVO FAULT and ELAC PITCH FAULT cannot be made in the sim.	No flyPad failure for the servos.	-

How to use

1. Open flyPad > Failures > ATA 27 Flight Controls and activate ELAC, SEC or FCDC failures.
2. Follow the ECAM and STATUS; an ELAC may need its pb OFF then ON after the failure is cleared.

How to test

Setup: on the ground, both engines running.

#	Do this	Expected result
1	C1: overhead HYD panel PTU pb OFF, then ENG 1 PUMP pb OFF.	Amber HYD G ENG 1 PUMP LO PR with the G SYS LO PR line (no separate HYD G SYS LO PR).
2	ATA 27: **ELAC 1**, then **SEC 1**; STS key.	ELAC 1 FAULT, SEC 1 FAULT, then amber F/CTL L ELEV FAULT and F/CTL L AIL FAULT; no R ELEV, no SERVO FAULT. INOP SYS lists L ELEV and L AIL.
3	Deactivate ELAC 1 and SEC 1; ENG 1 PUMP and PTU pb back ON.	Everything goes (ELAC 1 may need its pb OFF then ON).
4	C2: ATA 27: **ELAC 1**, then **ELAC 2**; STS key.	One amber F/CTL ELAC 1+2 FAULT (-ELAC 1 OFF THEN ON, -ELAC 2 OFF THEN ON, .IF BOTH ELAC FAULT : ...). STATUS: MAX SPEED 320 KT, ALTN LAW : PROT LOST, WHEN L/G DN : DIRECT LAW; INOP SYS F/CTL PROT, L+R AIL, ELAC 1+2.
5	Deactivate ELAC 2, then ELAC 1.	Back to F/CTL ELAC 1 FAULT only, then no alert.

27. FAC actuator failures, YAW DAMPER / RUD TRIM / RUD TRV LIM cautions

> **Aircraft:** A32NX | **Where:** flyPad > Failures > ATA 22 Auto Flight; overhead FLT CTL panel (FAC pbs); E/WD, STATUS | **Status:** Not sim-tested (unit tests: FacActuatorAlerts.spec.ts, 17 tests; native fac_actuator_failure_test.cpp).

> **NOTE**
> This feature and the next ones up to "DC BUS 2 takes over DC BUS 1 after 5 s" came to this branch later: the A32NX parts of the FCOM alert additions, the new voices, the exceedance marks fix and the DC BUS 1 transfer. **None of them is sim-tested**: they are checked by unit tests and Rust tests only. Three commits of that set change nothing for the user: a CHANGELOG move (the HYD and APU SHUT DOWN lines into the current list), a spec fix (the anti-ice cautions spec sets the normal fuel, cargo door and landing elevation state) and a Rust lint fix-up (an unused host stub of the systems WASM removed).

What it is for

The FCOM gap audit found the FAC actuator failures missing: the FAC inputs for its actuators were always "healthy". Most other P1 items (HYD single and dual system cautions, PTU FAULT, ELAC and elevator faults, F/CTL STATUS, APU AUTO / EMER SHUT DOWN) were already on this branch: see "HYD and APU shutdown alerts" and "F/CTL surface cautions and STATUS".

- Six new failures, ATA 22: the yaw damper actuators 1 and 2, the rudder trim actuator motors 1 and 2 and the rudder travel limiter actuator motors 1 and 2. FAC 1 drives the green yaw damper actuator and motor 1 of the trim and limiter actuators, FAC 2 the yellow actuator and motor 2. A failed actuator is reported to its FAC and the other FAC takes the function over (FAC 1 has the priority).
- New AUTO FLT cautions with their procedures and STATUS (A320 FCOM PRO-ABN-AUTO_FLT, the flight phase inhibitions are read on the FCOM PDF figures).

What the crew sees

Situation	What you should see
One actuator failed	AUTO FLT YAW DAMPER 1(2), AUTO FLT RUD TRIM 1(2) FAULT or AUTO FLT RUD TRV LIM 1(2). STATUS INOP SYS YAW DAMPER 1(2), RUD TRIM 1(2) or RUD TRV LIM 1(2). The other FAC keeps the function.
Both actuators of a function failed	AUTO FLT YAW DAMPER SYS, RUD TRIM SYS or RUD TRV LIM SYS with -FAC 1 ... OFF THEN ON and -FAC 2 ... OFF THEN ON. STATUS INOP SYS YAW DAMPER, RUD TRIM or RUD TRV LIM. Both yaw dampers lost: the ELACs go to alternate law (F/CTL ALTN LAW).
RUD TRV LIM SYS	RUD WITH CARE ABV 160 KT. When the limiter stays locked at high speed with the slats extended (design choice: limit below 24 degrees): MAX X WIND FOR LDG 15 KT, -AUTO BRK ... DO NOT USE, .AT LDG ROLL: -DIFF BRAKING ... AS RQRD, also on the STATUS.
Function lost by the FACs or the hydraulics (FAC 1+2 FAULT, FAC pbs OFF, HYD G+Y SYS LO PR)	No SYS caution: those alerts already list the lost function (design choice).

Limitations and why

Limitation	Why	What to do
No F/CTL GND SPLR or SPD BRK FAULT cautions.	They need a SEC model change: the fault bit and the lever status are constants in the generated SEC code.	-
A failed FAC reports no actuator fault.	The FAC 1(2) FAULT alert covers it.	-
A full MSFS restart is needed after an update.	The fly-by-wire WASM module changes.	Restart the sim after installing.

How to use

1. Open flyPad > Failures > ATA 22 Auto Flight and activate an actuator failure.
2. Follow the ECAM (FAC OFF THEN ON for a SYS caution) and read the STATUS page.
3. Deactivate the failure after the test.

How to test

#	Do this	Expected result

1	Cruise, AP ON. flyPad > Failures > ATA 22: **Yaw damper actuator 1 (FAC 1, green)**; STS key.	Amber AUTO FLT YAW DAMPER 1; STATUS INOP SYS YAW DAMPER 1; the yaw damper stays active through FAC 2.
2	Add **Yaw damper actuator 2 (FAC 2, yellow)**.	AUTO FLT YAW DAMPER SYS with -FAC 1 ... OFF THEN ON, -FAC 2 ... OFF THEN ON (no separate YAW DAMPER 1 / 2); F/CTL ALTN LAW; STATUS INOP SYS YAW DAMPER.
3	Deactivate both failures; FAC 1 and FAC 2 pb OFF then ON.	The cautions go.
4	Repeat with **Rudder trim actuator motor 1 (FAC 1)** and **Rudder trim actuator motor 2 (FAC 2)**.	AUTO FLT RUD TRIM 1 FAULT, then AUTO FLT RUD TRIM SYS; STATUS INOP SYS RUD TRIM.
5	Repeat with **Rudder travel limiter actuator motor 1 (FAC 1)** and **Rudder travel limiter actuator motor 2 (FAC 2)**.	AUTO FLT RUD TRV LIM 1, then AUTO FLT RUD TRV LIM SYS with RUD WITH CARE ABV 160 KT; STATUS INOP SYS RUD TRV LIM.

What PR testers should check

- One actuator failed: the 1(2) caution and the other FAC takes over.
- Both actuators failed: the SYS caution; for the yaw dampers, alternate law too.

28. ADR and IR failures, ADIRU power supplies, NAV ADR / IR / RA alerts

> **Aircraft:** A32NX (A380X ADIRU supplies too) | **Where:** flyPad > Failures > ATA 34 Navigation; overhead ADIRS panel; E/WD, STATUS | **Status:** Not sim-tested (unit tests: adirs.rs 17 Rust tests, AdirsAlerts.spec.ts 22, the failure list specs).

What it is for

- **ADR and IR fault failures** (ATA 34): the ADR (IR) outputs go to failure warning and its FAULT light comes on (A320 FCOM DSC-34-NAV-10-20). **Erroneous failures:** an ADR with an erroneous pitot reads a low speed (40 kt below), an ADR with an erroneous AOA probe a high angle of attack (10 degrees above), both with a valid status; an IR can have an erroneous heading. These values are design values (the FCOM gives none).
- **Per-ADIRU power supplies:** ADR 1 on AC ESS, ADR 2 on AC BUS 2, ADR 3 on AC BUS 1; the IRs fall back to the batteries (IR 2 and 3 for 5 min), with the ON BAT light and, on the ground, the external horn (FCOM). Power cuts up to 200 ms and the first 5 s after loading are ignored, so a flight loaded on the runway or in the air keeps its alignment.
- **New NAV alerts** (A320 FCOM PRO-ABN-NAV): ADR 1(2)(3) FAULT, ADR 1+2 (1+3)(2+3) FAULT, ADR 1+2+3 FAULT, ADR DISAGREE, IR 1(2)(3) FAULT, IR 1+2 (1+3)(2+3) FAULT, IR DISAGREE, RA 1 AND 2 FAULT, RA DEGRADED, and the RA 1(2) FAULT STATUS (CAT 2 ONLY; INOP SYS RA 1(2), CAT 3).
- The shared ADIRS model now takes the supplies of each aircraft, so the A380X ADIRU supplies (from the A380 FCOM ELEC INOP SYS lists) are on this branch too. The erroneous ADR model comes from a prerequisite commit of the shared ADIRS code (the A380X air data alerts that use it are on `core/failures-a380x`).

What the crew sees

Failure	E/WD	STATUS
ADR 1 fault	NAV ADR 1 FAULT: -AIR DATA SWTG ... CAPT 3, -ADR 1 P/B ... OFF	-
ADR 2 fault	NAV ADR 2 FAULT: -AIR DATA SWTG ... F/O 3, -ADR 2 P/B ... OFF, -BARO REF ... CHECK	-
Two ADRs	NAV ADR 1+2 (1+3)(2+3) FAULT with the switching and ATC/XPDR lines and the ADR P/B OFF lines	INOP SYS ADR 1+2 (1+3)(2+3)

Three ADRs	NAV ADR 1+2+3 FAULT: -AP+FD OFF, -A/THR OFF, -PROBE/WINDOW HEAT ON, -ADR 1+2+3 P/B OFF, -SPD FLY THE GREEN, -STBY INST MAY BE UNREL, -ALL ADR OFF PROC APPLY	INOP SYS ADR 1+2+3
Two remaining ADRs disagree	NAV ADR DISAGREE: .IF SPD DISAGREE: -ADR CHECK PROC APPLY; .IF NO SPD DISAGREE: AOA DISCREPANCY	Approach procedure (FLAP 3), IF NO SPD DISAGREE: RISK OF UNDUE STALL WARN
IR 1 (2) fault	NAV IR 1 (2) FAULT: -ATT HDG SWTG ... CAPT 3 (F/O 3), -ATC/XPDR ... SYS 2 (SYS 1)	INOP SYS IR 1 (2)
RA 1 and RA 2	NAV RA 1 AND 2 FAULT	INOP SYS RA 1+2

Limitations and why

Limitation	Why	What to do
No ELAC alternate law on ADR DISAGREE.	Not done yet.	-
No flashing IR FAULT light (ATT mode), no ADIRU and AVNCS external power panel light, no ATT HDG CAPT 3 supply switching.	Not done yet.	-
The ADR DISAGREE (16 kt, 3.6 degrees AOA in flight) and IR DISAGREE (5 degrees) thresholds and the RA DEGRADED threshold (15 ft or 10 %) are design choices.	The FCOM gives no value.	-
A full MSFS restart is needed after an update.	The systems WASM module changes.	Restart the sim after installing.

How to use

1. Open flyPad > Failures > ATA 34 Navigation and activate an ADR, IR or erroneous failure.
2. Follow the ECAM: AIR DATA / ATT HDG switching and the ADR or IR pb OFF on the overhead ADIRS panel.
3. Deactivate the failure and set the switching back to NORM.

How to test

#	Do this	Expected result
1	Cold and dark, BAT 1 and 2 ON, no external power; ADIRS IR 1, 2, 3 selectors to NAV.	The ON BAT light comes on and the external horn sounds on the ground (IRs on the batteries).
2	Cruise, AP ON. flyPad > Failures > ATA 34: **ADR 1 (air data lost, FAULT light)**.	ADR 1 FAULT light on the ADIRS panel; NAV ADR 1 FAULT with -AIR DATA SWTG ... CAPT 3, -ADR 1 P/B ... OFF.
3	Add **ADR 2 (air data lost, FAULT light)**; STS key.	NAV ADR 1+2 FAULT; STATUS INOP SYS ADR 1+2. Deactivate both.
4	ADR 3 pb OFF; then **ADR 1 airspeed erroneous (pitot)**.	Expected from the code: NAV ADR DISAGREE (ADR 1 reads 40 kt low against ADR 2). Deactivate, ADR 3 pb ON.

5	**IR 1 (inertial data lost, FAULT light)**.	IR 1 FAULT light; NAV IR 1 FAULT with -ATT HDG SWTG ... CAPT 3, -ATC/XPDR ... SYS 2; STATUS INOP SYS IR 1.
6	**RA 1** and **RA 2**.	NAV RA 1 AND 2 FAULT; STATUS INOP SYS RA 1+2.

What PR testers should check

- Cold and dark with the ADIRS in NAV: ON BAT light and the horn.
- Flights loaded on the runway and in cruise keep the IR alignment.
- ADR double and triple faults against the ELAC law.

29. alerts on existing data (CAB PR, THRUST LOCKED, fuel, WAI, doors, L/G, GPWS)

> **Aircraft:** A32NX | **Where:** E/WD, STATUS; overhead AIR COND, FUEL, ANTI ICE and GPWS panels; FCU A/THR pb; doors | **Status:** Not sim-tested (unit tests: 11 spec files, AirConditioningAlerts, AllEnginesFailure, AutoFlightStatus, DoorAlerts, FuelTransferAlerts, GpwsAlerts, LandingGearAlerts, ThrustLockedAlert, WingAntiIceAlerts, WingTankLowLevel and PaxOxygenMasks specs; 50 tests failing before the wiring).

What it is for

FCOM alerts whose data already existed in the sim but that the FWC did not show (A320 FCOM line numbers):

- **RAM AIR ON** green memo with the RAM AIR pb ON (DSC-21-10-50).
- **CAB PR LDG ELEV FAULT**: LDG ELEV selector at AUTO and no FMGS landing elevation, for example no destination in the flight plan; -LDG ELEV ... ADJUST. New STATUS lines of CAB PR EXCESS CAB ALT, SAFETY VALVE OPEN, SYS 1(2) FAULT and SYS 1+2 FAULT, with the MAN CAB PR CTL table.
- **ENG THRUST LOCKED**: the levers are not moved within 5 s after an involuntary A/THR disconnection or a disconnection by the FCU pb; the title flashes, the single chime and MASTER CAUT come again every 5 s; -THR LEVERS ... MOVE. Shown in flight phases 5, 6 and 7.
- **AUTO FLT:** STATUS of an involuntary A/THR OFF; FAC 1+2 FAULT now says RUD WITH CARE ABV 160 KT (was 140 KT).
- **FUEL:** the L+R WING TK LO LVL lines with LAND ASAP; **FUEL AUTO TRANSFER FAULT** and **FUEL CTR TK XFR OFF** (both CTR TK XFR pbs OFF in automatic mode).
- **WING A.ICE SYS FAULT** (WING ANTI ICE pb ON and one valve stays closed; with an engine bleed lost the -X BLEED ... OPEN variant) and **WING A.ICE L(R) HI PR** with THRUST LIM PENALTY.
- The passenger oxygen masks drop when the cabin altitude passes 14 000 ft.
- **DOOR** cautions for the four cabin doors, the two emergency exits and the two cargo doors; in flight .IF ABN CAB V/S: MAX FL ... 100/MEA-MORA, on the ground title only. Shown in phases 2, 3, 6 and 9.
- **L/G DOORS NOT CLOSED** (with MAX SPEED 250/.60 if the recycle fails) and **L/G SYS DISAGREE**, with the ATA 32 STATUS lines.
- **NAV GPWS FAULT** (-GPWS ... OFF, line kept after the pb is OFF) and **NAV GPWS TERR DET FAULT** (-GPWS TERR ... OFF).
- The STATUS of **ENG ALL ENGINES FAILURE** .

Limitations and why

Limitation	Why	What to do
No FMGC CAT 2/3 inop and AP OFF STATUS lines.	They need an FMGC model.	-

No WING A.ICE L(R) VALVE OPEN / OPEN ON GND.	No wing anti-ice valve failure in the sim.	-
No DOOR AVIONICS cautions.	The avionics doors are not in the sim.	-
The STATUS of ANTI ICE ENG 1 VALVE CLSD lists INOP SYS "WAI REGUL".	Known bug, the same: the WAI REGUL line and the ENG 1 A. ICE line have the same code.	See "Engine anti-ice valves, probe and window heat, anti-ice key events".

How to use

1. Nothing to set: the alerts come from the normal cockpit state.
2. Follow the ECAM and the STATUS page.

How to test

#	Do this	Expected result
1	At the gate, powered. Overhead RAM AIR pb ON.	Green RAM AIR ON memo on the E/WD. RAM AIR back OFF.
2	Engines running, no destination in the MCDU flight plan, pressurization LDG ELEV AUTO.	Amber CAB PR LDG ELEV FAULT with -LDG ELEV ... ADJUST. Enter a destination: the caution goes.
3	Engines running. Open the L FWD cabin door (flyPad or the MSFS door control).	Amber DOOR L FWD CABIN, title only (on the ground). Close the door.
4	Engines running, FUEL MODE SEL AUTO. CTR TK L XFR and CTR TK R XFR pbs OFF.	Amber FUEL CTR TK XFR OFF with -CTR TK L XFR ... ON, -CTR TK R XFR ... ON. Both pbs back ON.
5	Cruise, A/THR active, thrust levers in CL. Push the FCU A/THR pb; do not move the levers.	After 5 s amber ENG THRUST LOCKED, title flashing, with -THR LEVERS ... MOVE; the single chime and MASTER CAUT come again every 5 s. Move the levers: the caution goes.
6	Engines running, ATA 34: **EGPWC**.	Amber NAV GPWS FAULT with -GPWS ... OFF (and NAV GPWS TERR DET FAULT with -GPWS TERR ... OFF). Deactivate the failure.

What PR testers should check

- CAB PR LDG ELEV FAULT without a destination.
- FUEL AUTO TRANSFER FAULT with odd fuel loads at spawn.
- The DOOR cautions.
- ENG THRUST LOCKED flashing.
- The position of the RAM AIR ON memo.

30. PFD FMA messages, AVAIL, SD FUEL imbalance, SD BLEED crossbleed valve

> **Aircraft:** A32NX | **Where:** PFD FMA; E/WD N1 gauges; SD FUEL and BLEED pages | **Status:** Not sim-tested (unit tests: AvailIndication.spec.ts, FmaLine3Messages.spec.ts, CrossBleedValveIndication.spec.ts, FuelImbalanceIndication.spec.ts).

What it is for

- **PFD FMA line 3 messages** (A320 FCOM DSC-22_30-100): **DISCONNECT AP FOR LDG** (non-precision approach, AP still on at the minimum -50 ft, or 400 ft RA without a minimum), **FOR GA: SET TOGA** (approach vertical mode, below 1000 ft RA, all levers at or above MCT but below TOGA), **SET GREEN DOT SPEED** (engine out, selected

speed 10 kt or more away from green dot, not in ALT* / ALT) and **CHECK APPR SELECTION** (the FMS message that sets it is on `core/gps`, merged here).

- **AVAIL** on the E/WD N1 gauge after a start: steady for 10 s on the ground, pulsing for 60 s after a relight in flight; it goes when the lever leaves the idle detent.
- **SD FUEL:** the wing quantities pulse when the imbalance is above 1500 kg.
- **SD BLEED:** the crossbleed valve is amber in transit or when it disagrees with its selector.

Limitations and why

Limitation	Why	What to do
No MORE DRAG FMA message.	It needs VNAV outputs.	-
No WHEEL UP LOCK indication.	The gear model has no uplock state.	-
No bleed lines in ENG SHUT DOWN.	Not done yet.	-
DISCONNECT AP FOR LDG takes FINAL, FPA or V/S in the approach phase as a non-precision approach.	Design choice: the PFD does not receive the FMS approach type.	-

How to use

1. Nothing to set: read the PFD FMA, the E/WD and the SD pages as usual.

How to test

#	Do this	Expected result
1	On the ground, start engine 1.	At the end of the start AVAIL shows steady under N1 for 10 s, then goes.
2	Engines running or in cruise, with more than 1500 kg difference between the left and right wing tanks; SD FUEL page.	The quantities of the heavier wing pulse.
3	Engines running, SD BLEED page. flyPad > Failures > ATA 36: **X BLEED valve stuck closed**; X BLEED selector OPEN.	The crossbleed valve symbol is amber (disagree). Selector AUTO, deactivate the failure.
4	ILS approach, AP and A/THR ON, below 1000 ft RA. Thrust levers to MCT (not TOGA).	Amber FOR GA: SET TOGA on FMA line 3. Levers back to CL or TOGA.
5	Non-precision approach (FINAL APP), AP ON, BARO minimum set; continue below the minimum -50 ft.	Amber DISCONNECT AP FOR LDG on FMA line 3. Disconnect the AP.

What PR testers should check

- The width of the fuel imbalance underline.
- The AVAIL and fuel pulse animations.
- The colour of the MCDU CHECK APPR SELECTION message.

31. SD advisory mode

> **Aircraft:** A32NX | **Where:** SD (lower ECAM display); ECAM control panel | **Status:** Not sim-tested (unit tests: SdAdvisories.spec.ts, SdAdvisoryMode.spec.ts, FakeDmc.spec.ts).

What it is for

The ECAM advisory mode of the A320 (FCOM DSC-31-20, PRO-ABN-ABN-ADV): when a parameter drifts out of its normal range, its value pulses green and its system page is called on the SD by itself. A failure page has priority, and the crew can still select another page.

- Sources: APU EGT (above EGT MAX -33 C, not during the start), APU flap not closed 3 min after MASTER OFF, cabin V/S, cabin altitude and delta P (CAB PRESS page), IDG oil temperature (ELEC page), engine oil and vibrations (ENGINE page), fuel imbalance above 1500 kg in phases 2 and 6 (FUEL page).
- The values are the same that pulse on the SD pages, so the page called shows the pulsing value.

Limitations and why

Limitation	Why	What to do
With one ECAM display the page is called anyway; the ECP pb light does not pulse.	The single display flag is written by a display reconfiguration that is not on this branch; nothing writes it on this branch, so the advisory always behaves as with two displays.	-
No APU oil quantity, nacelle temperature, fuel temperature or CKPT OXY advisory.	No data for them in the sim, or the page does not pulse that value.	-

How to use

1. Nothing to set. When a page comes by itself, read the pulsing value; select another page with the ECP if needed.

How to test

#	Do this	Expected result
1	Cruise, AP ON, SD on the CRUISE page. Pressurization MODE SEL MAN; MAN V/S CTL UP in short taps until the cabin V/S is about +2000 ft/min (keep the cabin below 9550 ft).	The SD changes to the CAB PRESS page by itself; the V/S value pulses green.
2	MAN V/S CTL DN until the V/S is back to normal; MODE SEL back to AUTO.	The pulsing stops and the SD goes back to the flight phase page.

What PR testers should check

- The advisory page in normal operations: no page called without a reason.

32. CVR automatic power and CVR TEST

> **Aircraft:** A32NX | **Where:** Overhead RCDR panel (GND CTL and CVR TEST pbs) | **Status:** Not sim-tested (unit tests: CockpitVoiceRecorder.spec.ts, 11 tests).

What it is for

The cockpit voice recorder runs by itself as in the A320 FCOM (DSC-23-10-40): on the ground during the first 5 min after the electrical network is powered, on the ground with an engine running, in flight, and 5 min more after the last engine shutdown; on the ground the GND CTL pb runs it too. The CVR TEST pb, pressed and held, gives the test tone when the CVR runs and the parking brake is on.

Limitations and why

Limitation	Why	What to do

The CVR always stops 5 min after the last engine shutdown.	The CVR jack is not simulated.	GND CTL pb ON.
A full MSFS restart is needed after an update.	The sound file list (cvr_test) changes.	Restart the sim after installing.

How to use

1. Parking brake ON.
2. Press and hold the CVR TEST pb: the test tone is heard while the CVR runs.
3. After the first 5 min on the ground without an engine running, push RCDR GND CTL ON first.

How to test

#	Do this	Expected result
1	Cold and dark: BAT 1 and 2 ON, EXT PWR ON, parking brake ON. Within 5 min, press and hold CVR TEST.	The test tone is heard.
2	Wait more than 5 min (no engine, GND CTL OFF); press and hold CVR TEST.	No tone (the CVR stopped).
3	RCDR GND CTL pb ON; press and hold CVR TEST.	The test tone is heard again.

What PR testers should check

- CVR TEST without GND CTL in the first 5 min after power-up.

33. CPC landing elevation without FMGS data, EXCESS CAB ALT threshold

> **Aircraft:** A32NX | **Where:** SD CAB PRESS page; pressurization LDG ELEV selector; E/WD | **Status:** Not sim-tested (Rust tests of cabin_pressure_controller.rs).

What it is for

- With the LDG ELEV selector at AUTO and no FMGS landing elevation (no destination), the cabin pressure controller has no landing elevation: the SD CAB PRESS page shows XX for it (A320 FCOM DSC-21-20-30, PRO-ABN-CAB_PR). It goes with CAB PR LDG ELEV FAULT (see "Alerts on existing data").
- **CAB PR EXCESS CAB ALT** in cruise triggers above a cabin altitude of 9550 ft only, also with a high destination airport; in climb and descent above the higher of 9550 ft and 1000 ft above the airfield.

Limitations and why

Limitation	Why	What to do
Without FMGS data the controller keeps a landing elevation of 0 ft until the crew sets LDG ELEV.	The FCOM gives no value for this case.	Turn the LDG ELEV selector (procedure LDG ELEV ... ADJUST).
A full MSFS restart is needed after an update.	The systems WASM module changes.	Restart the sim after installing.

How to use

1. Enter a destination in the flight plan, or set the LDG ELEV selector by hand.

How to test

#	Do this	Expected result
1	Engines running, no destination in the flight plan, LDG ELEV AUTO, SD CAB PRESS page.	LDG ELEV shows XX; CAB PR LDG ELEV FAULT on the E/WD.
2	Turn the LDG ELEV selector out of AUTO.	The caution goes. Back to AUTO and enter a destination: the FMGS landing elevation replaces XX.

What PR testers should check

- No CAB PR EXCESS CAB ALT in cruise below 9550 ft cabin altitude with a high destination airport.

34. brake and steering control units (BSCU, ABCU), BRAKES alerts

> **Aircraft:** A32NX | **Where:** flyPad > Failures > ATA 32 Landing Gear; A/SKID & N/W STRG switch; autobrake pbs; E/WD, STATUS | **Status:** Not sim-tested (Rust: 15 test bed tests; unit tests: BrakeSystemAlerts.spec.ts 18, the failure list specs).

What it is for

- A new model of the BSCU with two channels and of the ABCU (A320 FCOM): channel 1 on AC BUS 1 and DC BUS 1, channel 2 on AC BUS 2 and DC BUS 2, the ABCU on DC ESS. The active channel changes at each gear DOWN selection and when it fails. Both channels lost: no antiskid, alternate braking limited to 1000 PSI, no nose wheel steering. ABCU lost too: parking brake only.
- The autobrake arms only with green pressure, the antiskid powered, no braking failure and one ADIRU. MAX activates only above 40 kt wheel speed and keeps braking below it.
- New cautions with their STATUS: **BRAKES A/SKID N/WS FAULT** (MAX BRK PR ... 1000PSI), **BRAKES ALTN BRK FAULT**, **BRAKES AUTO BRK FAULT**, **BRAKES NORM + ALTN FAULT** (PARK BRK ONLY), **BRAKES NORM BRK FAULT**, **BRAKES SYS 1(2) FAULT** and **WHEEL N/W STRG FAULT**.
- Six new failures, ATA 32. The brake and steering failure types come from a prerequisite commit of shared code (the A380X brake and steering failures that use them are on `core/failures-a380x`).

Limitations and why

Limitation	Why	What to do
The flight phase inhibitions are design choices.	The FCOM figures are missing from the FCOM text.	-
A unit lost only because its bus is unpowered gives no BRAKES caution.	Design choice: the ELEC bus alert already shows it with the same INOP SYS.	-
A full MSFS restart is needed after an update.	The systems WASM module changes.	Restart the sim after installing.

How to use

1. Open flyPad > Failures > ATA 32 Landing Gear and activate a BSCU, ABCU, braking, autobrake or steering failure.
2. Follow the ECAM and the STATUS; brake with care.

How to test

#	Do this	Expected result

1	On the ground, engines running, parking brake OFF. flyPad > Failures > ATA 32: **BSCU channel 1**.	BRAKES SYS 1 FAULT; braking and steering still work on channel 2.
2	Add **BSCU channel 2**; taxi slowly.	BRAKES A/SKID N/WS FAULT with MAX BRK PR ... 1000PSI; no nose wheel steering; the brakes give about 1000 PSI at most.
3	Add **ABCU (alternate braking)**; stop with the parking brake.	BRAKES NORM + ALTN FAULT with PARK BRK ONLY; STATUS PARK BRK ONLY. Deactivate the three failures.
4	**Nose wheel steering**.	WHEEL N/W STRG FAULT; no steering with the tiller.
5	In flight, autobrake LO armed. **Autobrake**.	BRAKES AUTO BRK FAULT.
6	Take-off run, autobrake MAX armed; reject above 40 kt, then repeat below 40 kt.	Above 40 kt the autobrake brakes to a stop; below 40 kt it does not activate.

What PR testers should check

- MAX rejected take-off above and below 40 kt.
- Taxi with BSCU channels 1 and 2 failed: no steering, about 1000 PSI.
- ABCU failure: parking brake only.
- AUTO BRK FAULT after arming LO in flight and failing the autobrake.

35. APU automatic shutdowns, cargo and avionics smoke, cargo extinguishing

> **Aircraft:** A32NX | **Where:** flyPad > Failures > ATA 26 Fire Protection and ATA 49 Airborne Auxiliary Power; overhead APU, CARGO SMOKE, VENTILATION and EMER ELEC panels; E/WD, STATUS | **Status:** Not sim-tested (Rust tests of the ECB, smoke detection, cargo extinguishing and the A320 fire tests; unit tests: SmokeAlerts.spec.ts, the failure list specs).

What it is for

- **APU automatic shutdowns** by the ECB (A320 FCOM DSC-49-20 FAULT light): ECB failure, overspeed, low oil pressure, high oil temperature, and the loss of its DC supply above N 70 % (confirmed 1 s). The MASTER SW FAULT light comes on and APU AUTO SHUT DOWN shows (see "HYD and APU shutdown alerts"). Four new failures, ATA 49.
- **Cargo smoke detection** (FWD and AFT, two loops each on the SDCU channels 1 and 2): **SMOKE FWD (AFT) CARGO SMOKE** with -CAB FANS ... OFF, .IF FWD (AFT) CARG CLOSED: -AGENT ... DISCH, and the ground lines; the red SMOKE light of the CARGO SMOKE panel. One loop failed is tolerated; both loops failed: **SMOKE FWD (AFT) CRG DET FAULT**, STATUS INOP SYS FWD (AFT) CRG DET.
- **Cargo extinguishing** : one bottle with two discharge heads; the cargo DISCH pbs now work and their DISCH light comes on when the bottle is empty. A bottle at low pressure gives **SMOKE FWD CRG BTL 1 FAULT** and **SMOKE AFT CRG BTL 1 FAULT**.
- **Avionics smoke** (ventilation extract duct detector): **AVIONICS SMOKE** after 5 s with -SMOKE/FUMES PROC ... APPLY; the SMOKE light of the GEN 1 LINE pb and the FAULT lights of the BLOWER and EXTRACT pbs come on.
- Nine new failures, ATA 26. The CARGO SMOKE TEST pb still shows the FWD CARGO SMOKE warning, as before.

Limitations and why

Limitation	Why	What to do
Other APU shutdown causes (inlet flap, slow start, EGT overtemperature, sensors...) do not exist.	No model behind them.	-

The -FWD (AFT) ISOL VALVE ... OFF line never shows.	The A320 here has no cargo ventilation.	-
The cargo bottle squibs cannot fail; no bottle diverter or control alerts.	Not modelled.	-
A full MSFS restart is needed after an update.	The systems WASM module and the cockpit XML (lights) change.	Restart the sim after installing.

How to use

1. Open flyPad > Failures > ATA 26 Fire Protection (smoke, detection loops, bottle) or ATA 49 Airborne Auxiliary Power (APU).
2. Follow the ECAM: for cargo smoke, CAB FANS OFF and the cargo DISCH pb of the compartment (AGENT DISCH).

How to test

#	Do this	Expected result
1	On the ground, APU running (AVAIL). flyPad > Failures > ATA 49: **APU overspeed (automatic shutdown)**.	The APU shuts down; APU MASTER SW FAULT light; APU AUTO SHUT DOWN with -MASTER SW ... OFF. MASTER SW OFF, deactivate the failure.
2	Repeat with **APU low oil pressure (automatic shutdown)**, **APU high oil temperature (automatic shutdown)** and **APU electronic control box (ECB)**.	The same automatic shutdown with the FAULT light.
3	Cruise. ATA 26: **Aft cargo smoke**.	Red SMOKE AFT CARGO SMOKE with -CAB FANS ... OFF, .IF AFT CARG CLOSED: -AGENT ... DISCH; red SMOKE light on the CARGO SMOKE panel.
4	Push the AFT DISCH pb.	After the discharge the DISCH lights come on and the AGENT line goes. Deactivate the failure.
5	ATA 26: **Forward cargo smoke detection loop 1**, then **Forward cargo smoke detection loop 2**.	Nothing with one loop; with both, amber SMOKE FWD CRG DET FAULT; STATUS INOP SYS FWD CRG DET.
6	ATA 26: **Cargo fire bottle low pressure**.	SMOKE FWD CRG BTL 1 FAULT and SMOKE AFT CRG BTL 1 FAULT.
7	ATA 26: **Avionics smoke (ventilation extract duct)**; wait 5 s.	AVIONICS SMOKE with -SMOKE/FUMES PROC ... APPLY; GEN 1 LINE SMOKE light, BLOWER and EXTRACT FAULT lights. Deactivate the failure.

What PR testers should check

- On batteries only, the APU shuts down when its DC supply goes (but not with the APU GEN or EXT PWR on).
- No APU shutdown on electrical power transfers.
- The DISCH, AGENT and BTL light emissives.
- The smoke failures from the flyPad.

36. A380X smoke detection function (systems part)

> **Aircraft:** A380X | **Where:** A380X overhead CARGO SMOKE panel; flyPad > Failures > ATA 26 (with core/a380x-elec-fuel-alerts) | **Status:** Not sim-tested (6 Rust tests in fire_and_smoke_protection.rs).

What it is for

The A380X Smoke Detection Function of the CIDS for the cargo compartments (FWD, AFT, bulk), the avionics bays and the IFE bay, each with two smoke detectors (A380 FCOM DSC-26-50, DSC-26-60). On the CARGO SMOKE panel the AGENT TO FWD / AFT pbs show SMOKE and DISCH, and the BTL 1 and BTL 2 lights come on when a bottle has discharged: bottle 1 first (about one minute), then bottle 2 slowly. It lives on this branch because it extends the shared fire model of both aircraft, which is only here.

> **CAUTION**
> The A380X SMOKE, SMOKE DET FAULT and CARGO BTL FAULT alerts, the A380X flyPad failure list and the simvar docs are on `core/a380x-elec-fuel-alerts`. **Merge both branches** for the A380X smoke alerts; this branch alone only has the systems side and the panel lights.

Limitations and why

Limitation	Why	What to do
The crew rests, lavatories, CWS / RCC and showers have no smoke detection; the SDF cannot fail as a computer.	Not modelled. The NSS has its own zone since 2026-10-18 (see the next feature).	-
Detector 1 is on the DC ESS bus and detector 2 on the DC 2 bus.	Design choice: the FCOM gives no supply.	-
A full MSFS restart is needed after an update.	The A380X systems WASM module and the cockpit XML change.	Restart the sim after installing.

How to use

1. With `core/a380x-elec-fuel-alerts` merged: open the A380X flyPad > Failures > ATA 26 and activate a smoke or detector failure.
2. Follow the ECAM; push AGENT TO FWD or AFT on the CARGO SMOKE panel for a cargo smoke.

How to test

#	Do this	Expected result
1	A380X in cruise, both branches merged. flyPad > Failures > ATA 26: **Aft cargo smoke**.	The SMOKE light of the AGENT TO AFT pb comes on (and the SMOKE alert of core/a380x-elec-fuel-alerts).
2	Push AGENT TO AFT.	BTL 1 light after about one minute, then BTL 2 later. Deactivate the failure.
3	**Forward cargo smoke detector 1**.	The SMOKE DET FAULT alert of the forward compartment (alerts on core/a380x-elec-fuel-alerts).

What PR testers should check

- The A380X smoke failures from the flyPad and the CARGO SMOKE panel lights.

37. A380X NSS smoke zone and the SMOKE light of the NSS MASTER sw

> **Aircraft:** A380X | **Where:** NSS MASTER sw (overhead NSS panel); flyPad > Failures > ATA 26 (with core/a380x-elec-fuel-alerts) | **Status:** Not sim-tested on the core branch. Develop (83f832d77): Rust tests only (2 new tests in fire_and_smoke_protection.rs), no sim test recorded.

What it is for

The A380X smoke detection function gets a tenth zone, the Network Server System (NSS), with two smoke detectors and the SMOKE logic of the avionics bays (both detectors, or one failed and the other one detecting). NSS smoke has no ECAM alert: it lights the SMOKE light of the NSS MASTER sw, which stays on as long as smoke is detected, also with the NSS power cut off (A380 FCOM DSC-26-50-10, DSC-46-20-20-87). The failure mapping (26049 to 26051) is in the A380X systems module of this branch.

> **CAUTION**
> The flyPad failures that trigger it (**Network server system (NSS) smoke**, **... smoke detector 1**, **... smoke detector 2**) are on `core/a380x-elec-fuel-alerts`. **Merge both branches**: this branch alone has the zone and the light but no way to fail it from the flyPad.

Limitations and why

Limitation	Why	What to do
A failed NSS smoke detector shows nothing.	The FCOM has no NSS SMOKE DET FAULT alert; the fault is only kept in L:A32NX_SMOKE_NSS_DET_FAULT.	-
A full MSFS restart is needed after an update.	The A380X systems WASM module and the cockpit XML change.	Restart the sim after installing.

How to use

1. With `core/a380x-elec-fuel-alerts` merged: flyPad > Failures > ATA 26, activate **Network server system (NSS) smoke**.
2. Watch the SMOKE light of the NSS MASTER sw.

How to test

#	Do this	Expected result
1	A380X at the gate, EXT PWR on, both branches merged. flyPad > Failures > ATA 26: **Network server system (NSS) smoke**.	The SMOKE light of the NSS MASTER sw comes on; no ECAM alert.
2	NSS MASTER sw OFF.	The SMOKE light stays on.
3	Deactivate the failure, NSS MASTER sw back on.	The SMOKE light goes off.

38. Relight EGT watch in the shared engine start (no change on the A32NX)

> **Aircraft:** A32NX (shared code with the A380X) | **Where:** None on the A32NX | **Status:** Not sim-tested on the core branch. Develop (30af1a232): Rust tests only, no sim test recorded.

What it is for

The shared engine start sequence of the FADEC can now watch the EGT of an in-flight relight against a relight EGT limit of the aircraft schedule and report START FAULT EGT OVERLIMIT until the ENG MASTER is set OFF (A380 FCOM ENG START FAULT). The A32NX has no published relight EGT limit: its schedule sets none, so nothing changes on the A32NX. The A380X schedule (865 °C) and its use are on `core/failures-a380x`, which carries the same shared engine_start.rs change.

Limitations and why

Limitation	Why	What to do

No relight EGT watch on the A32NX.	No relight EGT limit is published for the A320.	-

How to use

1. Nothing to set.

How to test

#	Do this	Expected result
1	A32NX in flight: shut down engine 1 and relight it as in "Engine flameout and seizure failures, in-flight relight".	The relight runs as before; no new ENG 1 START FAULT EGT OVERLIMIT.

39. BLEED page HP valve and bleed valve disagree

> **Aircraft:** A32NX | **Where:** SD BLEED page; flyPad > Failures > ATA 36 Pneumatic | **Status:** Not sim-tested (Rust ata_36_failures tests, 2; unit tests: EngineBleedValveIndication.spec.ts).

What it is for

On the SD BLEED page (A320 FCOM): the HP valve is amber when it is closed with an HP valve fault, and the engine bleed valve is amber when it has been out of its commanded position for 5 s. This extends "Pneumatic failures with BMC leak and bleed monitoring". The A380X bleed monitoring and leak alerts are on `core/a380x-elec-fuel-alerts`.

Limitations and why

Limitation	Why	What to do
A full MSFS restart is needed after an update.	The systems WASM module changes.	Restart the sim after installing.

How to use

1. Open flyPad > Failures > ATA 36 Pneumatic, activate a valve failure and watch the SD BLEED page.

How to test

#	Do this	Expected result
1	Engines running at idle, SD BLEED page. flyPad > Failures > ATA 36: **Engine 1 HP valve stuck closed**.	ENG 1 HP VALVE FAULT; the engine 1 HP valve symbol is amber, closed.
2	Deactivate it. **Engine 1 bleed valve stuck closed** with ENG 1 BLEED pb ON; wait 5 s.	The engine 1 bleed valve symbol turns amber (closed while commanded open). Deactivate the failure.

What PR testers should check

- No false bleed valve disagree on a fast thrust increase at take-off.
- Leak failures from the flyPad (a pylon leak needs high power to be detected).

40. HP fuel valve, FADEC fault failures, pre-start motoring

> **Aircraft:** A32NX | **Where:** flyPad > Failures > ATA 73 Engine - Fuel And Control and ATA 78 Exhaust; ENG panel (FAULT lights); E/WD | **Status:** Not sim-tested (Rust tests of the systems and a320 crates; unit tests: EngineMonitoringAlerts.spec.ts, the failure list specs).

What it is for

- **HP fuel valve stuck** (A320 FCOM PRO-ABN-ENG): stuck closed on a stopped engine there is no fuel and no light-up: **ENG 1(2) HP FUEL VALVE** with HP FUEL VALVE NOT OPEN, -MAN START ... OFF, -ENG MASTER ... OFF; stuck open on a running engine: HP FUEL POS SWT FAULT at the shutdown. The valve position must disagree with its command for 2 s.
- **FADEC fault failures** with crew awareness alerts: **ENG 1(2) FADEC ALTERNATOR**, **OVSPD PROT FAULT**, **PROBES FAULT**, **SENSOR FAULT** (on the ground -THR LEVER ... IDLE, -ENG MASTER ... OFF; in flight AVOID RAPID THR CHANGES, also on the STATUS), **ONE TLA FAULT** and **REV SWITCH FAULT** (the reverser still works). 14 new failures, ATA 73 and 78.
- The **ENG FAULT light** also comes on for a thrust control malfunction (both FADEC channels or both lever resolvers lost) and an abnormal HP valve position.
- **Pre-start motoring:** an automatic ground start of a hot engine dry cranks it for up to 60 s first, by its thermal state (the schedule is a design choice, the FCOM gives none). See also "Engine ignition and start failures, FCOM start sequence".

Limitations and why

Limitation	Why	What to do
Most FADEC faults have no effect on the engine model.	They are crew awareness alerts; the other ENG alerts need engine models.	-
No approach idle.	No FCOM value.	-
A full MSFS restart is needed after an update.	The systems WASM module changes.	Restart the sim after installing.

How to use

1. Open flyPad > Failures > ATA 73 Engine - Fuel And Control (or ATA 78 Exhaust for the reverser switch) and activate a failure.
2. Follow the ECAM. Most of these alerts are inhibited from take-off power to 1500 ft and in approach, so test them on the ground or in cruise.

How to test

#	Do this	Expected result
1	On the ground, APU BLEED ON, engine 1 shut down. flyPad > Failures > ATA 73: **Engine 1 HP fuel valve stuck**; ENG MODE IGN/START, ENG MASTER 1 ON.	No light-up (no fuel flow); amber ENG 1 HP FUEL VALVE with HP FUEL VALVE NOT OPEN and -ENG MASTER 1 ... OFF; ENG 1 FAULT light. MASTER OFF, deactivate.
2	Engines running on the ground. **Engine 1 FADEC alternator**.	Amber ENG 1 FADEC ALTERNATOR, title only. Deactivate.
3	**Engine 1 core sensors (PS3)**.	ENG 1 SENSOR FAULT with -THR LEVER 1 ... IDLE, -ENG MASTER 1 ... OFF (on the ground). Deactivate.
4	**Thrust lever 1 one resolver**, then ATA 78: **Reverser 1 permission switch**.	ENG 1 ONE TLA FAULT; then ENG 1 REV SWITCH FAULT. Deactivate both.
5	Shut engine 1 down after a long run, then start it again at once with the automatic start.	Dry cranking (N2 steady, no fuel) for up to 60 s before the ignition and fuel.

What PR testers should check

- Hot restart motoring.

41. gravity feed ceiling, LP valve and FQI failures

> **Aircraft:** A32NX | **Where:** flyPad > Failures > ATA 28 Fuel; overhead FUEL panel; E/WD | **Status:** Not sim-tested (Rust: gravity_feed 16 and engine_lp_valve 5 tests; unit tests: FuelLpValveFqiAlerts.spec.ts, FuelPumpsAndValves.spec.ts).

What it is for

- **Gravity feed ceiling** (A320 FCOM GRAVITY FUEL FEEDING): FL 360 after more than 30 min above FL 300, FL 260 otherwise, and the higher of FL 150 and take-off elevation + 7000 ft if FL 300 was never passed. An engine fed by gravity (both pumps of its wing tank off or failed, and no pump pushing through the X FEED) above the ceiling is starved after 3 min (design choice) and flames out; it stays out until ENG MASTER OFF below the ceiling, then MASTER ON relights it inside the relight envelope. This extends the gravity feed of "FCOM fixes: FADEC relight, automatic ignition, start faults, gravity feed, IR alignment".
- **LP fuel valve jammed** failures: jammed open the engine still shuts down with its ENG MASTER, and **FUEL ENG 1(2) LP VALVE OPEN** shows; jammed closed the engine is starved.
- **FQI channel** failures: **FUEL FQI CH 1(2) FAULT** .

Limitations and why

Limitation	Why	What to do
The 3 min before the starvation and the pressure altitude as flight level are design choices.	The FCOM gives neither a time nor a partial thrust loss.	Descend below the ceiling within 3 min.
Restoring the pumps or descending alone does not relight a starved engine.	As the FCOM relight procedure: ENG MASTER OFF then ON.	-
No fuel alerts that need temperature, IDG, overflow or inerting models.	Not modelled.	-
A full MSFS restart is needed after an update.	The systems WASM module changes.	Restart the sim after installing.

How to use

1. With all wing tank pumps of one side lost, follow the ECAM and descend below the gravity feed ceiling.
2. Open flyPad > Failures > ATA 28 Fuel for the LP valve and FQI failures.

How to test

#	Do this	Expected result
1	On the ground, engines running. flyPad > Failures > ATA 28: **ENG 1 LP fuel valve jammed in place**; ENG MASTER 1 OFF.	Engine 1 shuts down; amber FUEL ENG 1 LP VALVE OPEN. Deactivate.
2	**FQI channel 1**.	Amber FUEL FQI CH 1 FAULT. Deactivate.
3	Cruise at FL 330 less than 30 min after passing FL 300, X FEED closed. **L TK pump 1** and **L TK pump 2**; wait 3 min.	Engine 1 is fed by gravity above the FL 260 ceiling and flames out after 3 min (ENG 1 FAIL).
4	Descend below FL 260; ENG MASTER 1 OFF then ON inside the relight envelope; deactivate the pump failures.	Engine 1 relights.

What PR testers should check

- ELEC EMER CONFIG in cruise: pumps off, flameout after 3 min above the ceiling unless descending.

- No false FUEL alert.

42. engine anti-ice valves, probe and window heat, anti-ice key events

> **Aircraft:** A32NX (A380X anti-ice systems and cockpit part too) | **Where:** flyPad > Failures > ATA 30 Ice And Rain Protection; overhead ANTI ICE panel; E/WD, STATUS | **Status:** Not sim-tested (Rust tests engine_anti_ice, probe_window_heat, A380 ice_rain_protection and oxygen; unit tests: EngineAntiIceAlerts, ProbeWindowHeatAlerts, AntiIceCautions, FadecWingAntiIce specs, the failure list specs).

What it is for

- **Engine anti-ice valve** (shared model, A320 FCOM DSC-30-30): it opens with the pb ON (or without power) once the engine supplies HP air above 10 psig, in 2 s, with the FAULT light in transit; the MSFS engine anti-ice follows the valve. New cautions **ANTI ICE ENG 1(2) VALVE CLSD** (AVOID ICING CONDITIONS) and **ANTI ICE ENG 1(2) VALVE OPEN** (THRUST LIM PENALTY) with STATUS. VALVE CLSD needs the engine running.
- **Probe and window heat computers** and all 14 probe and window heat cautions with their STATUS (DSC-30-40-10, DSC-30-50-10): ANTI ICE ALL PITOT, CAPT + F/O PITOT, CAPT + STBY PITOT, F/O + STBY PITOT, CAPT (F/O) (STBY) PROBES, the single PITOT, L STAT, R STAT and AOA cautions, CAPT (F/O) TAT, L + R WINDSHIELD, L (R) WINDSHIELD and L (R) WINDOW.
- **MSFS anti-ice key events** (ANTI_ICE_*_ENGn, ANTI_ICE_ON / OFF / SET / TOGGLE, TOGGLE_STRUCTURAL_DEICE) now move the cockpit pbs on both aircraft, so key bindings and hardware panels keep working; the sim variables follow the valves.
- About 25 new A32NX failures, ATA 30. Because the key event mapping serves both aircraft, the A380X anti-ice systems and cockpit side (ENG and WING ANTI ICE pb behaviour, oxygen, FADEC wing anti-ice data) is on this branch too; the A380X ECAM alerts, flyPad failures and SD oxygen page are on `core/a380x-elec-fuel-alerts`.

Limitations and why

Limitation	Why	What to do
ANTI ICE ENG 1 VALVE CLSD lists INOP SYS "WAI REGUL" instead of "ENG 1 A. ICE".	Known bug, the same: two STATUS entries have the same code (300300002) and the later one, WAI REGUL, wins. This branch keeps that order so its specs agree; one of them needs a new code.	-
No ADR probe heat bits; no wing anti-ice indication on the SD BLEED page.	Not done yet.	-
The key event mapping tests cannot run on the host.	They need the WASM target; they compile only.	-
A full MSFS restart is needed after an update.	The systems WASM modules and the cockpit XML change.	Restart the sim after installing.

How to use

1. Use the overhead ANTI ICE pbs, or the MSFS anti-ice key bindings: both move the pbs.
2. Open flyPad > Failures > ATA 30 Ice And Rain Protection to fail a valve, a heat computer or a heater.

How to test

#	Do this	Expected result

1	On the ground, engines running. flyPad > Failures > ATA 30: **Engine 1 anti-ice valve stuck closed**; ANTI ICE ENG 1 pb ON; STS key.	FAULT light on the pb; amber ANTI ICE ENG 1 VALVE CLSD with AVOID ICING CONDITIONS. STATUS INOP SYS shows "WAI REGUL" (known bug, see above). Pb OFF, deactivate.
2	**Engine 2 anti-ice valve stuck open**; ANTI ICE ENG 2 pb stays OFF.	ANTI ICE ENG 2 VALVE OPEN with THRUST LIM PENALTY; STATUS THRUST LIM PENALTY. Deactivate.
3	**CAPT pitot heating**.	ANTI ICE CAPT PITOT with -AIR DATA SWTG ... CAPT 3; STATUS INOP SYS CAPT PITOT. Add **F/O pitot heating**: one ANTI ICE CAPT + F/O PITOT with its procedure.
4	**CAPT probe heat computer**.	ANTI ICE CAPT PROBES with -AIR DATA SWTG ... CAPT 3.
5	**Left windshield heating**, then **Right windshield heating**.	ANTI ICE L WINDSHIELD, then one ANTI ICE L + R WINDSHIELD; STATUS INOP SYS WSHLD HEAT. Deactivate all.
6	Bind a key to the MSFS engine 1 anti-ice toggle and press it.	The ANTI ICE ENG 1 pb moves; no loop and no repeated toggling.

What PR testers should check

- Key binding and hardware panel toggles of ENG and WING anti-ice (no loop, no toggle storm at load).
- A32NX ENG A.ICE FAULT light before about 25 % N2.
- A380X wing anti-ice pb test and take-off inhibition (with core/a380x-elec-fuel-alerts).

43. SFCC flap and slat channel failures

> **Aircraft:** A32NX | **Where:** flyPad > Failures > ATA 27 Flight Controls; E/WD, STATUS, SD F/CTL | **Status:** Not sim-tested (Rust SFCC channel tests; unit tests: SlatFlapChannelAlerts.spec.ts, the failure list specs).

What it is for

Four new failures, ATA 27: the flap and slat channels of SFCC 1 and 2. A failed channel stops its motor, so the surfaces move at half speed (A320 FCOM), and sets its fault bit. New crew awareness cautions **F/CTL FLAP SYS 1(2) FAULT** and **F/CTL SLAT SYS 1(2) FAULT**. STATUS: FLAP SYS 1 FAULT gives APPR PROC: GPWS FLAP MODE ... OFF; FLAP SYS 1(2) FAULT ENG 1(2) APPR IDLE ONLY and FLAPS SLOW; SLAT SYS FAULT SLATS SLOW. The A380X part is on `core/a380x-elec-fuel-alerts`.

Limitations and why

Limitation	Why	What to do
No F/CTL FLAPS (SLATS) FAULT / LOCKED, tip brake or flap attach sensor alerts.	The FCOM text gives no trigger for the first; the model has no tip brakes nor attach sensors.	-
An unpowered SFCC gives no channel caution.	The ELEC bus alerts cover it.	-
A full MSFS restart is needed after an update.	The systems WASM module changes.	Restart the sim after installing.

How to use

1. Open flyPad > Failures > ATA 27 Flight Controls and activate an SFCC channel failure.
2. Expect slow flaps or slats; read the STATUS before the approach.

How to test

#	Do this	Expected result
1	Cruise or on the ground with engines running. flyPad > Failures > ATA 27: **SFCC 1 flap channel**; STS key.	Amber F/CTL FLAP SYS 1 FAULT; STATUS APPR PROC: GPWS FLAP MODE ... OFF, ENG 1 APPR IDLE ONLY, FLAPS SLOW.
2	Select flaps 1 then 2 (below their speeds).	The flaps move at about half the normal speed.
3	**SFCC 2 slat channel**.	F/CTL SLAT SYS 2 FAULT; STATUS SLATS SLOW. Deactivate both.

What PR testers should check

- Half-speed flaps and slats with a channel failure.

44. GPWS self-test, TCAS computer failure, reactive windshear

> **Aircraft:** A32NX | **Where:** Glareshield GPWS pb; PFD; ND; flyPad > Failures > ATA 34 Navigation | **Status:** Not sim-tested (Rust egpws tests, 3 new; unit tests: TcasFault.spec.ts 5, ReactiveWindshearDetection.spec.ts 13).

What it is for

- **GPWS self-test** (A320 FCOM, 48990): on the ground, pressing the GPWS pb runs the test: GLIDESLOPE with the amber light, PULL UP with the red light, then the terrain test pattern with TERRAIN TERRAIN PULL UP. Not with GPWS SYS OFF; it stops at lift-off. The sequence is a design choice (Honeywell EGPWS short test).
- **TCAS computer** failure (ATA 34): out of standby, amber TCAS on the PFD and the ND, NAV TCAS FAULT, no traffic and no RA.
- **Reactive windshear** (FAC 1 and 2): red WINDSHEAR on both PFDs for at least 15 s. The shared detector and PFD warning serve both aircraft; the A380X PRIM part is on `core/a380x-elec-fuel-alerts`. The WINDSHEAR voice is in "Voices: sidestick priority, intermediate RA callouts, WINDSHEAR".

Limitations and why

Limitation	Why	What to do
The windshear threshold (F-factor 0.13, filtered) is a design choice.	The FCOM gives no formula.	-
No REAC W/S DET FAULT alert, no SRS escape guidance, no predictive windshear, no FLS modes.	They need models, terrain or wind data.	-
A full MSFS restart is needed after an update.	The systems WASM module (EGPWS) changes.	Restart the sim after installing.

How to use

1. GPWS test: on the ground, GPWS SYS ON, press the GPWS pb on the glareshield.
2. TCAS: flyPad > Failures > ATA 34 Navigation, **TCAS computer**.
3. Windshear: nothing to set; follow the FCOM windshear procedure if WINDSHEAR shows.

How to test

#	Do this	Expected result
1	On the ground, powered, GPWS SYS ON. Press the GPWS pb on the glareshield.	GLIDESLOPE with the amber light, PULL UP with the red light, then TERRAIN TERRAIN PULL UP (the terrain test pattern step).
2	GPWS SYS pb OFF; press the GPWS pb again.	No test.
3	TCAS mode AUTO (not STBY). flyPad > Failures > ATA 34: **TCAS computer**.	Amber TCAS on the PFD and the ND; NAV TCAS FAULT; no traffic. Deactivate.

What PR testers should check

- No false WINDSHEAR in gusts and turns.
- The GPWS test lights and aurals.

45. NAV FM/GPS POS DISAGREE and NAV LS TUNING DISAGREE

> **Aircraft:** A32NX | **Where:** E/WD, STATUS | **Status:** Not sim-tested (unit tests: NavPositionAlerts.spec.ts).

What it is for

- **NAV FM/GPS POS DISAGREE** (A320 FCOM): the FM and GPS positions disagree; amber with MASTER CAUT and single chime, -A/C POS ... CHECK. The FMS computes the condition: the FM position model is on `core/gps`, merged here. Inhibited for the first 15 s of phase 5.
- **NAV LS TUNING DISAGREE** : MMR 1 and MMR 2 tuned differently for 3 s (design choice); crew awareness, STATUS INOP SYS CAT 2; shown in phases 6 and 7 only.

Limitations and why

Limitation	Why	What to do
LS TUNING DISAGREE has no AP / FD effect (APPR arming).	That belongs to the flight guidance.	-

How to use

1. Nothing to set. On FM/GPS POS DISAGREE check the aircraft position (MCDU POSITION MONITOR, ND).

How to test

#	Do this	Expected result
1	Fly a normal flight with an ILS approach tuned by the FMS.	No NAV LS TUNING DISAGREE and no NAV FM/GPS POS DISAGREE.
2	There is no flyPad failure that makes the FM and GPS positions disagree on purpose.	The caution comes only from the FM position model of core/gps (L:A32NX_FM_GPS_POS_DISAGREE): check that it never shows in normal flights.

What PR testers should check

- No nuisance LS TUNING DISAGREE while the FMS tunes the two receivers.

46. Voices: sidestick priority, intermediate RA callouts, WINDSHEAR

> **Aircraft:** A32NX | **Where:** Sidestick take over pbs; FWC voices | **Status:** Not sim-tested (unit tests: SidestickPriorityAurals, IntermediateHeightCallout, FwsVoiceCallouts specs).

What it is for

- **DUAL INPUT** every 5 s while both sidesticks are deflected, and **PRIORITY LEFT / PRIORITY RIGHT** once the other sidestick has been deactivated by the take over pb for 1 s (A320 FCOM). The 1 s is a design choice so that a short press to disconnect the AP does not call it.
- **Intermediate radio height callout:** below 410 ft, when no height callout came for 11 s (above 50 ft) or 4 s (below 50 ft), the present height is said, rounded down to a callout word, then every 4 s.
- **WINDSHEAR** x3 for the reactive windshear warning.
- The new clips are synthetic voices made with a public-domain voice (tools/fws-voice).

Limitations and why

Limitation	Why	What to do
DUAL INPUT cannot sound.	The F/O sidestick sends no input to the flight controls in the sim; the logic is ready.	-
The voices wait while a GPWS or TCAS aural plays.	As the radio height callouts do (design choice).	-
A full MSFS restart is needed after an update.	New sound files and sound.xml.	Restart the sim after installing.

How to use

1. Bind a key or button to the captain sidestick take over (AP disconnect) pb; press and hold it.
2. Listen to the callouts in a slow final approach.

How to test

#	Do this	Expected result
1	In flight, AP OFF. Press and hold the captain take over pb for 3 s.	PRIORITY LEFT once, after about 1 s.
2	Short press of the take over pb.	No PRIORITY voice.
3	Final approach: level off or descend very slowly between 300 and 200 ft RA for more than 15 s.	About 11 s after the last callout: TWO HUNDRED (or THREE HUNDRED), then every 4 s.

What PR testers should check

- The WAV volume against the other voices, and the retrigger.
- The intermediate callouts are not too chatty.

47. FWC aural priority: stall warning, TCAS voices, C chord in a TCAS RA

> **Aircraft:** A32NX | **Where:** FWC aurals; GPWS voices; altitude alert | **Status:** Not sim-tested on the A32NX (the same stall logic was flown on the A380X on 2026-11-05; the TCAS part needs conflicting traffic). Unit tests: FwsSoundManager.spec (the PseudoFWC test-bed specs stay on develop).

What it is for

- **Stall warning below 1 500 ft:** the cricket and STALL also sound below 1 500 ft radio height, in alternate and direct law.
- **The stall is the top aural:** while it sounds the other FWC sounds (chimes, cavalry charge, C chord, callouts) wait, a continuous one stops and sounds again after it, and the GPWS voices are silenced (the EGPWC audio inhibit input, which was never set).
- **TCAS voices first:** the radio height callouts, PITCH PITCH and the sidestick priority voices wait while a TCAS voice message is said (new `L:A32NX_TCAS_AURAL_ACTIVE`).
- **C chord in a TCAS RA:** the altitude alert C chord is inhibited during any TCAS RA, also when the AP/FD TCAS mode is not available (A320 FCOM DSC-31-40: "In the case of a TCAS RA").

Limitations and why

Limitation	Why	What to do
A stall warning longer than 60 s sets the GPWS fault.	The EGPWC audio inhibit monitor of FlyByWire flags an inhibit held for more than 60 s.	-
DUAL INPUT cannot sound.	The F/O sidestick sends no input in the sim.	-
A one-shot sound already playing is not cut by the stall.	It ends first.	-

How to use

1. Nothing to set: on the STALL aural fly the stall recovery; the other aurals come back after it.

How to test

#	Do this	Expected result
1	A32NX in flight, ELAC 1 and 2 OFF (alternate law), below 1 500 ft radio height: slow down until the stall warning.	Cricket + STALL; no chime, C chord or GPWS voice during it.
2	With a master caution chime pending (for example a failure from the flyPad) during the stall.	The chime comes after the stall ends.
3	TCAS RA with traffic (needs conflicting traffic), AP/FD TCAS mode not engaged, near the FCU altitude.	No C chord during the RA; radio height callouts wait while a TCAS voice plays.

48. Engine exceedance marks only on displayed values

> **Aircraft:** A32NX | **Where:** E/WD N1, N2 and EGT gauges | **Status:** Not sim-tested (unit tests: EngineLimits.spec.ts).

What it is for

The red N1, N2 and EGT exceedance marks and the N2 red cross latched on any value, also while the gauge showed XX (FADEC not powered) and on invalid numbers. FCOM: a value that is not available shows XX instead (A320 FCOM). They now only latch on valid, displayed values and keep the highest value. The red crosses seen in flight came from the old CFM56 N2 limit of 105 %, already replaced by the LEAP-1A 116.5 % (see "Engine stall, EGT overtemperature, overspeed and vibration failures"). The A380X part is on `core/a380x-elec-fuel-alerts`.

Limitations and why

Limitation	Why	What to do

The marks follow the FADEC power flag (L:A32NX_FADEC_POWERED_ENGn) for the validity.	A value without FADEC power is shown as XX, not as a value (FCOM).	-

How to use

1. Nothing to set.

How to test

#	Do this	Expected result
1	Cold and dark, then power up and start both engines normally.	No red mark and no red cross on the N1, N2 and EGT gauges.
2	Before the start, on batteries only (FADEC not powered: XX on the gauges); then power up and start.	No red mark appears while XX shows or when the values come.

What PR testers should check

- No red exceedance mark or cross after a normal flight.

49. DC BUS 2 takes over DC BUS 1 after 5 s

> **Aircraft:** A32NX | **Where:** SD ELEC page; flyPad > Failures > ATA 24 Electrical Power | **Status:** Not sim-tested (Rust: 3 new test bed tests; the TR 1 fault distribution test now waits 5 s).

What it is for

A320 FCOM DSC-24-10-30-30, FAILURE OF AC BUS 1 : "DC BUS 2 supplies DC BUS 1 and DC BAT BUS automatically after 5 s." The model switched at once. Now, after a TR 1 or AC BUS 1 loss, DC BUS 1 and the DC BAT BUS stay unpowered for 5 s, then DC BUS 2 supplies them. The DC ESS BUS stays on the ESS TR as before. A failed DC BUS 1 busbar still transfers at once, and the TR 2 case stays immediate (no FCOM delay). This changes the AC BUS 1 FAULT case of "ELEC bus, ESS bus and TR alerts" and of the recovery section below.

Limitations and why

Limitation	Why	What to do
A full MSFS restart is needed after an update.	The systems WASM module changes.	Restart the sim after installing.

How to use

1. Nothing to set; after an AC BUS 1 or TR 1 loss, wait 5 s for DC BUS 1 to come back.

How to test

#	Do this	Expected result
1	Cruise, SD ELEC page. flyPad > Failures > ATA 24: **TR 1**.	DC BUS 1 and DC BAT BUS amber for about 5 s, then green again (supplied through DC BUS 2). Deactivate.
2	**AC 1**.	ELEC AC BUS 1 FAULT; DC BUS 1 and DC BAT BUS dark for 5 s, then back. Deactivate.

What PR testers should check

- After a TR 1 or AC BUS 1 failure, DC BUS 1 and DC BAT BUS are dark for 5 s, then recover.
- No odd ECAM flicker in that window.

50. Recovery after electrical failures: FAC reset, AC BUS 1 transfer, SEC test, IDG

> **Aircraft:** A32NX | **Where:** Overhead FLT CTL panel (FAC, SEC pbs), ELEC panel (IDG pbs); flyPad > Failures > ATA 24 | **Status:** FAC reset tested in the sim by the user: OK (DC BUS 1+2 FAULT before the DC ESS fix, FAC 1 and FAC 2 OFF THEN ON gave normal law back; on the DC ESS fix: only FAC 2 FAULT, cleared by FAC 2 OFF THEN ON). AC BUS 1 transfer: OK (before the DC ESS fix). SEC power-up test and IDG: not sim-tested on this branch.

What it is for

Removing a flyPad electrical failure gives the power back, but in flight some computers stay failed until the crew resets them, as on the aircraft. This section says what stays failed after the failures of this branch and how to recover it. Every statement was checked against the code of this branch and the A320 FCOM.

FAC FAULT stays after the power comes back

Each FAC watches its power supply (`Fac.cpp`). A cut of 10 to 200 ms stops it and it restarts by itself; after a cut of more than **200 ms** its memory is lost and it latches FAULT. On the ground (nose gear compressed) it comes back by itself; **in flight only its pushbutton brings it back: FAC OFF then ON**. This is the FCOM procedure: AUTO FLT FAC 1(2) FAULT "FAC (AFFECTED) ... OFF THEN ON" and AUTO FLT FAC 1+2 FAULT "FAC 1 ... OFF THEN ON, FAC 2 ... OFF THEN ON" . FAC 1 is supplied by the DC ESS (shed) bus, FAC 2 by DC BUS 2.

Situation	What you should see
DC BUS 1+2 FAULT (flyPad *DC 1 + DC 2*), then the failures removed	The ESS TR keeps the DC ESS BUS powered (this branch, FCOM DSC-24-10-30-30), so FAC 1, AP 1 and normal law stay. FAC 2 lost its power: **AUTO FLT FAC 2 FAULT** stays after DC BUS 2 is back (the FCOM INOP SYS of DC BUS 1+2 FAULT lists FAC 2, AP 2 and FCU 2). Recover with **FAC 2 OFF then ON**.
DC EMER CONFIG (flyPad *DC 1 + DC 2 + DC ESS*), then the failures removed	Both FACs lost their power: **AUTO FLT FAC 1+2 FAULT** stays (alternate law, no AP, no A/THR, rudder with care). Recover with **FAC 1 OFF then ON, then FAC 2 OFF then ON**; normal law, AP and A/THR come back.
Any other FAC FAULT after a power cut in flight	Same action: the affected FAC OFF then ON. If it does not come back, leave it OFF (FCOM).

AC BUS 1 FAULT: a few seconds on the AC ESS BUS transfer

AC BUS 1 normally supplies the AC ESS BUS. With AC BUS 1 lost, the AC ESS FEED automatic switching moves the AC ESS BUS to AC BUS 2 (FCOM); on this branch that takes **3 s** (`AC_ESS_FEED_TO_AC_BUS_2_DELAY_IN_SECONDS`). FCOM: "the AC ESS BUS and the DC ESS BUS will automatically recover, due to the fact that the AC BUS 2 will automatically supply the AC ESS BUS" . During the transfer the captain PFD, the captain ND and the E/WD, which read the AC ESS BUS, go blank for **up to 3 s** and then come back by themselves.

> **NOTE**
> Observation, not rechecked: in an earlier sim test the AP disconnected (AP OFF warning) during this transfer. That test was flown before this branch's fix of the DC ESS power break on DC bus changes, which also dropped the AP and A/THR. If the AP drops, re-engage it once the displays are back.

SEC power-up test after a power cut

After a power cut of more than 20 ms the SEC runs its 30 s power-up test when its power comes back, also in flight (`Sec.cpp`, the FlyByWire logic): its FAULT light comes on during the first 2.5 s and again from 27 to 29.5 s, and the

E/WD can show F/CTL SEC FAULT for a moment. No action: the SEC is back after the test. (A fork change that skips this test in flight is only, not on this branch.) The ELACs and FCDCs restart by themselves when their power comes back.

IDG disconnection is final

The IDG pushbutton disconnects the drive from the engine for good: on the aircraft "only maintenance personnel can reconnect it" (FCOM). On this branch the drive is never reconnected for the rest of the flight, in the air or on the ground (`engine_generator.rs`); an IDG oil overheat above 200 °C also disconnects it by itself (thermal disconnect). That engine generator stays lost: the other generator, or the APU GEN, supplies its side.

> **NOTE**
> Disconnect the IDG only with the engine running or windmilling, and do not hold the IDG pushbutton for more than 3 s (FCOM).

How to use

1. After an electrical failure is over, read the E/WD: an **AUTO FLT FAC 1(2) FAULT** or **FAC 1+2 FAULT** that stays means the FAC latched. Overhead FLT CTL panel: that **FAC pushbutton OFF, then ON** (FAC 1 first when both).
2. Check the FAULT light goes off and the E/WD alert goes; re-engage the AP and the A/THR (FCU).
3. With AC BUS 1 lost: wait for the captain displays to come back (3 s), then re-engage the AP if it dropped.
4. A short F/CTL SEC FAULT about 28 s after the power came back is the SEC power-up test: no action.
5. Before pushing an IDG pushbutton, remember the generator is lost for the rest of the flight.

How to test

#	Do this	Expected result
1	A32NX in cruise, AP 1 and A/THR on. flyPad > Failures > ATA 24: activate *DC 1*, then *DC 2*.	ELEC DC BUS 1+2 FAULT. The DC ESS BUS stays powered: normal law, AP 1 and A/THR stay engaged. AUTO FLT FAC 2 FAULT shows (FAC 2 on DC BUS 2).
2	flyPad: deactivate *DC 1* and *DC 2*. Wait 30 s.	DC BUS 1 and 2 are back; AUTO FLT FAC 2 FAULT stays, FAC 2 FAULT light on.
3	Overhead FLT CTL panel: **FAC 2** pushbutton OFF, then ON.	The FAC 2 FAULT light goes off, the alert clears; AP 2 is available again.
4	flyPad: activate *DC 1*, *DC 2* and *DC ESS* (DC EMER CONFIG), wait 10 s, then deactivate the three.	AUTO FLT FAC 1+2 FAULT stays after the failures are removed; alternate law, no AP, no A/THR.
5	FAC 1 OFF then ON, then FAC 2 OFF then ON.	Both FAULT lights off, the alert clears, normal law; AP and A/THR can be engaged again.
6	flyPad: activate *AC 1*.	ELEC AC BUS 1 FAULT; the captain PFD, ND and the E/WD blank for up to 3 s and come back (AC ESS BUS on AC BUS 2). If the AP dropped, re-engage it.
7	Within about 30 s after a DC bus came back, watch the SEC pushbuttons.	A SEC FAULT light may come on briefly (power-up test, around 27 s) and go off by itself.

51. Crew procedures: manual cabin pressure, ignition indications, engine fire, A/THR without V2

> **Aircraft:** A32NX | **Where:** Overhead CABIN PRESS panel, ENG FIRE pbs; pedestal ENG MODE selector; FCU | **Status:** Manual cabin V/S tested in the sim by the user: OK (- with short taps: cabin V/S at most 2,070 ft/min). Engine fire run-on: OK. IGNITION memo: FAILED (no memo with continuous ignition on), open. A/THR with V2: OK; the no-V2 procedure is not sim-tested.

What it is for

Procedures and indications that surprised during the sim tests: how to fly the cabin pressure by hand, why the SD shows no IGN with the continuous ignition on, why an engine keeps running after its FIRE pushbutton, and how to get the A/THR without a V2. Checked against the code of this branch and the A320 FCOM.

Manual cabin pressure control (MAN V/S CTL)

When both pressure controllers are lost (CAB PR SYS 1+2 FAULT: MODE SEL MAN, MAN V/S CTL AS RQRD, FCOM; in a DC EMER CONFIG test the cabin climbed at about 5,500 ft/min with CAB PR EXCESS CAB ALT), the crew drives the outflow valve with the MAN V/S CTL switch: **UP opens** the valve (cabin climbs), **DN closes** it (cabin descends) (FCOM). The manual motor, on the DC BAT BUS, moves the valve over its full travel in **55 s** (`pressure_valve.rs`). In cruise the valve is only a few percent open, so the cabin V/S reacts strongly: about **1,800 ft/min per second** of input in the model (an estimate).

- Use **short taps of 0.2 to 0.5 s**, then wait and watch the cabin V/S on the SD CAB PRESS page: "only short inputs should be applied" (FCOM), and the valve change can take 10 s to show.
- Never hold the switch to drive the valve to a stop in cruise: the cabin climbs or dives by thousands of ft/min.
- Targets from the STATUS page: climb 500 ft/min, descent 300 ft/min; cabin altitude by flight level (FL390 8,000 ft ... FL250 3,000 ft); in final approach MAN V/S CTL FULL UP, delta P zero before opening the doors.
- To recover: **MODE SEL back to AUTO** once a controller works again (e.g. after the electrical failure is removed): the automatic system takes the valve back. MAN for at least 10 s then AUTO also selects the other system (FCOM).

IGNITION memo and the SD IGN indication

The green **IGNITION** memo shows "when continuous ignition is activated on any engine" (FCOM): ENG MODE selector at IGN/START with the engine running, the FADEC automatic ignition, or the engine anti-ice on this branch. The SD ENGINE page shows the igniters (IGN A / B), the start valve and the PSI only with the ENG MODE selector at **IGN/START or CRANK** (`StartIndications.ts`; the ENGINE page comes at the start sequence or at CRANK, FCOM). So with the selector at NORM and the continuous ignition on (engine anti-ice), the SD shows no IGN letters: the E/WD memo is the indication.

> **NOTE**
> Known issue: in a sim test the IGNITION memo did not show although the continuous ignition variable was on. The cause is not found yet.

An engine keeps running after its ENG FIRE pushbutton

The ENG FIRE pushbutton closes the LP fuel valve (FCOM); the engine then runs on the fuel left between the valve and the nozzles and stops by itself, "after a time delay up to 2 min 30 s" (FCOM, on this branch through the LP valve model of `core/fire`). How long it runs on depends on the engine state (thrust, airport altitude): in the model about 1 min 40 s at a sea-level idle, never more than 2 min 30 s. The FCOM ENG FIRE procedure sets the **ENG MASTER OFF first**, which closes the LP and HP valves and stops the combustion without run-on, then pushes the FIRE pushbutton.

A/THR without a V2

The A/THR arms at FLX/TOGA only when SRS engages, and SRS needs a V2: "If V2 is not inserted, the speed reference system (SRS) will not engage for takeoff" (FCOM). Without V2 the A/THR stays off at take-off, V/S engages 5 s after lift-off and the current speed becomes the FCU target. This is the FlyByWire FMGC code (`FmgcComputer.cpp`, unchanged from FlyByWire master), the same on every branch.

1. Normal: enter V2 on the MCDU PERF TAKE OFF page (with the other speeds): the A/THR arms at FLX/TOGA.
2. Without V2, at ACC ALT (FCOM TAKEOFF WITH NO V2 ENTRY): push the **A/THR** pushbutton on the FCU, set the thrust levers to the **CL** detent, push the SPD knob for a managed speed (or select the climb speed and pull it).

3. If the A/THR still does not engage (seen after a quick runway start without V2): push or pull the FCU SPD knob first so that a speed target exists, then push the A/THR pushbutton.

How to use

1. Manual pressure: MODE SEL MAN, short MAN V/S CTL taps (0.2-0.5 s) towards the STATUS target V/S, watch the SD CAB PRESS page; MODE SEL AUTO when a controller is back.
2. Continuous ignition: look for the green IGNITION memo on the E/WD; the SD IGN letters only show at IGN/START or CRANK.
3. Engine fire: follow the ECAM order (THR LEVER IDLE, ENG MASTER OFF, ENG FIRE pb PUSH, AGENT 1 after 10 s).
4. Take-off: always enter V2; if it was not entered, follow the no-V2 steps above at ACC ALT.

How to test

#	Do this	Expected result
1	A32NX in cruise at FL250 or above, cabin pressure AUTO. Overhead CABIN PRESS panel: MODE SEL **MAN**. Open the SD CAB PRESS page.	MAN shows in the MODE SEL pushbutton; the outflow valve stays where it was.
2	MAN V/S CTL: one short tap (0.2-0.5 s) UP, then wait 10 s.	The cabin V/S increases (climb) by a few hundred ft/min (about 1,800 ft/min per second of input); a longer input gives a much larger change.
3	Short taps DN until the cabin V/S is near 0, then MODE SEL **AUTO**.	The automatic system takes the valve back; the cabin V/S returns to its normal schedule.
4	Engines running, ENG MODE selector at NORM. Overhead ANTI ICE panel: ENG 1 pb ON; SD ENGINE page.	E/WD green IGNITION memo (known issue: it did not show in a sim test). The SD shows no IGN letters (selector at NORM).
5	ENG MODE selector to IGN/START.	The SD ENGINE page shows the IGN letters, the start valves and the PSI.
6	On the ground, engines at idle, no fire: ENG MASTER 1 left ON, push the ENG 1 FIRE pushbutton.	The SD FUEL LP valve 1 closes (amber crossline); engine 1 keeps running, then stops by itself within 2 min 30 s.
7	Take-off with V2 entered, thrust levers FLX or TOGA.	FMA: A/THR armed (blue), SRS engaged.
8	Take-off without V2 (PERF page V2 empty), thrust levers FLX or TOGA. At ACC ALT: A/THR pushbutton, thrust levers CL, SPD knob pushed.	No A/THR at take-off; V/S engages 5 s after lift-off; after the pushbutton the A/THR is active in the CL detent.